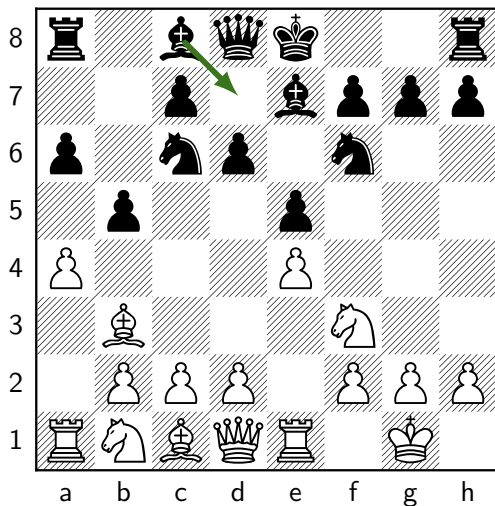


Melanie-BOTS (CCRL engines) — Ruy Lopez, Closed

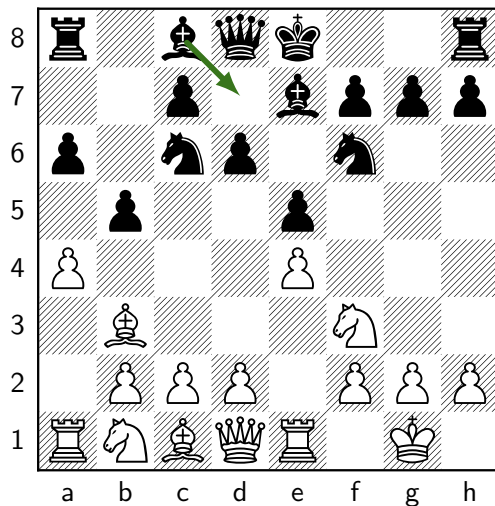
Melanie (Deep_Vision)

Abstract

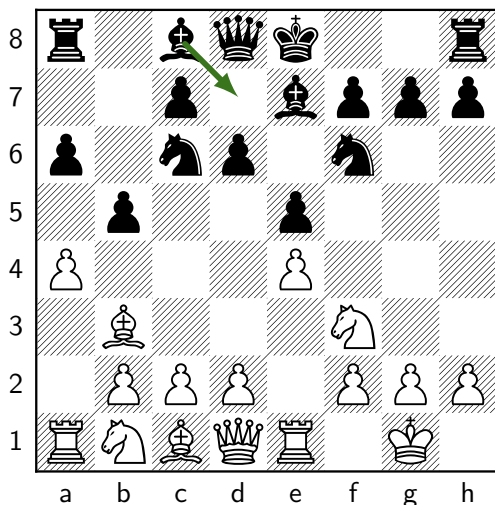
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



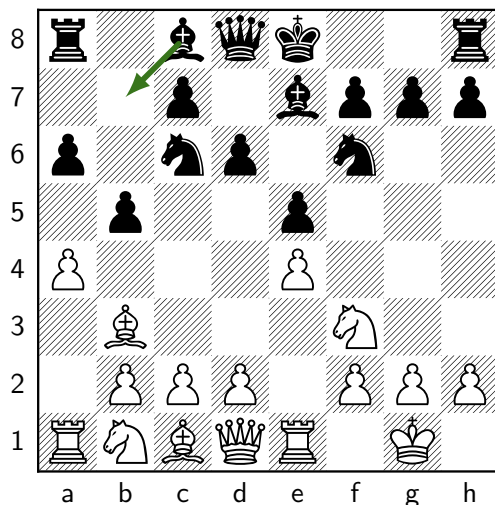
(a)



(b)

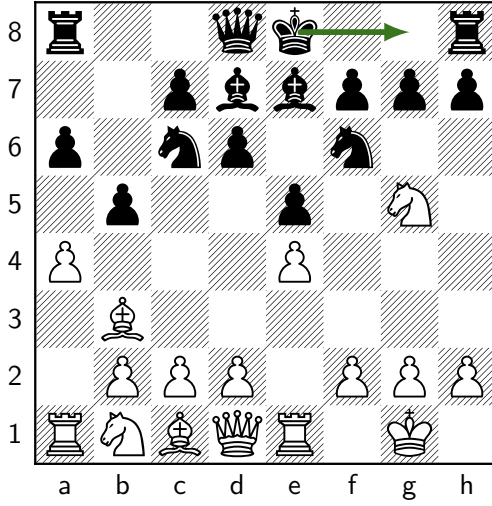


(c)

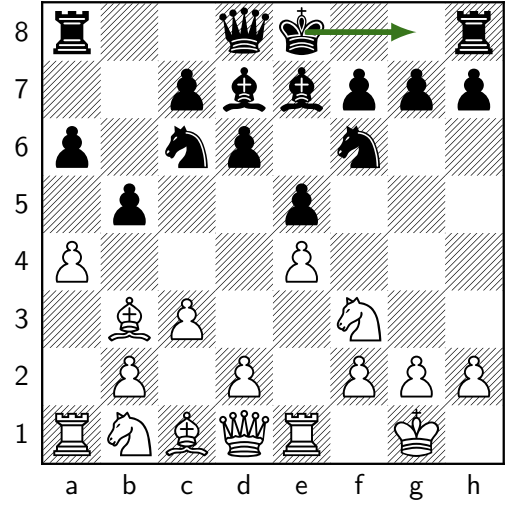


(d)

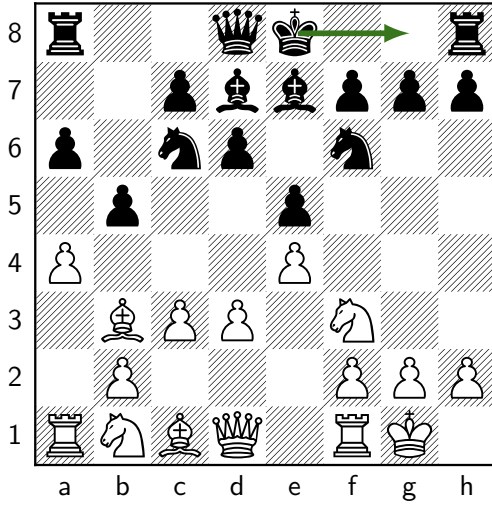
Figure 1: **Move 8, Black: Bd7.** (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#693` (ply 15) [structural distance $d=0$, QAP obj=102]: the master played Bd7; under μ our *Bc1* plays the role of that game’s *Bc1*; transferred move c8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#694` (ply 15) [structural distance $d=0$, QAP obj=102]: the master played Bd7; under μ our *Bc1* plays the role of that game’s *Bc1*; transferred move c8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3035` (ply 15) [structural distance $d=0$, QAP obj=102]: the master played Bb7; under μ our *Bc1* plays the role of that game’s *Bc1*; transferred move c8b7. Synthesis: 2 retrieved analogs transferred to Bd7 (summed QAP objective 204, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



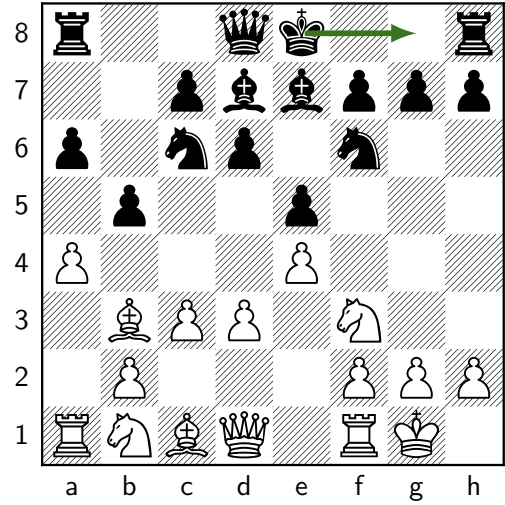
(a)



(b)

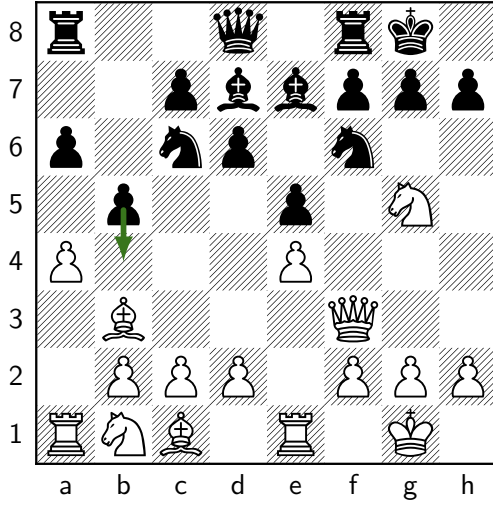


(c)

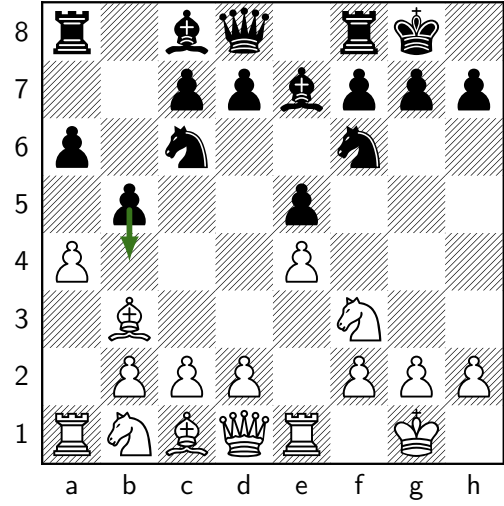


(d)

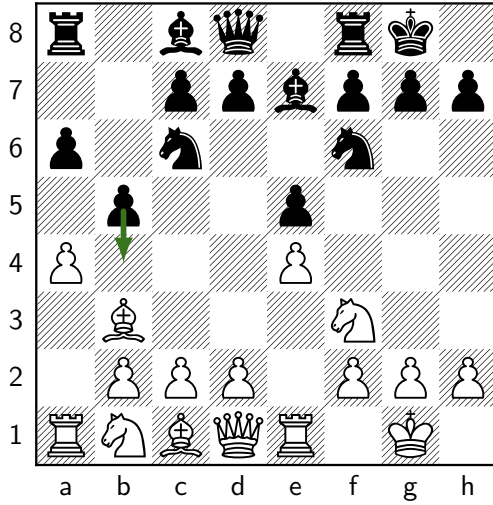
Figure 2: **Move 9, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#693` (ply 17) [structural distance $d=72$, QAP obj=70]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pe5* as their *Ra8*, our *nb1* as their *Pg7*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2193` (ply 17) [structural distance $d=69$, QAP obj=69]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pd6* as their *Ra8*, our *bc1* as their *Pg7*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2917` (ply 17) [structural distance $d=69$, QAP obj=69]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pd6* as their *Ra8*, our *bc1* as their *Pg7*; transferred move e8g8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to 0-0 (summed QAP objective 209, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



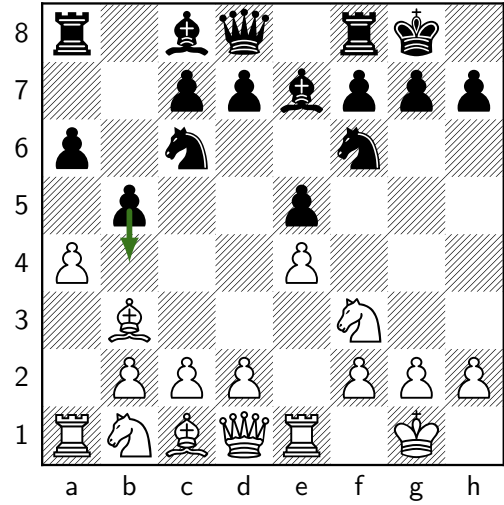
(a)



(b)

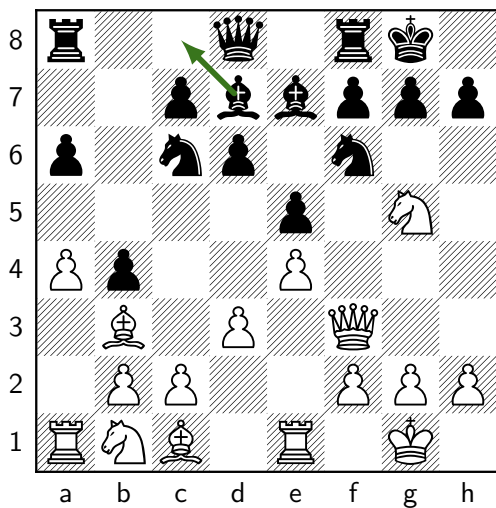


(c)

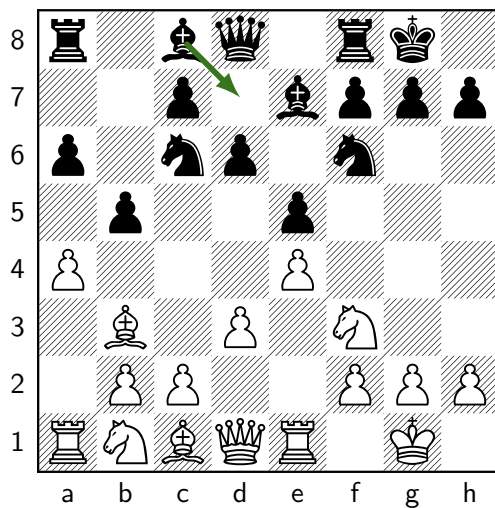


(d)

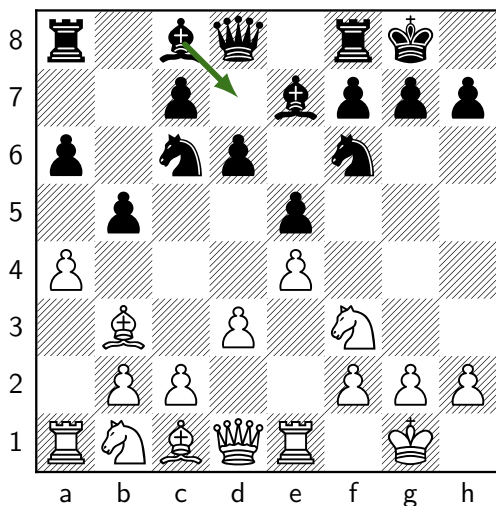
Figure 3: **Move 10, Black: b4.** (a) Our position; Melanie plays **b4** (**b5b4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1188` (ply 15) [structural distance $d=84$, QAP obj=96]: the master played **b4**; under μ our Pb_4 plays the role of that game’s Pb_4 ; cross-type roles: our Be_7 as their Ra_8 , our Pc_7 as their Bc_8 ; transferred move **b5b4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1189` (ply 15) [structural distance $d=84$, QAP obj=96]: the master played **b4**; under μ our Pb_4 plays the role of that game’s Pb_4 ; cross-type roles: our Be_7 as their Ra_8 , our Pc_7 as their Bc_8 ; transferred move **b5b4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2558` (ply 15) [structural distance $d=84$, QAP obj=96]: the master played **b4**; under μ our Pb_4 plays the role of that game’s Pb_4 ; cross-type roles: our Be_7 as their Ra_8 , our Pc_7 as their Bc_8 ; transferred move **b5b4** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **b4** (summed QAP objective 576, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



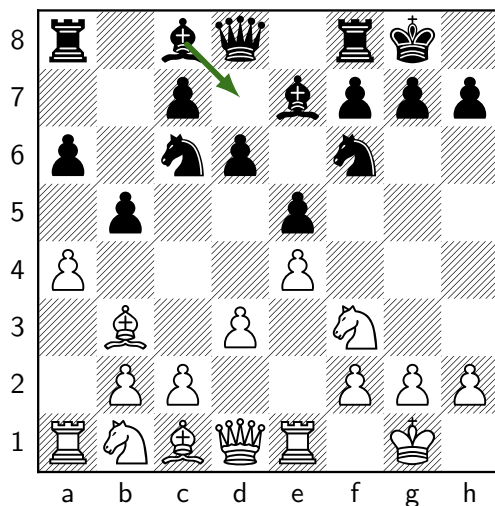
(a)



(b)

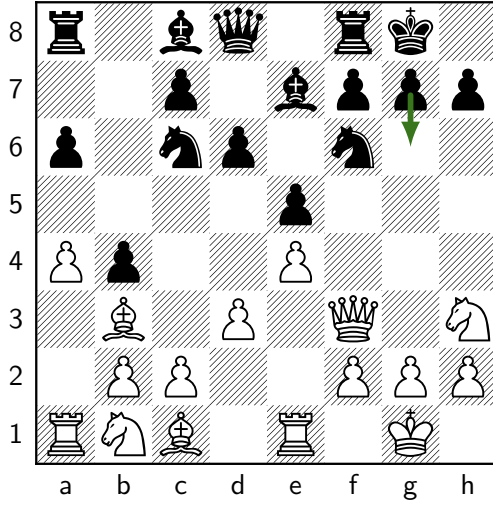


(c)

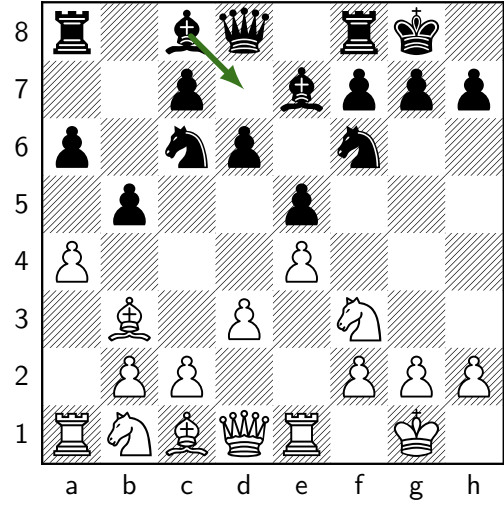


(d)

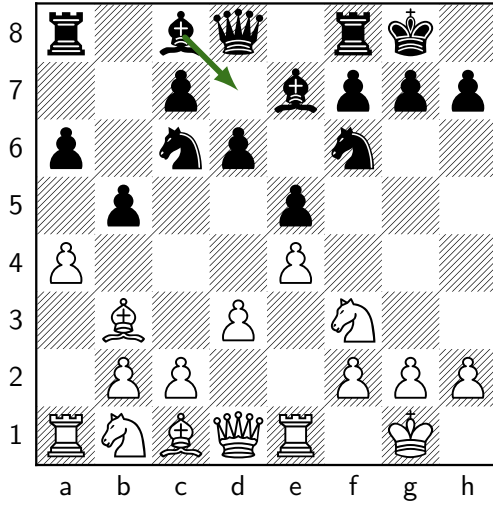
Figure 4: **Move 11, Black: Bc8.** (a) Our position; Melanie plays Bc8 (d7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3956` (ply 17) [structural distance $d=74$, QAP obj=70]: the master played Bd7; under μ our Bd2 plays the role of that game’s Bc1; cross-type roles: our pa4 as their Ra8, our Ra8 as their Be7; transferred move d7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#290` (ply 17) [structural distance $d=74$, QAP obj=70]: the master played Bd7; under μ our Bd2 plays the role of that game’s Bc1; cross-type roles: our pa4 as their Ra8, our Ra8 as their Be7; transferred move d7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#291` (ply 17) [structural distance $d=74$, QAP obj=70]: the master played Bd7; under μ our Bd2 plays the role of that game’s Bc1; cross-type roles: our pa4 as their Ra8, our Ra8 as their Be7; transferred move d7c8 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Bc8 (summed QAP objective 906, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



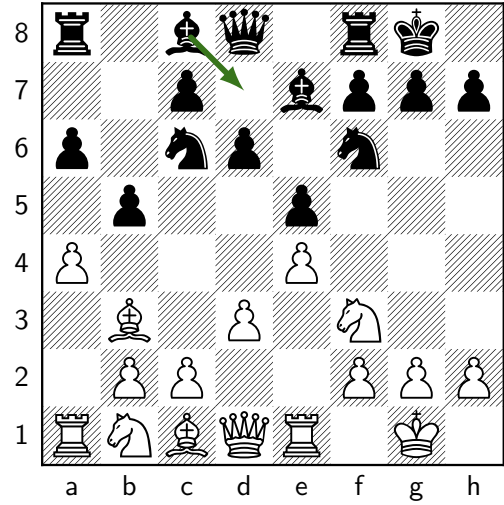
(a)



(b)

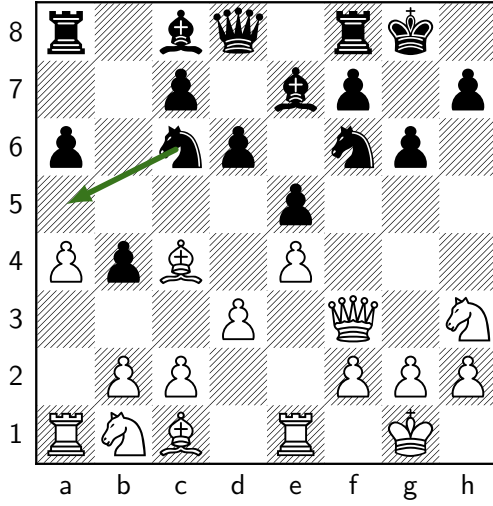


(c)

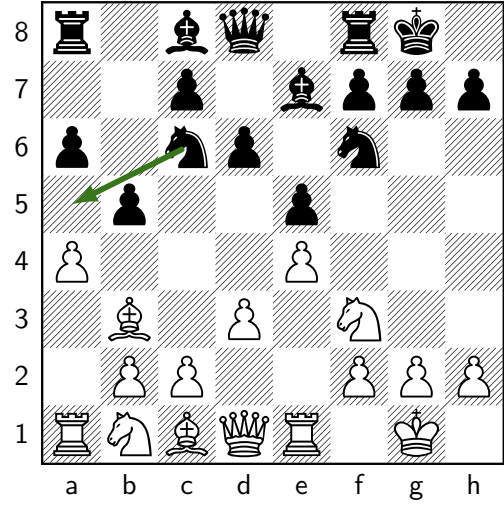


(d)

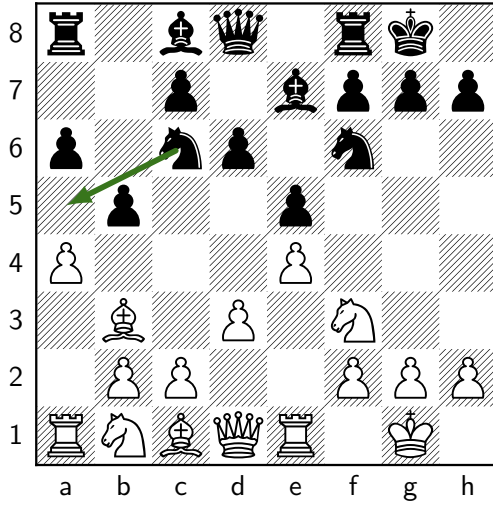
Figure 5: **Move 12, Black: g6 (g7g6).** (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3956` (ply 17) [structural distance $d=50$, QAP obj=66]: the master played Bd7; under μ our Pg2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Ra8, our pb2 as their Qd8; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#290` (ply 17) [structural distance $d=50$, QAP obj=66]: the master played Bd7; under μ our Pg2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Ra8, our pb2 as their Qd8; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#291` (ply 17) [structural distance $d=50$, QAP obj=66]: the master played Bd7; under μ our Pg2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Ra8, our pb2 as their Qd8; transferred move g7g6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to g6 (summed QAP objective 661, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



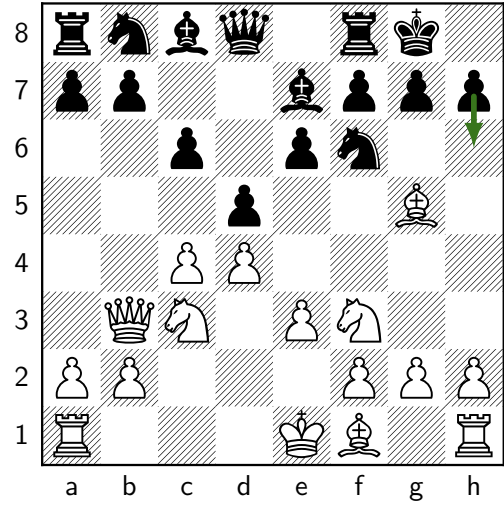
(a)



(b)

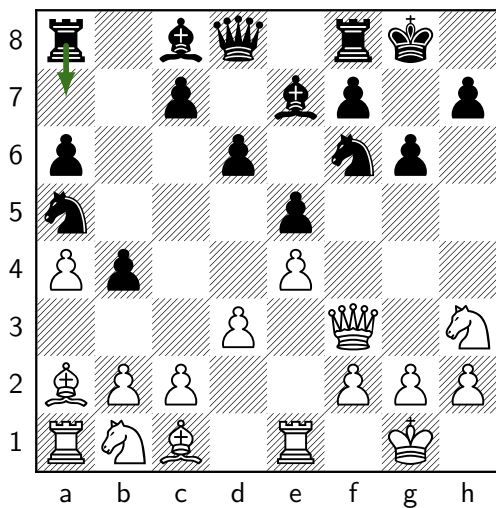


(c)

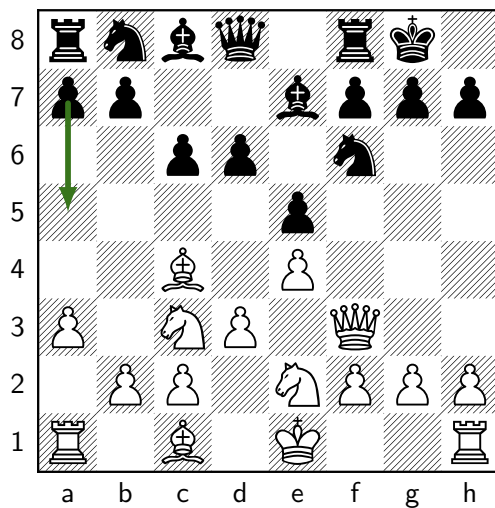


(d)

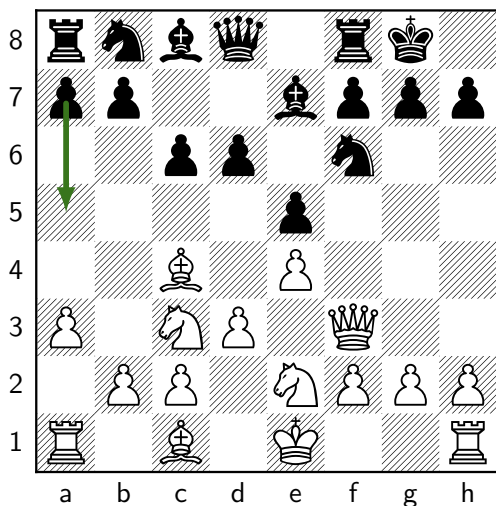
Figure 6: **Move 13, Black: Na5.** (a) Our position; Melanie plays Na5 (c6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3957` (ply 17) [structural distance $d=58$, QAP obj=64]: the master played Na5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *Ph7* as their *Ra8*, our *ph2* as their *Bc8*; transferred move c6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2621` (ply 17) [structural distance $d=58$, QAP obj=64]: the master played Na5; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *Ph7* as their *Ra8*, our *ph2* as their *Bc8*; transferred move c6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3307` (ply 13) [structural distance $d=81$, QAP obj=42]: the master played h6; under μ our *Qd1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Pb4* as their *Ra8*, our *pb2* as their *Nb8*; transferred move d8e8. Synthesis: 2 retrieved analogs transferred to Na5 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



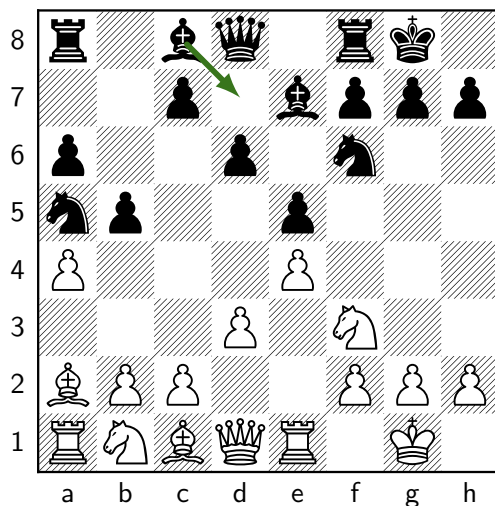
(a)



(b)

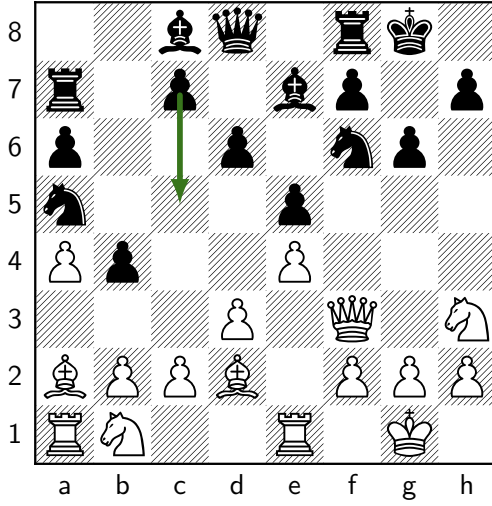


(c)

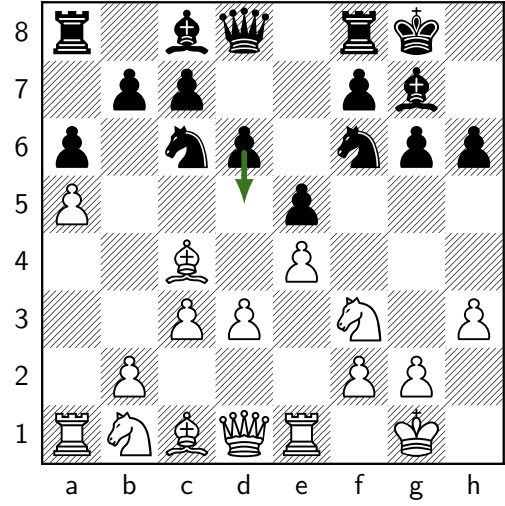


(d)

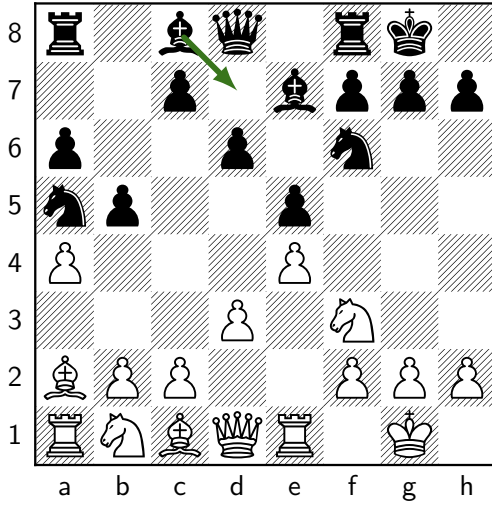
Figure 7: **Move 14, Black: Ra7**. (a) Our position; Melanie plays Ra7 (a8a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#160` (ply 13) [structural distance $d=64$, QAP obj=80]: the master played a5; under μ our *Ra1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *nh3* as their *Ra8*, our *pc2* as their *Nb8*; transferred move a8a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1107` (ply 13) [structural distance $d=64$, QAP obj=80]: the master played a5; under μ our *Ra1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *nh3* as their *Ra8*, our *pc2* as their *Nb8*; transferred move a8a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3957` (ply 19) [structural distance $d=63$, QAP obj=70]: the master played Bd7; under μ our *Pd3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pe5* as their *Qd8*, our *Qd8* as their *Pg7*; transferred move d6d5. Synthesis: 2 retrieved analogs transferred to Ra7 (summed QAP objective 159, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



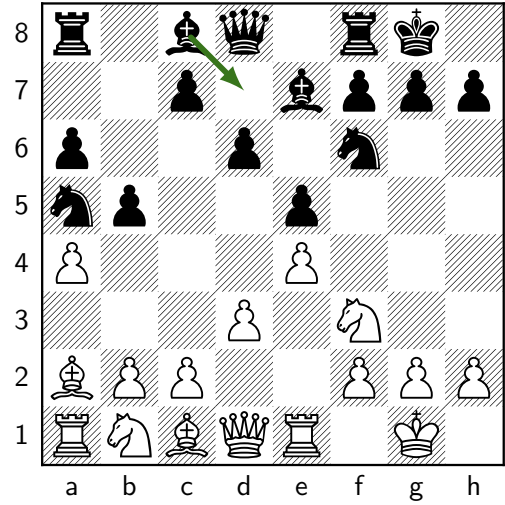
(a)



(b)

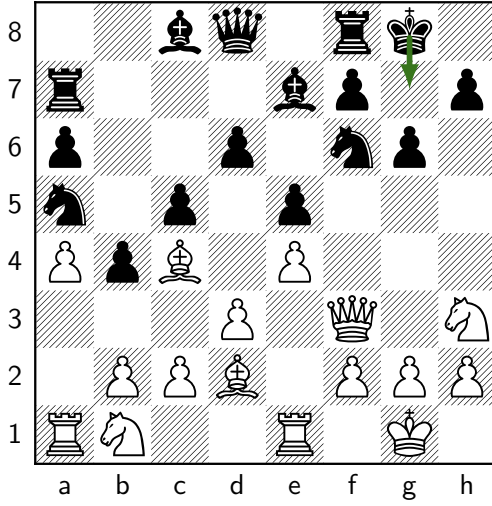


(c)

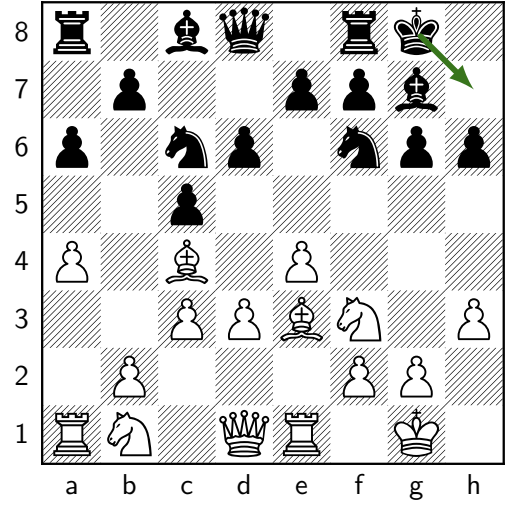


(d)

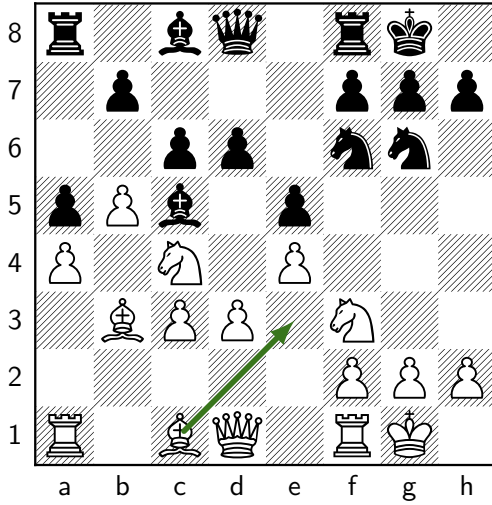
Figure 8: **Move 15, Black: c5.** (a) Our position; Melanie plays c5 (c7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3888` (ply 19) [structural distance $d=73$, QAP obj=61]: the master played d5; under μ our $Pd3$ plays the role of that game’s $Pd3$; cross-type roles: our $Pe5$ as their $Ra8$, our $Pg6$ as their $Bc8$; transferred move d6d5. (c) Retrieved from `ccrl_top_p0000#3957` (ply 19) [structural distance $d=56$, QAP obj=57]: the master played Bd7; under μ our $Pg3$ plays the role of that game’s $Bc1$ (a **cross-type** correspondence); cross-type roles: our $Na5$ as their $Ra8$, our $Pc7$ as their $Qd8$; transferred move g6g5. (d) Retrieved from `ccrl_top_p0002#2621` (ply 19) [structural distance $d=56$, QAP obj=57]: the master played Bd7; under μ our $Pg3$ plays the role of that game’s $Bc1$ (a **cross-type** correspondence); cross-type roles: our $Na5$ as their $Ra8$, our $Pc7$ as their $Qd8$; transferred move g6g5. Synthesis: no retrieved analog transferred to c5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



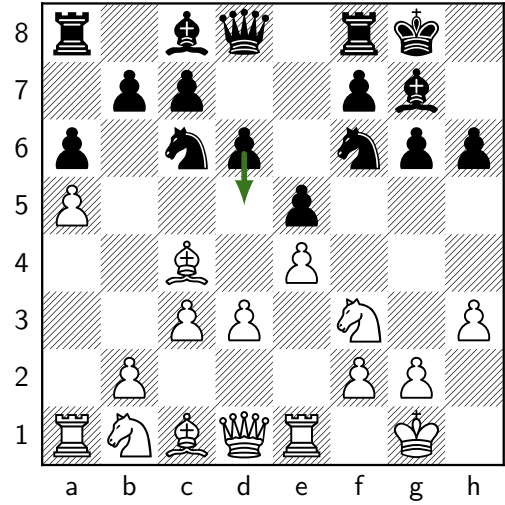
(a)



(b)

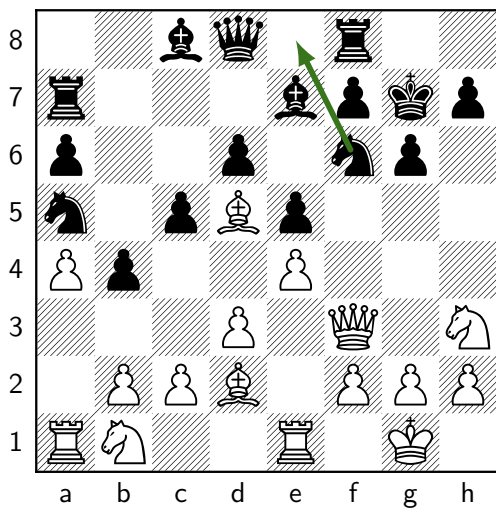


(c)

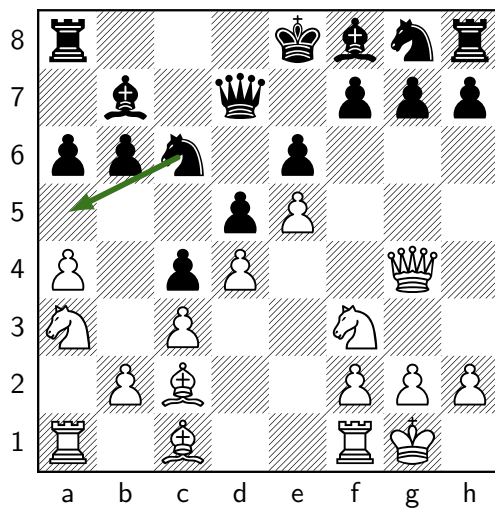


(d)

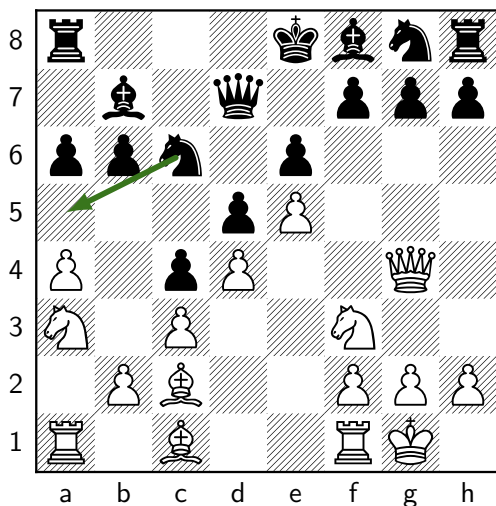
Figure 9: **Move 16, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1479` (ply 19) [structural distance $d=77$, QAP obj=57]: the master played Kh7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our pd3 as their Bc8, our ra1 as their Pb7; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1827` (ply 24) [structural distance $d=92$, QAP obj=66]: the master played Be3; under μ our Pd3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Be7 as their Qd1, our qf3 as their Pg2; transferred move d6d5. (d) Retrieved from `ccr1_top_p0000#3888` (ply 19) [structural distance $d=66$, QAP obj=60]: the master played d5; under μ our Pg3 plays the role of that game’s Pd3; cross-type roles: our pf2 as their Ra8, our pa4 as their Bc8; transferred move g6g5. Synthesis: 1 retrieved analog transferred to Kg7 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



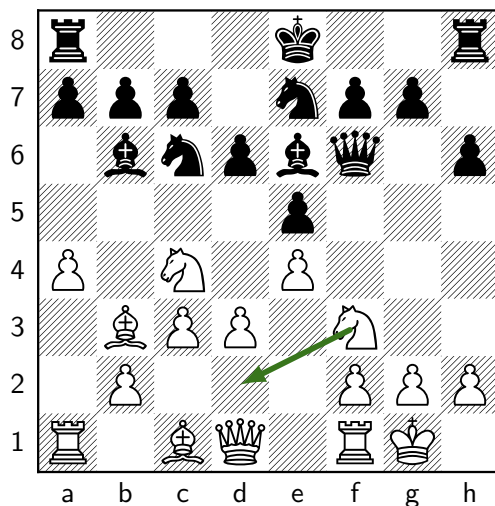
(a)



(b)

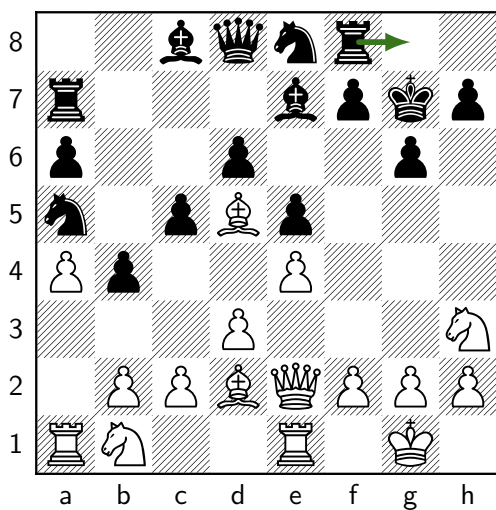


(c)

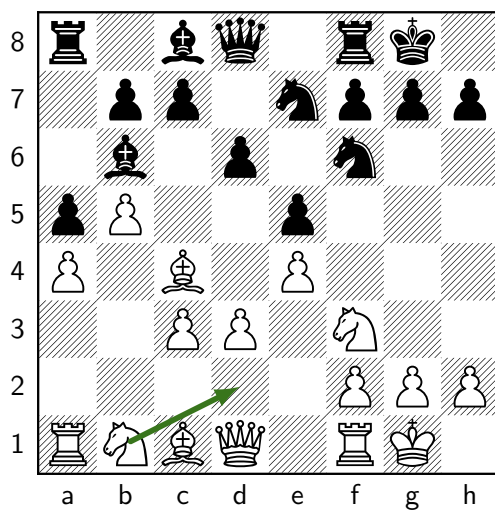


(d)

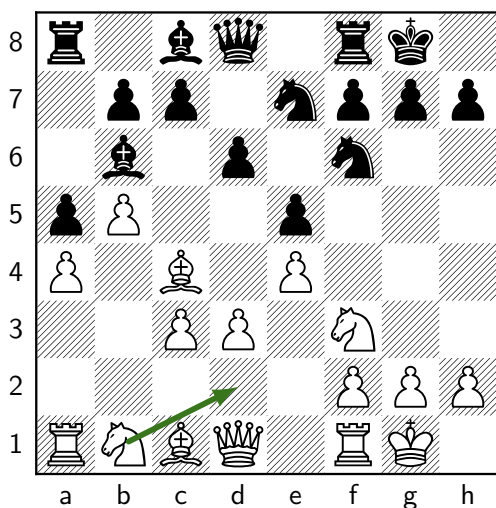
Figure 10: **Move 17, Black: Ne8.** (a) Our position; Melanie plays Ne8 (f6e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1078` (ply 25) [structural distance $d=107$, QAP obj=73]: the master played Na5; under μ our $Nf3$ plays the role of that game’s $Nc3$; cross-type roles: our $qf3$ as their $Bf8$, our $Pc5$ as their $Rh8$; transferred move f6e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1079` (ply 25) [structural distance $d=107$, QAP obj=73]: the master played Na5; under μ our $Nf3$ plays the role of that game’s $Nc3$; cross-type roles: our $qf3$ as their $Bf8$, our $Pc5$ as their $Rh8$; transferred move f6e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3027` (ply 20) [structural distance $d=109$, QAP obj=116]: the master played Nfd2; under μ our $Nf3$ plays the role of that game’s $Nf3$; cross-type roles: our $pb2$ as their $Bc1$, our $Pg6$ as their $Rf1$; transferred move f6d7. Synthesis: 2 retrieved analogs transferred to Ne8 (summed QAP objective 146, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



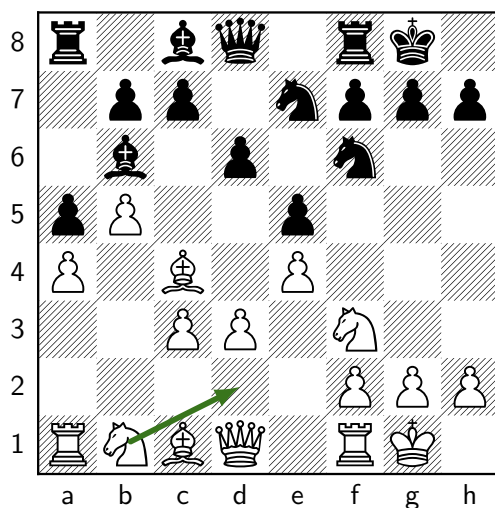
(a)



(b)

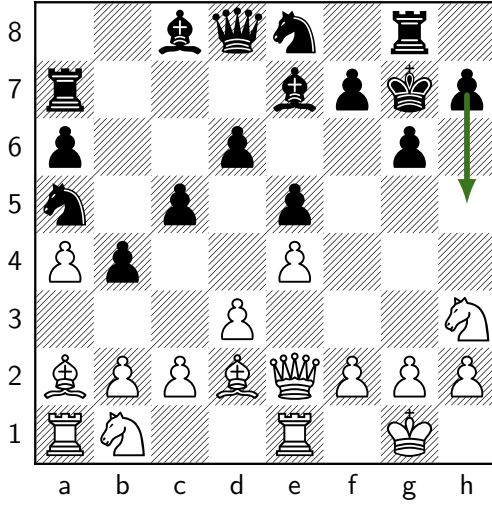


(c)

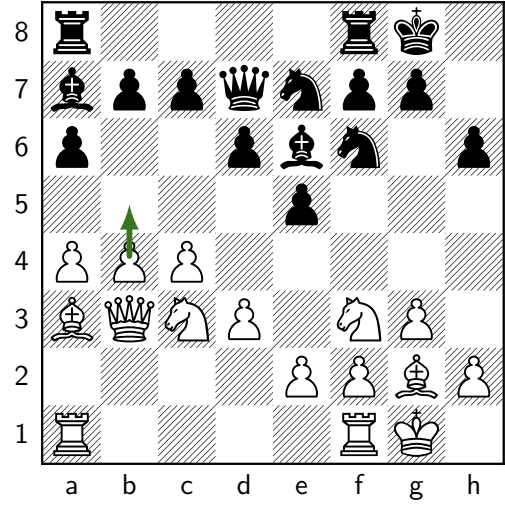


(d)

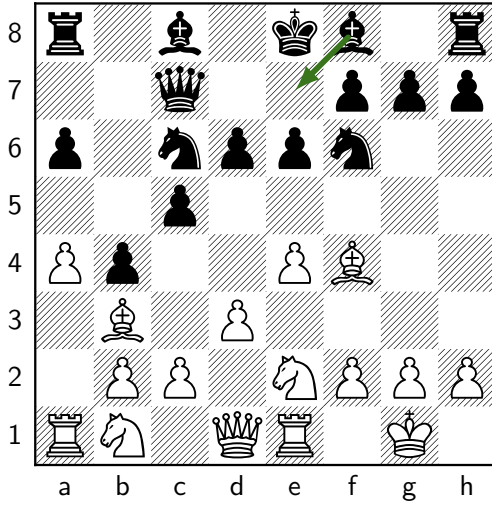
Figure 11: **Move 18, Black: Rg8**. (a) Our position; Melanie plays Rg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2200` (ply 18) [structural distance $d=65$, QAP obj=70]: the master played Nbd2; under μ our Rf1 plays the role of that game's Nb1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ra1, our Na5 as their Bc1; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#79` (ply 18) [structural distance $d=65$, QAP obj=70]: the master played Nbd2; under μ our Rf1 plays the role of that game's Nb1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ra1, our Na5 as their Bc1; transferred move f8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1486` (ply 18) [structural distance $d=65$, QAP obj=70]: the master played Nbd2; under μ our Rf1 plays the role of that game's Nb1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ra1, our Na5 as their Bc1; transferred move f8g8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rg8 (summed QAP objective 279, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



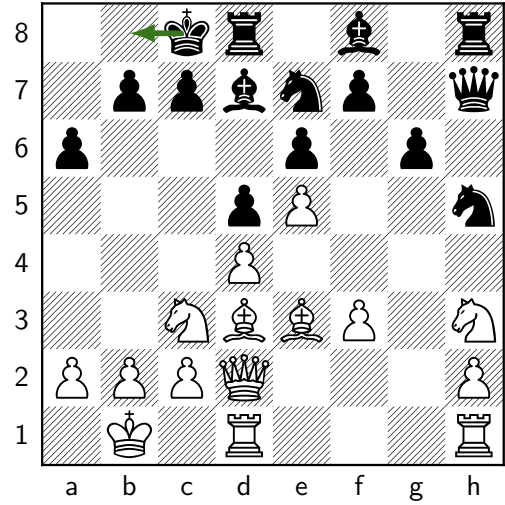
(a)



(b)

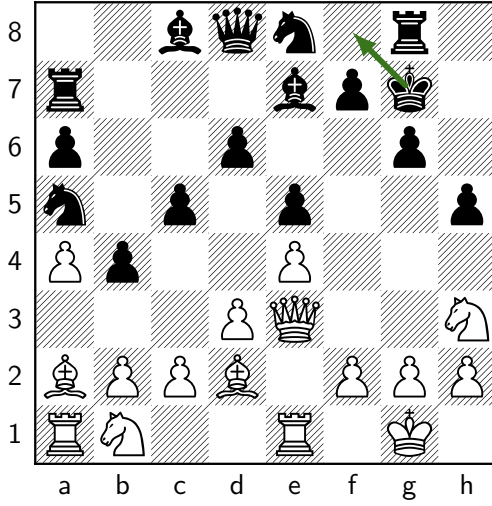


(c)

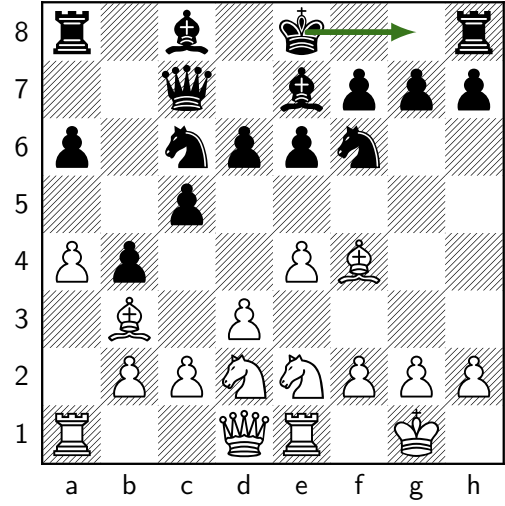


(d)

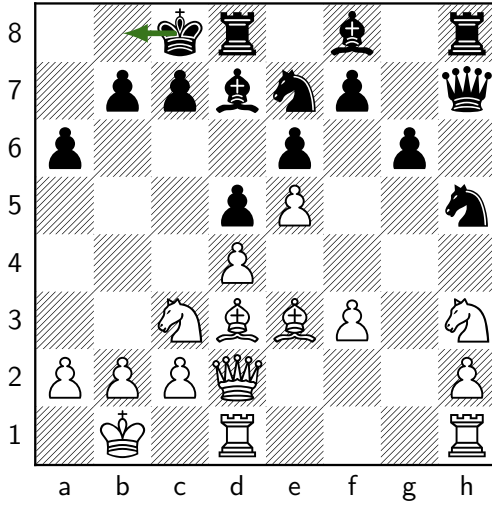
Figure 12: **Move 19, Black: h5**. (a) Our position; Melanie plays h5 (h7h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#1270 (ply 24) [structural distance $d=84$, QAP obj=70]: the master played b5; under μ our *Ph2* plays the role of that game’s *Pb4*; cross-type roles: our *pa4* as their *Ra1*, our *Na5* as their *Bg2*; transferred move h7h5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#957 (ply 23) [structural distance $d=74$, QAP obj=62]: the master played Be7; under μ our *Pc4* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Ra8*, our *ph2* as their *Bc8*; transferred move c5c4. (d) Retrieved from ccr1_top_p0001#58 (ply 35) [structural distance $d=81$, QAP obj=59]: the master played Kb8; under μ our *Kg2* plays the role of that game’s *Kc1*; cross-type roles: our *Ne8* as their *Rd8*, our *Qd8* as their *Bf8*; transferred move g7h8. Synthesis: 1 retrieved analog transferred to h5 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



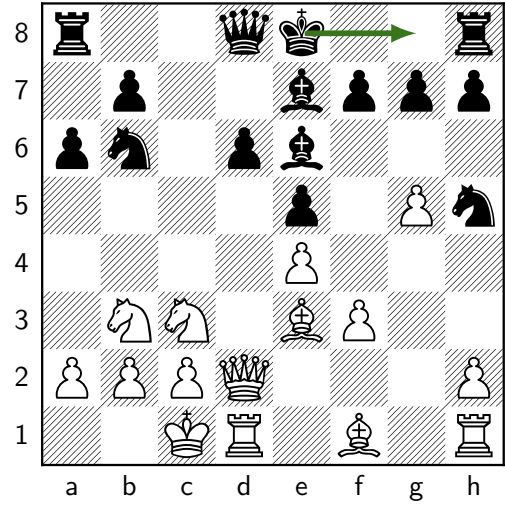
(a)



(b)

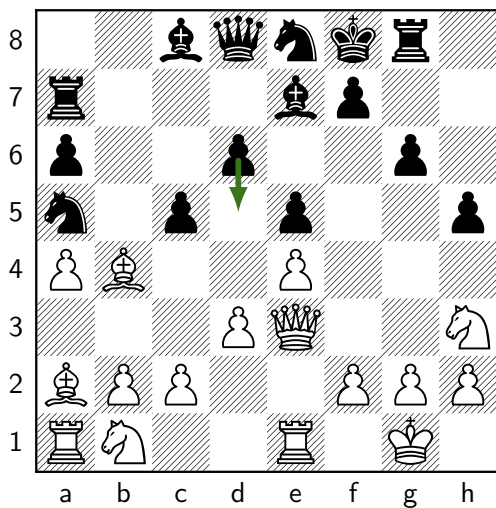


(c)

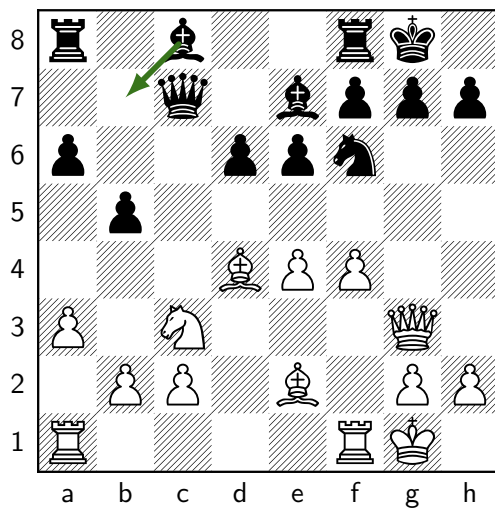


(d)

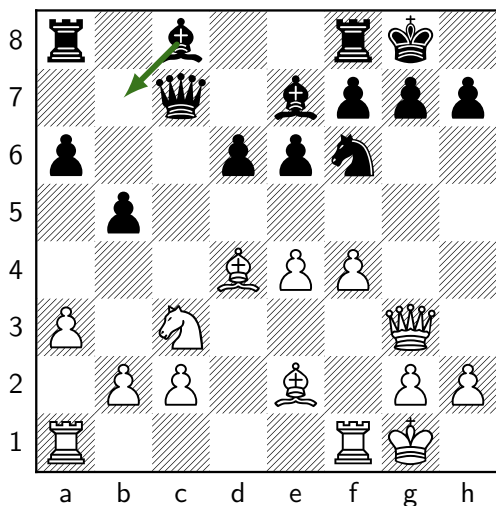
Figure 13: **Move 20, Black: Kf8**. (a) Our position; Melanie plays Kf8 (g7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#957 (ply 25) [structural distance $d=94$, QAP obj=67]: the master played 0-0; under μ our Kg2 plays the role of that game’s Ke1; cross-type roles: our pd3 as their Ra8, our Ne8 as their Bc8; transferred move g7f8 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#58 (ply 35) [structural distance $d=102$, QAP obj=56]: the master played Kb8; under μ our Kg2 plays the role of that game’s Kc1; cross-type roles: our Qd8 as their Bf8, our Bc8 as their Rh8; transferred move g7h7. (d) Retrieved from ccr1_top_p0003#1175 (ply 23) [structural distance $d=105$, QAP obj=53]: the master played 0-0; under μ our Kg2 plays the role of that game’s Ke1; cross-type roles: our Qd8 as their Ra8, our Rg8 as their Qd8; transferred move g7h7. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



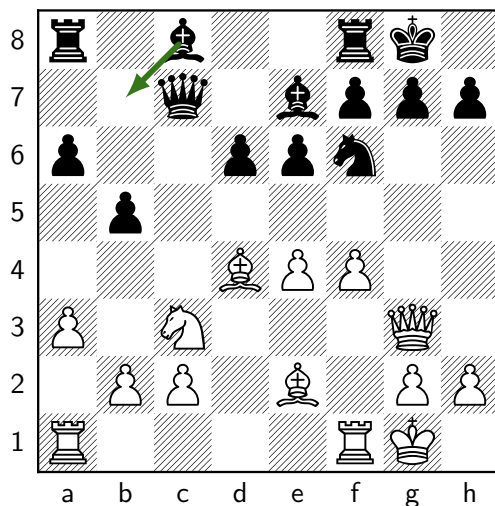
(a)



(b)

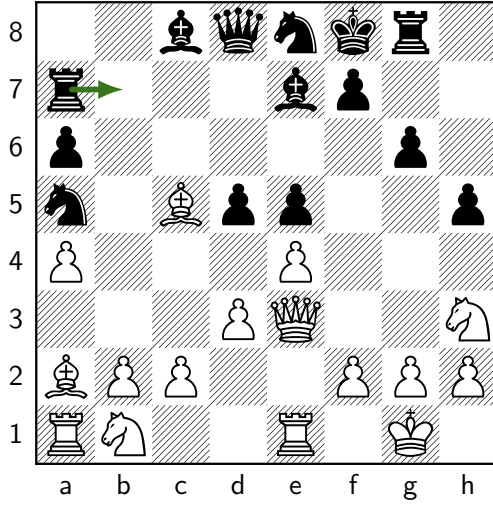


(c)

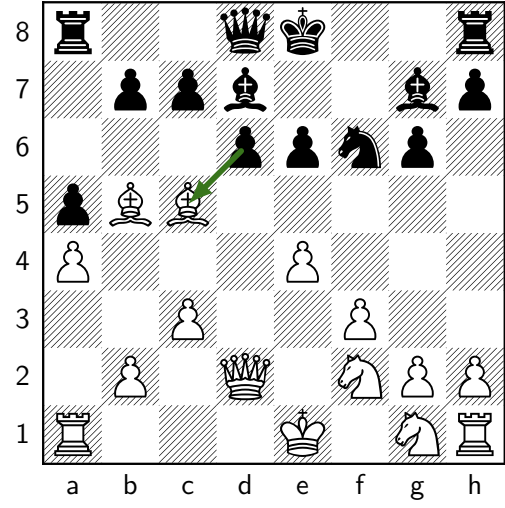


(d)

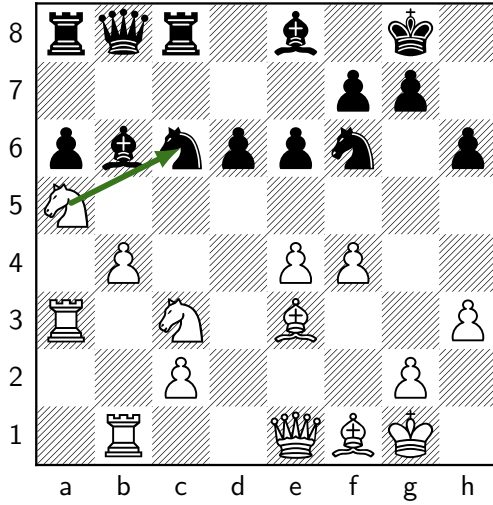
Figure 14: **Move 21, Black: d5**. (a) Our position; Melanie plays d5 (d6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2821` (ply 29) [structural distance $d=93$, QAP obj=31]: the master played Bb7; under μ our *Pd3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *pd3* as their *Qc7*; transferred move d6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#834` (ply 29) [structural distance $d=93$, QAP obj=31]: the master played Bb7; under μ our *Pd3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *pd3* as their *Qc7*; transferred move d6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#209` (ply 25) [structural distance $d=93$, QAP obj=31]: the master played Bb7; under μ our *Pd3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *pd3* as their *Qc7*; transferred move d6d5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to d5 (summed QAP objective 344, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



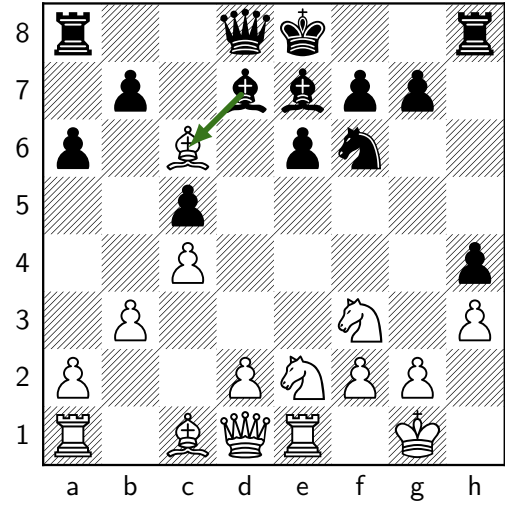
(a)



(b)

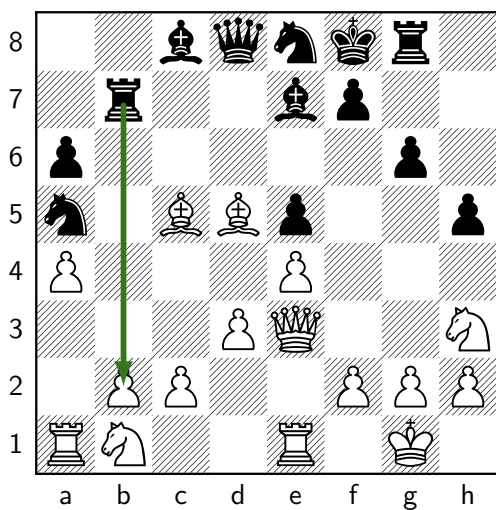


(c)

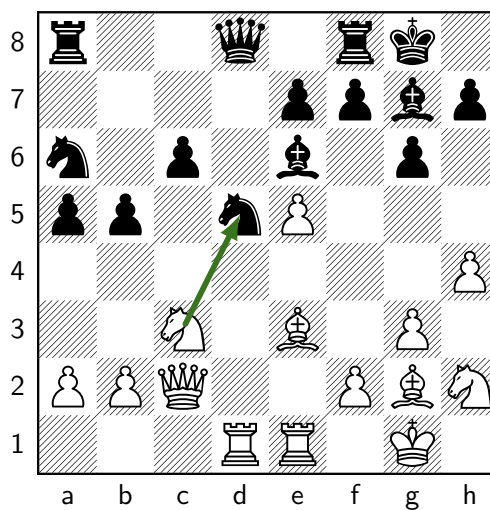


(d)

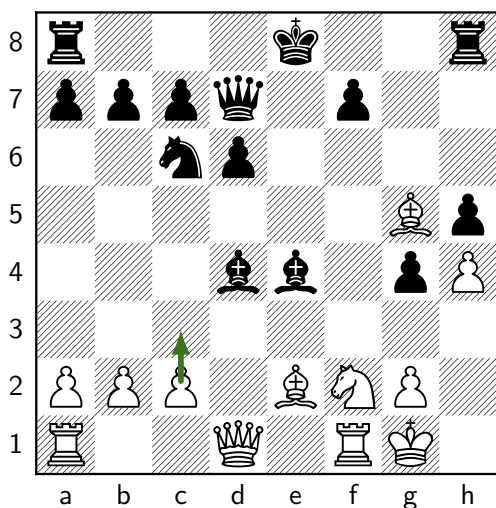
Figure 15: **Move 22, Black: Rb7.** (a) Our position; Melanie plays Rb7 (a7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2129` (ply 27) [structural distance $d=111$, QAP obj=70]: the master played `dxcc5`; under μ our *Pd4* plays the role of that game’s *Pd3*; cross-type roles: our *pf2* as their *Ra8*, our *Ne8* as their *Rh8*; transferred move `d5d4`. (c) Retrieved from `ccr1_top_p0003#1545` (ply 44) [structural distance $d=101$, QAP obj=68]: the master played `Nxc6`; under μ our *Ra2* plays the role of that game’s *Na5* (a **cross-type** correspondence); cross-type roles: our *pf2* as their *Rb1*, our *Rg8* as their *Bf1*; transferred move `a7a8`. (d) Retrieved from `ccr1_top_p0000#1838` (ply 27) [structural distance $d=113$, QAP obj=66]: the master played `Bxc6`; under μ our *Be2* plays the role of that game’s *Bd2*; cross-type roles: our *Ne8* as their *Ra8*, our *Rg8* as their *Qd8*; transferred move `e7c5`. Synthesis: no retrieved analog transferred to Rb7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



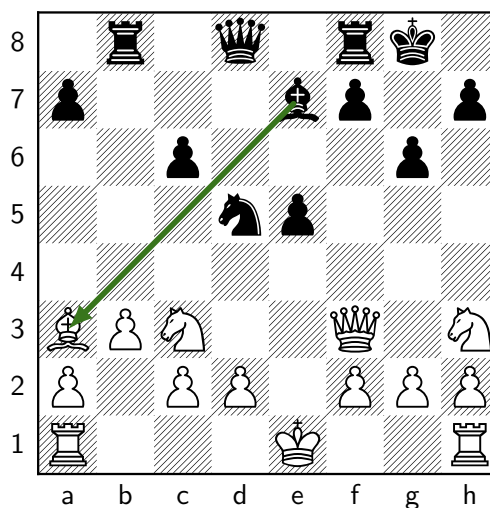
(a)



(b)

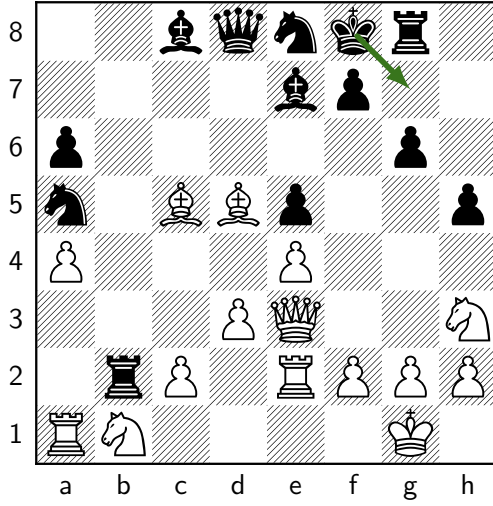


(c)

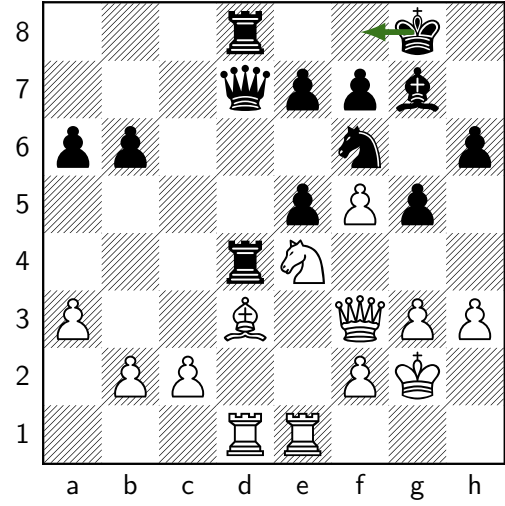


(d)

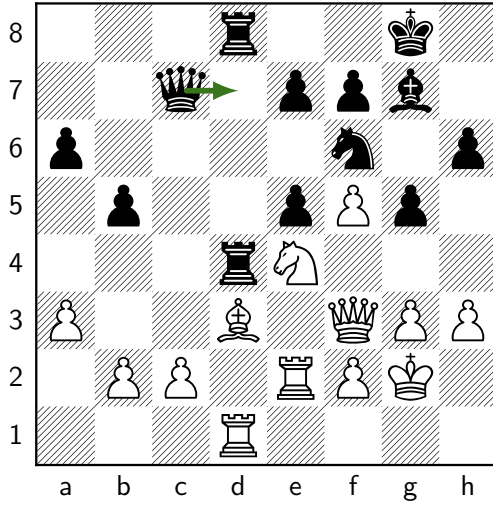
Figure 16: **Move 23, Black: Rxb2.** (a) Our position; Melanie plays $Rxb2$ (b7b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#654` (ply 40) [structural distance $d=135$, QAP obj=66]: the master played $Nxd5$; under μ our $Rb2$ plays the role of that game’s $Nc3$ (a **cross-type** correspondence); cross-type roles: our $Qd8$ as their $Rd1$, our $Bc8$ as their $Re1$; transferred move b7b6. (c) Retrieved from `ccr1_top_p0003#2058` (ply 28) [structural distance $d=137$, QAP obj=59]: the master played c3; under μ our $Ne1$ plays the role of that game’s $Pc2$ (a **cross-type** correspondence); cross-type roles: our $Qd8$ as their $Ra1$, our $Be7$ as their $Qd1$; transferred move e8f6. (d) Retrieved from `ccr1_top_p0000#2536` (ply 31) [structural distance $d=140$, QAP obj=58]: the master played $Bxa3$; under μ our $Be2$ plays the role of that game’s $Be2$; cross-type roles: our $Bc8$ as their $Rb8$, our $Ne8$ as their $Pa7$; transferred move e7c5. Synthesis: no retrieved analog transferred to $Rxb2$ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



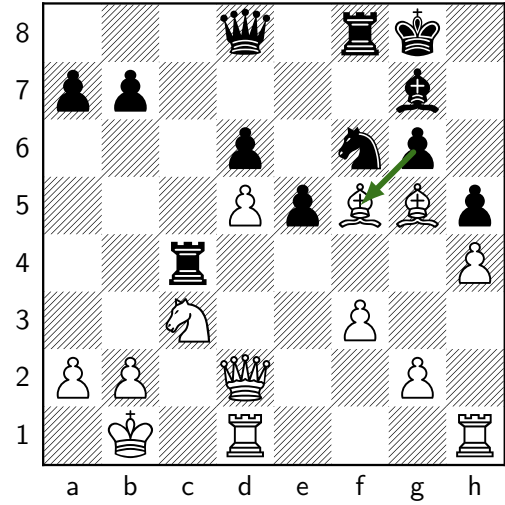
(a)



(b)

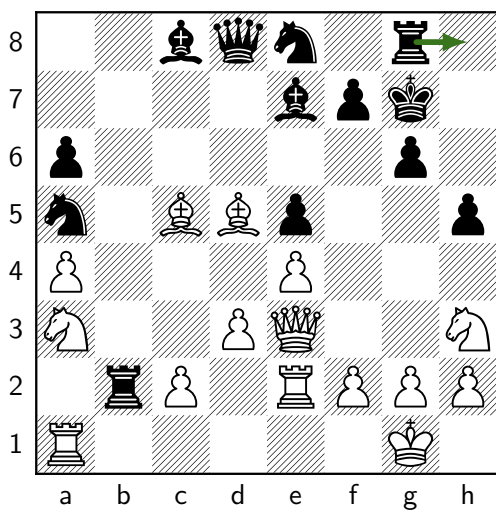


(c)

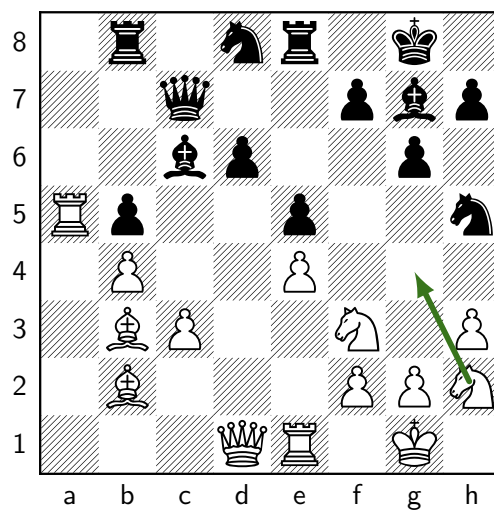


(d)

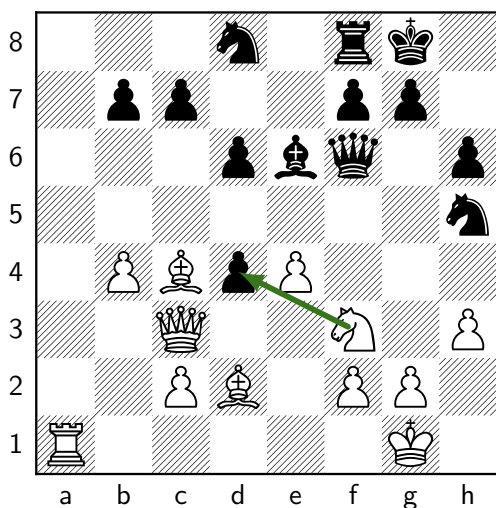
Figure 17: **Move 24, Black: Kg7.** (a) Our position; Melanie plays Kg7 (f8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2096` (ply 47) [structural distance $d=148$, QAP obj=64]: the master played Kf8; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Be7 as their Rd8, our Bc8 as their Pe7; transferred move f8g7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2574` (ply 57) [structural distance $d=148$, QAP obj=64]: the master played Qd7; under μ our Na4 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Pe7, our Ne8 as their Bg7; transferred move a5b7. (d) Retrieved from `ccrl_top_p0000#1259` (ply 37) [structural distance $d=144$, QAP obj=61]: the master played gxf5; under μ our Rb7 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pa7, our Ne8 as their Bg7; transferred move b2b8. Synthesis: 1 retrieved analog transferred to Kg7 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



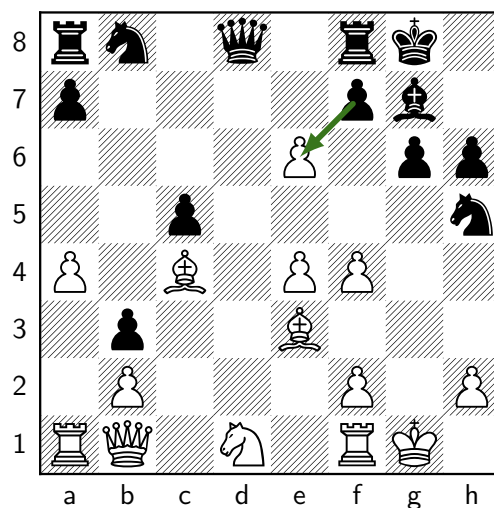
(a)



(b)

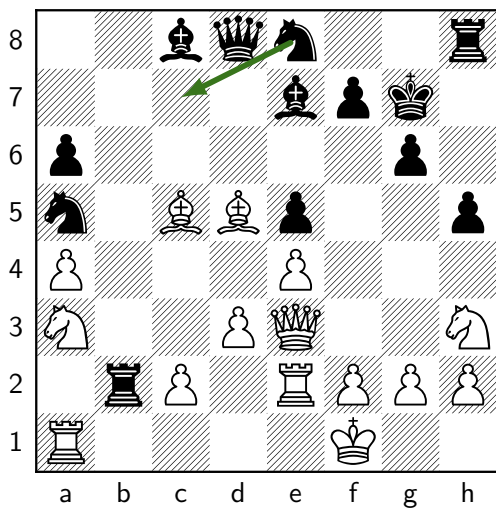


(c)

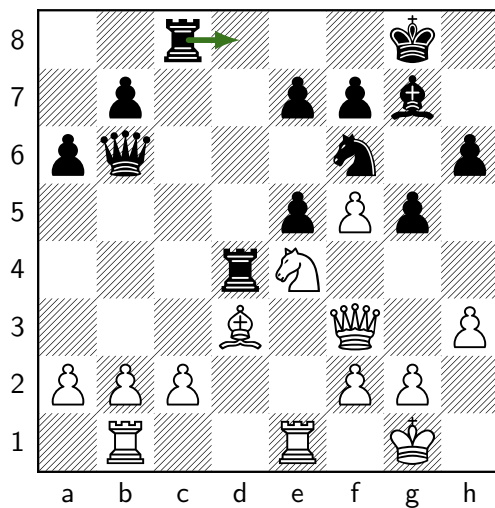


(d)

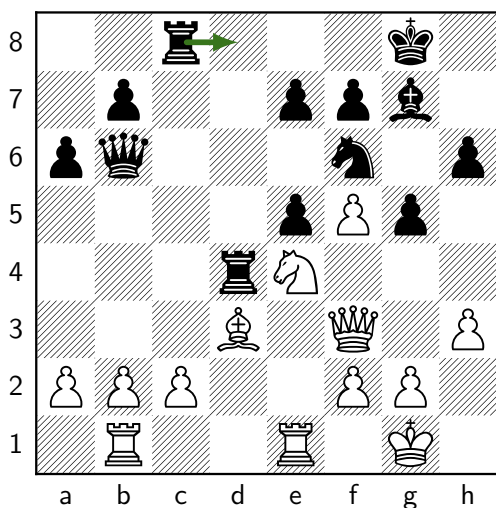
Figure 18: **Move 25, Black: Rh8**. (a) Our position; Melanie plays Rh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1055` (ply 46) [structural distance $d=133$, QAP obj=33]: the master played Ng4; under μ our *Rg1* plays the role of that game's *Nh2* (a **cross-type** correspondence); cross-type roles: our *Ne8* as their *Re1*, our *pd3* as their *Bb2*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#314` (ply 34) [structural distance $d=128$, QAP obj=26]: the master played Nxd4; under μ our *Rg1* plays the role of that game's *Nf3* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Ra1*, our *pd3* as their *Bd2*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3161` (ply 35) [structural distance $d=137$, QAP obj=51]: the master played fxe6; under μ our *Be2* plays the role of that game's *Pf2* (a **cross-type** correspondence); cross-type roles: our *Ne8* as their *Ra8*, our *ph2* as their *Nb8*; transferred move e7c5. Synthesis: 2 retrieved analogs transferred to Rh8 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



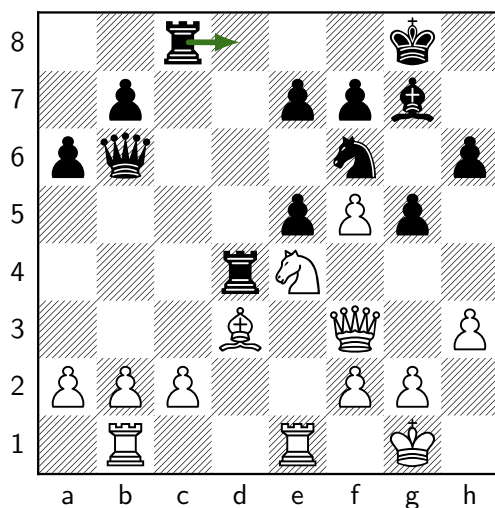
(a)



(b)

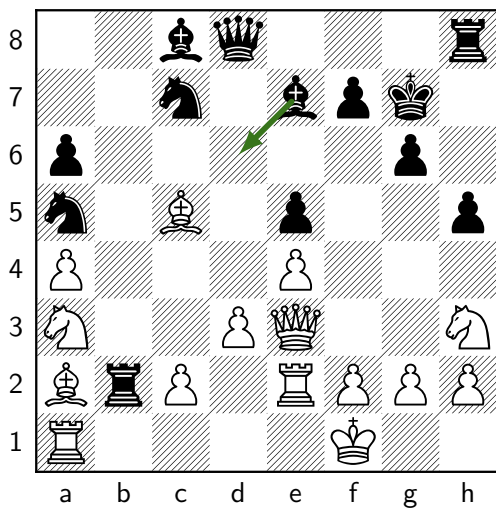


(c)

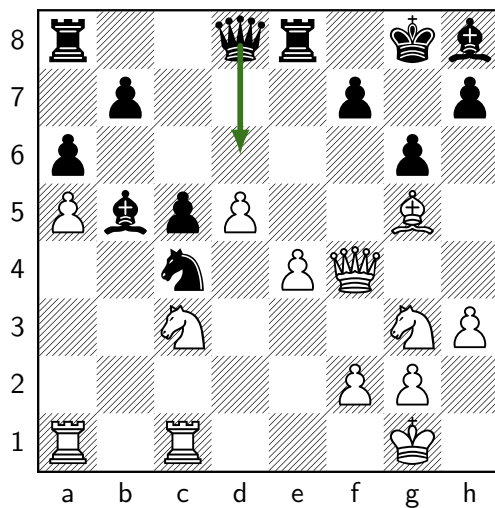


(d)

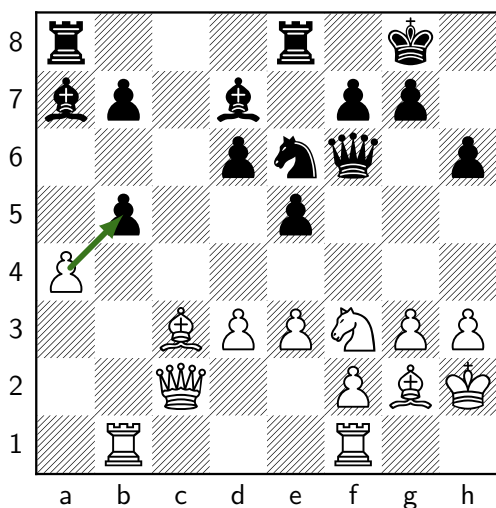
Figure 19: **Move 26, Black: Nc7.** (a) Our position; Melanie plays Nc7 (e8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2096` (ply 39) [structural distance $d=109$, QAP obj=41]: the master played Rcd8; under μ our Ne1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Pe7, our Na5 as their Bg7; transferred move e8c7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#115` (ply 39) [structural distance $d=109$, QAP obj=41]: the master played Rcd8; under μ our Ne1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Pe7, our Na5 as their Bg7; transferred move e8c7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#516` (ply 39) [structural distance $d=109$, QAP obj=41]: the master played Rcd8; under μ our Ne1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Pe7, our Na5 as their Bg7; transferred move e8c7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nc7 (summed QAP objective 164, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



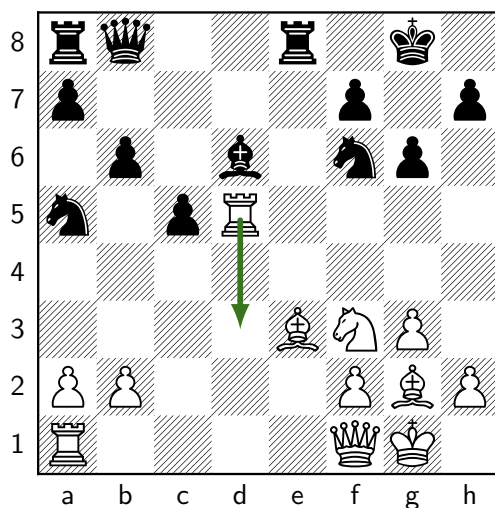
(a)



(b)

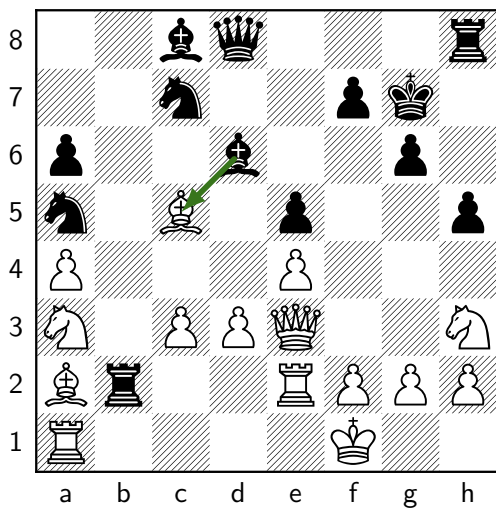


(c)

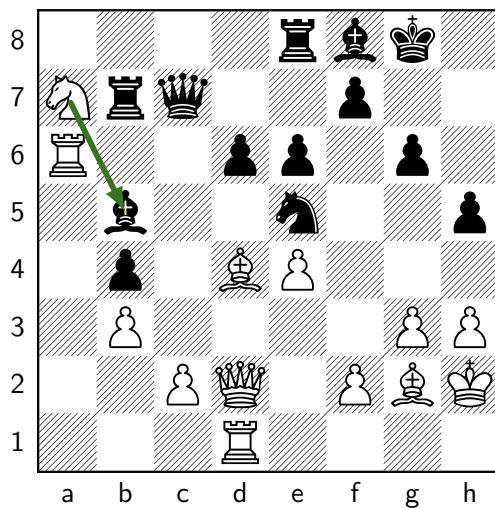


(d)

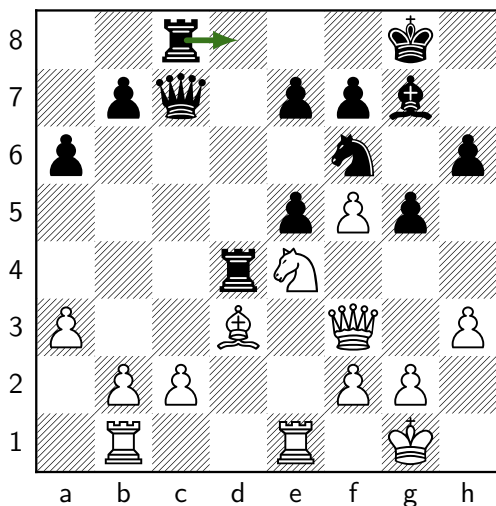
Figure 20: **Move 27, Black: Bd6.** (a) Our position; Melanie plays Bd6 (e7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1876` (ply 45) [structural distance $d=101$, QAP obj=36]: the master played Qd6; under μ our *Be2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Pe5* as their *Ra8*, our *Qd8* as their *Re8*; transferred move e7d6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1710` (ply 42) [structural distance $d=106$, QAP obj=29]: the master played axb5; under μ our *Be2* plays the role of that game’s *Pa4* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Rf1*, our *Pe5* as their *Bg2*; transferred move e7d6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1833` (ply 42) [structural distance $d=106$, QAP obj=61]: the master played Rd3; under μ our *Rb7* plays the role of that game’s *Rd5*; cross-type roles: our *Na5* as their *Qf1*, our *pg2* as their *Bg2*; transferred move b2b8. Synthesis: 2 retrieved analogs transferred to Bd6 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



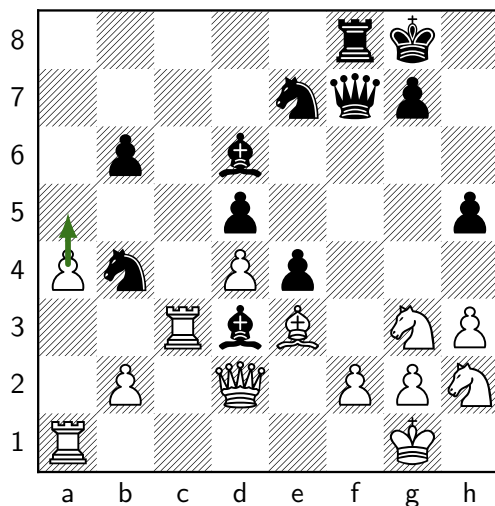
(a)



(b)

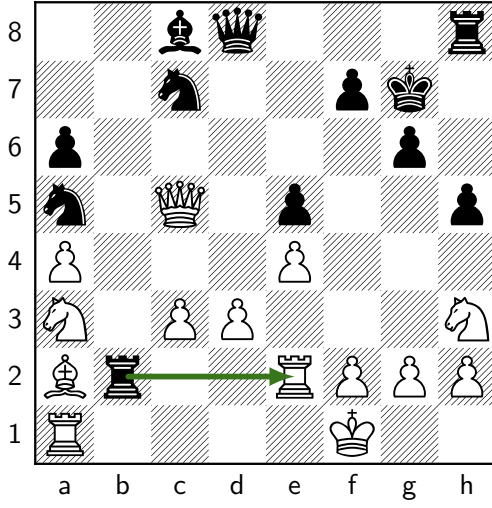


(c)

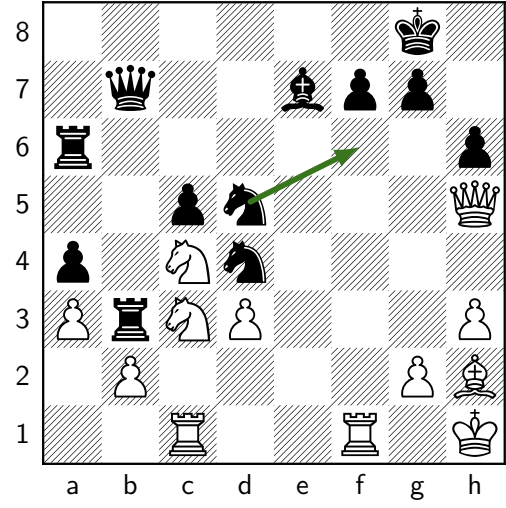


(d)

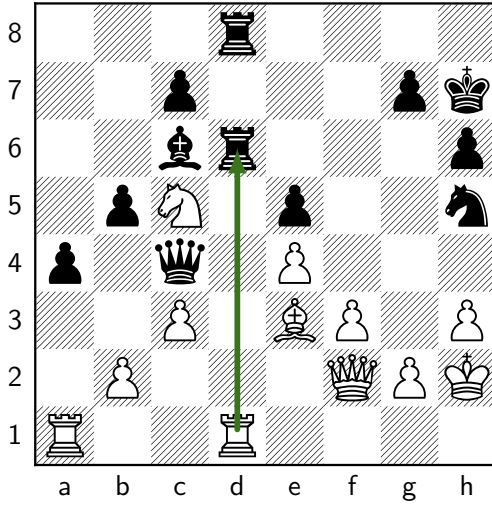
Figure 21: **Move 28, Black: Bxc5.** (a) Our position; Melanie plays Bxc5 (d6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#734` (ply 48) [structural distance $d=103$, QAP obj=34]: the master played Nxb5; under μ our Bd3 plays the role of that game’s Na7 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Rd1, our Nc7 as their Pc2; transferred move d6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2574` (ply 41) [structural distance $d=103$, QAP obj=51]: the master played Rcd8; under μ our Bc1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Nc7 as their Qc7, our Qd8 as their Pe7; transferred move c8b7. (d) Retrieved from `ccr1_top_p0003#2006` (ply 58) [structural distance $d=107$, QAP obj=44]: the master played a5; under μ our Na4 plays the role of that game’s Pa4 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Ra1, our Rb2 as their Qd2; transferred move a5c6. Synthesis: 1 retrieved analog transferred to Bxc5 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



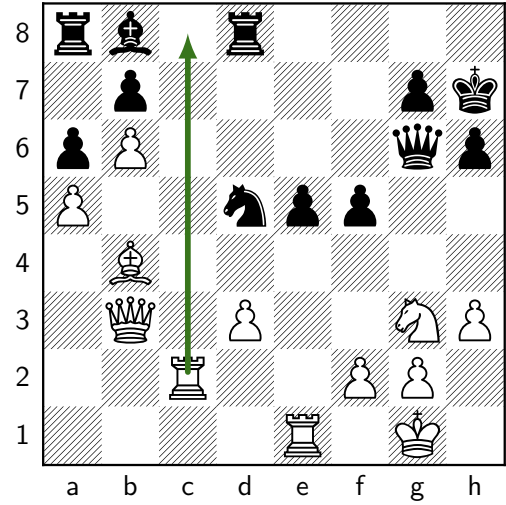
(a)



(b)

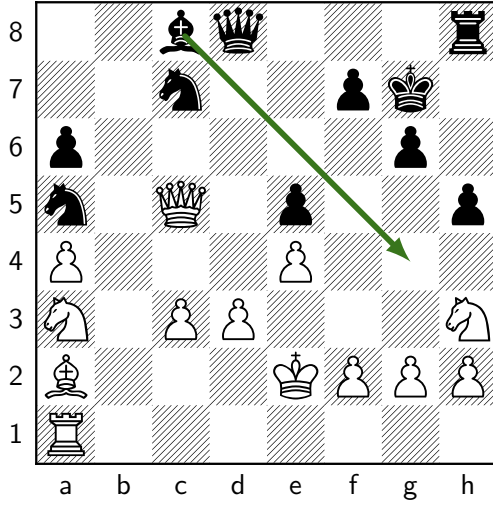


(c)

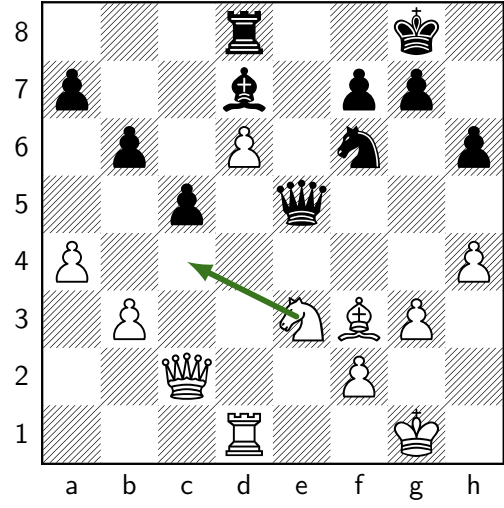


(d)

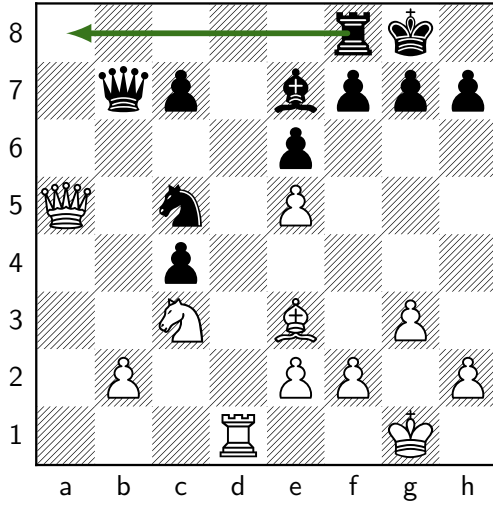
Figure 22: **Move 29, Black: Rxe2.** (a) Our position; Melanie plays **Rxe2** (**b2e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1854` (ply 55) [structural distance $d=90$, QAP obj=47]: the master played **Nf6**; under μ our **Rb7** plays the role of that game’s **Nd4** (a **cross-type** correspondence); cross-type roles: our **Pa6** as their **Be7**, our **Bc8** as their **Ra6**; transferred move **b2b7**. (c) Retrieved from `ccr1_top_p0002#700` (ply 52) [structural distance $d=90$, QAP obj=38]: the master played **Rxd6**; under μ our **Rb7** plays the role of that game’s **Rd1**; cross-type roles: our **Pf7** as their **Ra1**, our **Rh8** as their **Qf2**; transferred move **b2b4**. (d) Retrieved from `ccr1_top_p0002#388` (ply 60) [structural distance $d=84$, QAP obj=29]: the master played **Rc8**; under μ our **Ph4** plays the role of that game’s **Rc2** (a **cross-type** correspondence); cross-type roles: our **pe4** as their **Re1**, our **re2** as their **Qb3**; transferred move **h5h4**. Synthesis: no retrieved analog transferred to **Rxe2** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



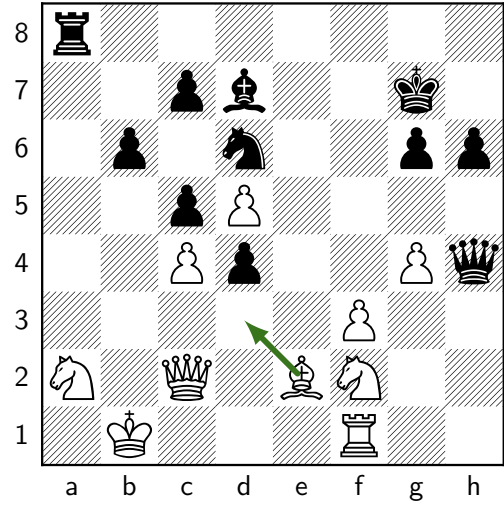
(a)



(b)

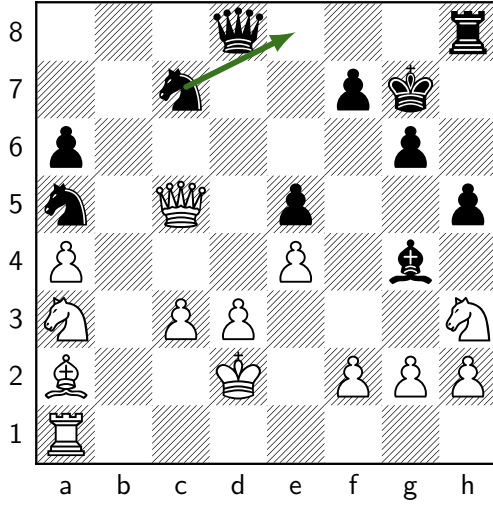


(c)

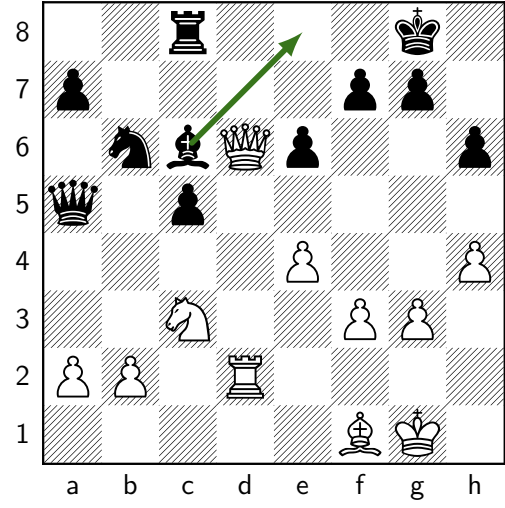


(d)

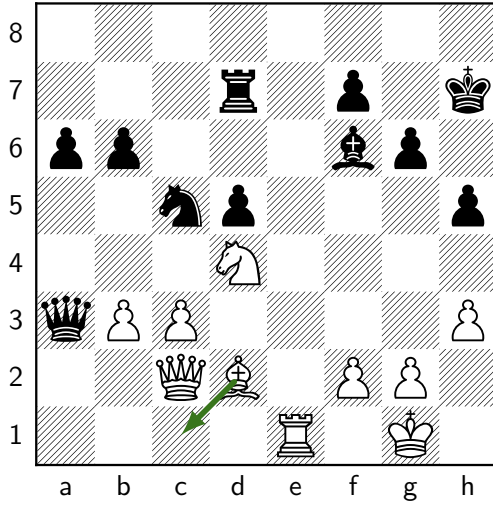
Figure 23: **Move 30, Black: Bg4+.** (a) Our position; Melanie plays Bg4+ (c8g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2511` (ply 74) [structural distance $d=80$, QAP obj=52]: the master played Nc4; under μ our Nc2 plays the role of that game’s Ne3; cross-type roles: our ba2 as their Rd1, our na3 as their Qc2; transferred move c7e6. (c) Retrieved from `ccr1_top_p0003#277` (ply 43) [structural distance $d=82$, QAP obj=49]: the master played Ra8; under μ our Pf2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Nc7 as their Pc7, our Pe5 as their Be7; transferred move f7f5. (d) Retrieved from `ccr1_top_p0001#459` (ply 70) [structural distance $d=81$, QAP obj=47]: the master played Bd3; under μ our Pf2 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rf1, our pg2 as their Qc2; transferred move f7f5. Synthesis: no retrieved analog transferred to Bg4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



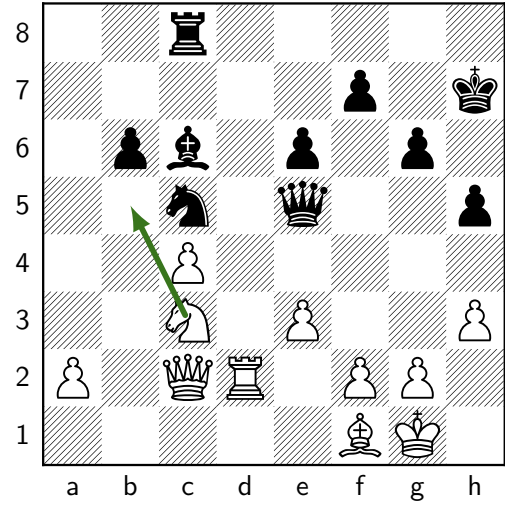
(a)



(b)

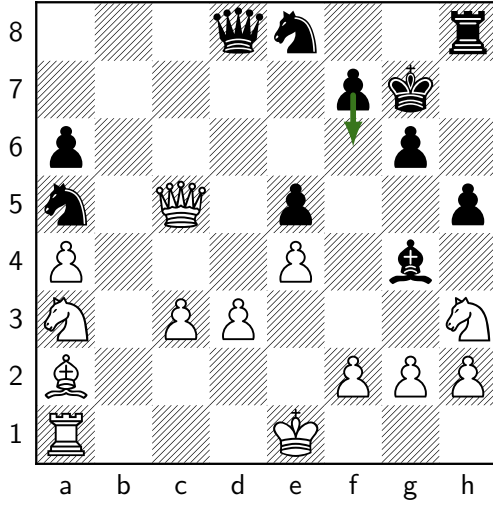


(c)

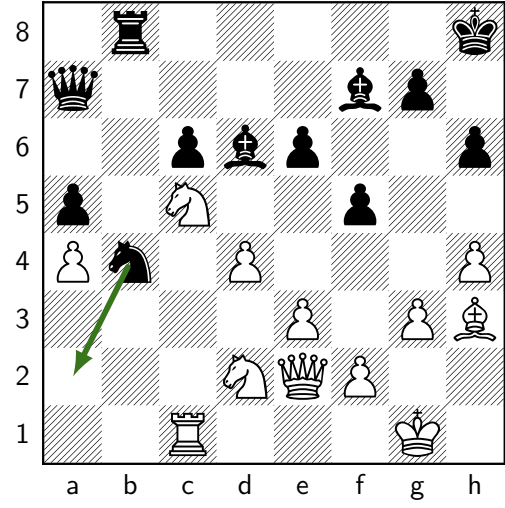


(d)

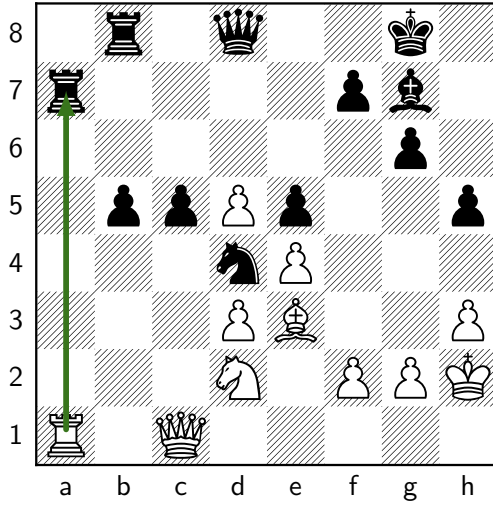
Figure 24: **Move 31, Black: Ne8.** (a) Our position; Melanie plays Ne8 (c7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#373` (ply 57) [structural distance $d=80$, QAP obj=46]: the master played Be8; under μ our Nc2 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rc8, our Rh8 as their Pg7; transferred move c7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3166` (ply 74) [structural distance $d=86$, QAP obj=42]: the master played Bc1; under μ our Nc2 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Re1, our Rh8 as their Pf2; transferred move c7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#262` (ply 64) [structural distance $d=83$, QAP obj=47]: the master played Nb5; under μ our Nc2 plays the role of that game’s Nc3; cross-type roles: our Pf7 as their Bf1, our Na5 as their Pe3; transferred move c7a8. Synthesis: 2 retrieved analogs transferred to Ne8 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



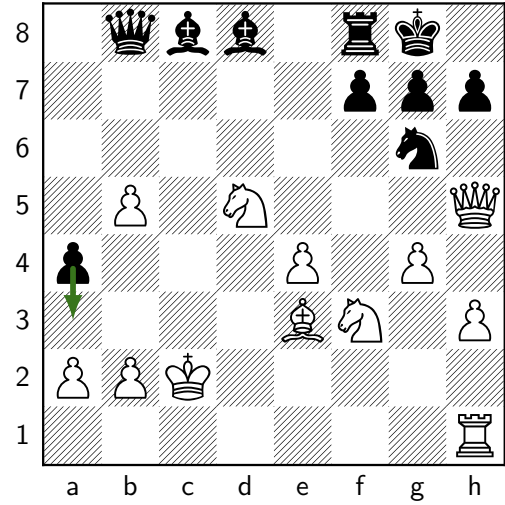
(a)



(b)

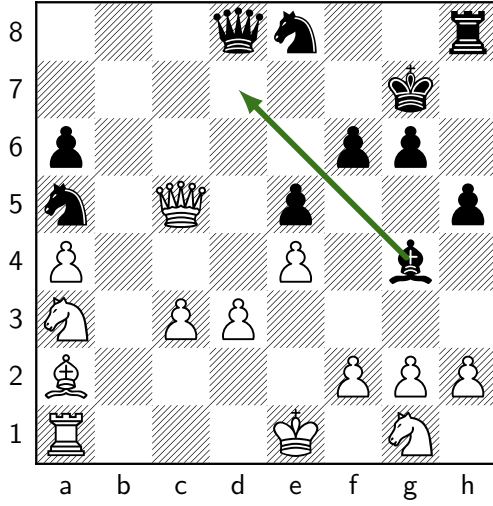


(c)

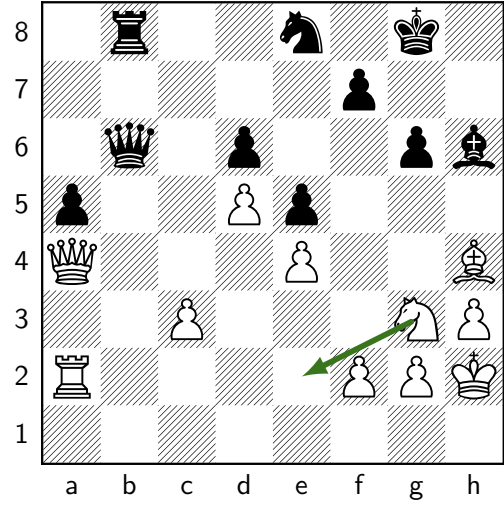


(d)

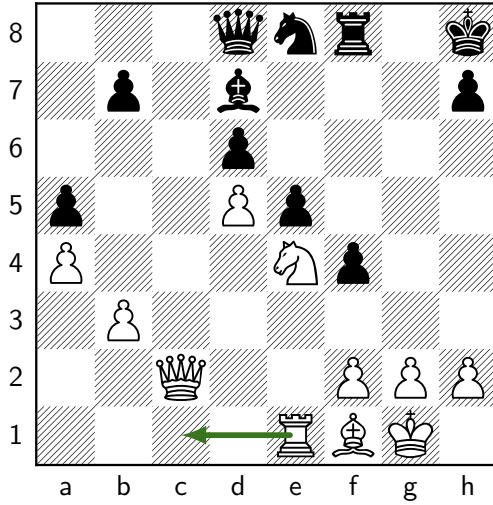
Figure 25: **Move 32, Black: f6.** (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#376` (ply 69) [structural distance $d=73$, QAP obj=31]: the master played Na2; under μ our Qd1 plays the role of that game’s Nb5 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Rb8, our Bg4 as their Qa7; transferred move d8h4. (c) Retrieved from `ccr1_top_p0000#2519` (ply 64) [structural distance $d=71$, QAP obj=29]: the master played Rxa7; under μ our Na4 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Nd2, our Ne8 as their ph5; transferred move a5b3. (d) Retrieved from `ccr1_top_p0000#2285` (ply 63) [structural distance $d=65$, QAP obj=25]: the master played a3; under μ our Pg3 plays the role of that game’s Pa5; cross-type roles: our Na5 as their Qb8, our Pe5 as their Bc8; transferred move g6g5. Synthesis: no retrieved analog transferred to f6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



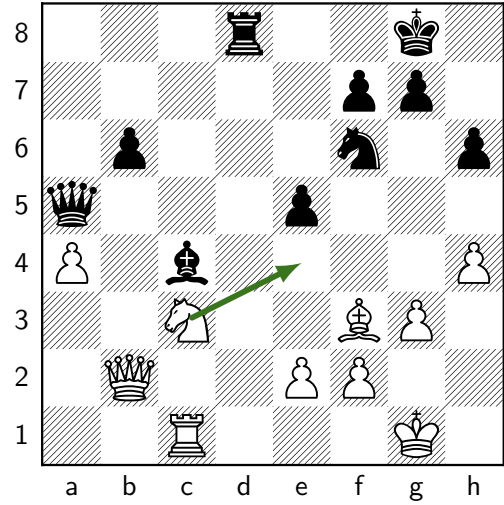
(a)



(b)

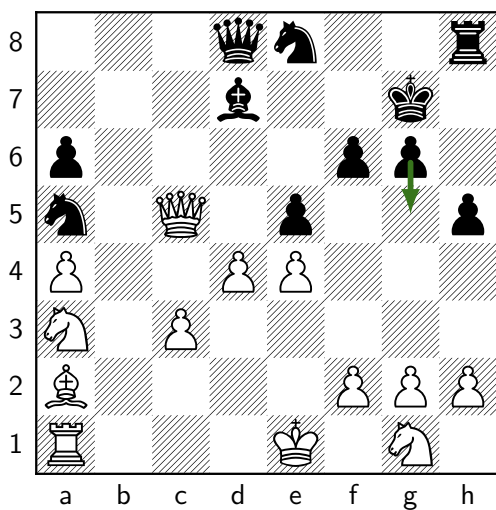


(c)

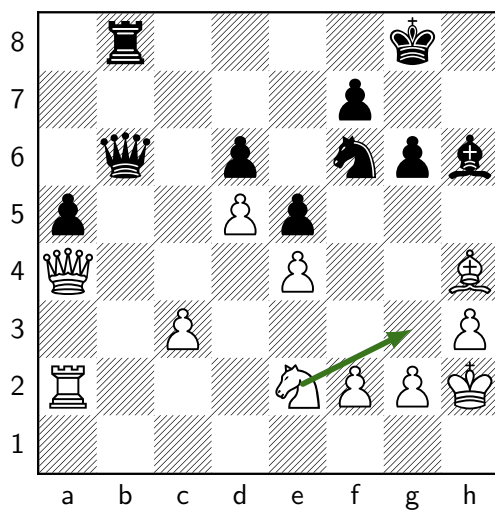


(d)

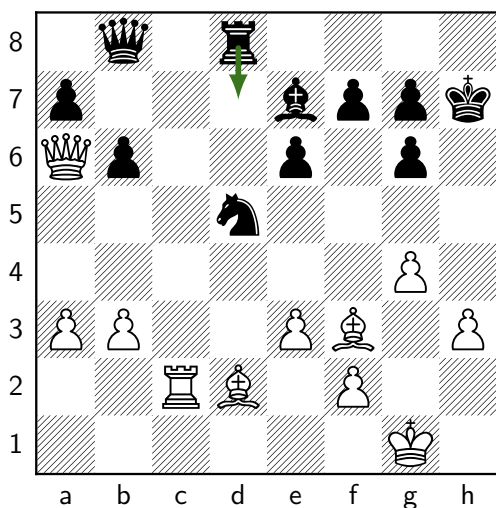
Figure 26: **Move 33, Black: Bd7.** (a) Our position; Melanie plays Bd7 (*g4d7*). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from *ccr1_top_p0003#548* (ply 76) [structural distance $d=64$, QAP obj=24]: the master played *Ne2*; under μ our *Bg5* plays the role of that game’s *Ng3* (a **cross-type** correspondence); cross-type roles: our *na3* as their *Ra2*, our *Ne8* as their *Ph3*; transferred move *g4d7* (**endorses this move**). (c) Retrieved from *ccr1_top_p0001#316* (ply 48) [structural distance $d=89$, QAP obj=10]: the master played *Rc1*; under μ our *Bg5* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rh8* as their *Bf1*, our *Pg6* as their *Qc2*; transferred move *g4d7* (**endorses this move**). (d) Retrieved from *ccr1_top_p0001#2936* (ply 52) [structural distance $d=89$, QAP obj=37]: the master played *Ne4*; under μ our *Na4* plays the role of that game’s *Nc3*; cross-type roles: our *Ne8* as their *Rc1*, our *pe4* as their *Bf3*; transferred move *a5c6*. Synthesis: 2 retrieved analogs transferred to Bd7 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



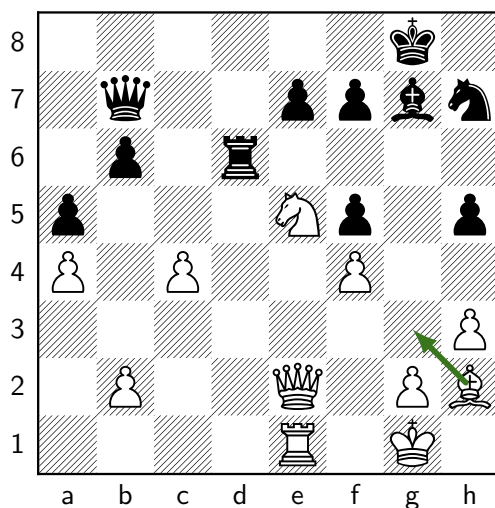
(a)



(b)

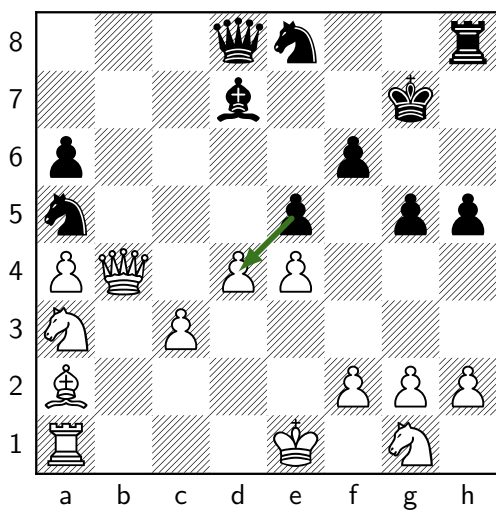


(c)

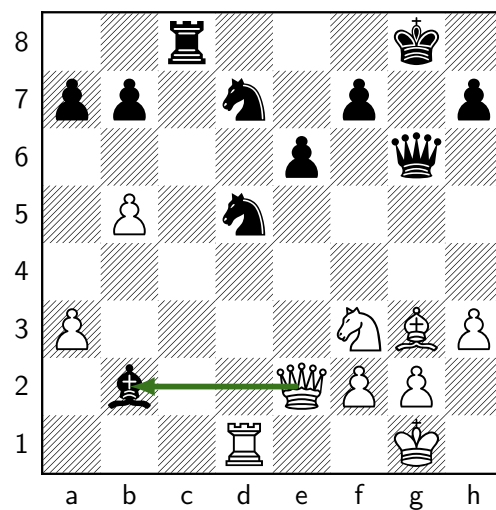


(d)

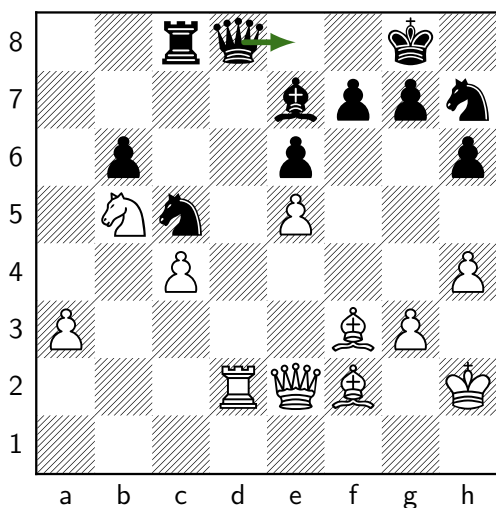
Figure 27: **Move 34, Black: g5**. (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#548 (ply 74) [structural distance $d=93$, QAP obj=35]: the master played Ng3; under μ our Pg3 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our na3 as their Ra2, our ba2 as their Pc3; transferred move g6g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#679 (ply 55) [structural distance $d=93$, QAP obj=53]: the master played Rd7; under μ our Ne1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pf6 as their Qb8, our Rh8 as their Pf7; transferred move e8c7. (d) Retrieved from ccr1_top_p0002#1405 (ply 52) [structural distance $d=85$, QAP obj=41]: the master played Bg3; under μ our Pf3 plays the role of that game’s Bh2 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Re1, our Rh8 as their Pb2; transferred move f6f5. Synthesis: 2 retrieved analogs transferred to g5 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



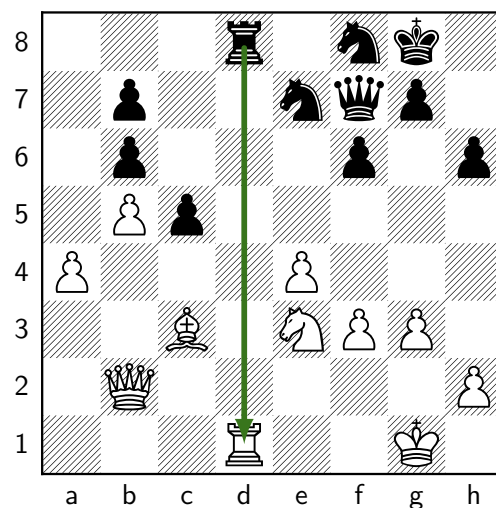
(a)



(b)

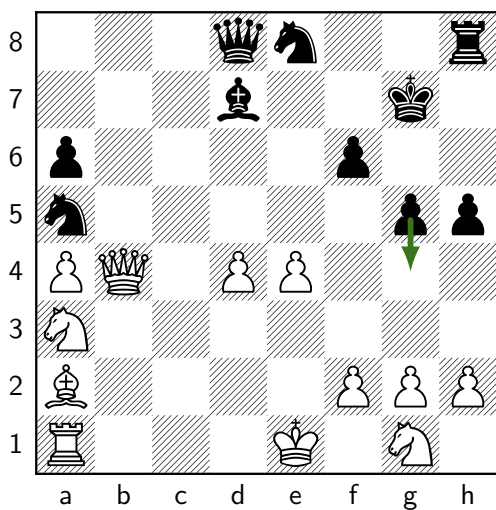


(c)

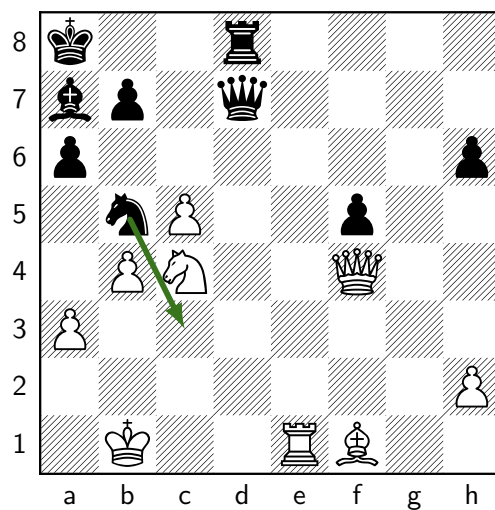


(d)

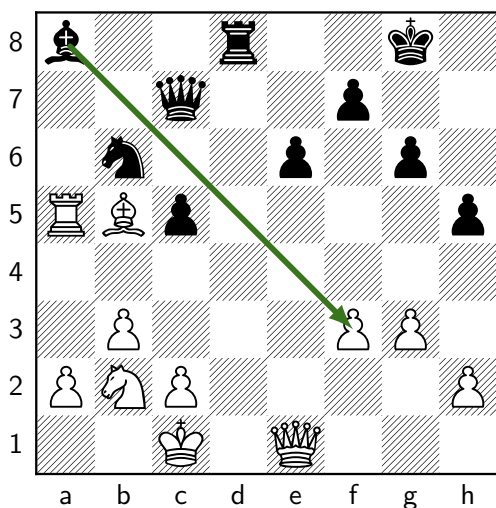
Figure 28: **Move 35, Black: exd4**. (a) Our position; Melanie plays **exd4** (e5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2672` (ply 64) [structural distance $d=82$, QAP obj=20]: the master played **Qxb2**; under μ our *Pe4* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *Pf6* as their *Rd1*, our *pd4* as their *bb2*; transferred move **e5d4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2277` (ply 79) [structural distance $d=87$, QAP obj=20]: the master played **Qe8**; under μ our *Pe4* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Pf6* as their *Rc8*, our *pc3* as their *Be7*; transferred move **e5d4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#103` (ply 63) [structural distance $d=82$, QAP obj=12]: the master played **Rxd1+**; under μ our *Pe4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Pf6* as their *Nf8*, our *na3* as their *Pb7*; transferred move **e5d4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **exd4** (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



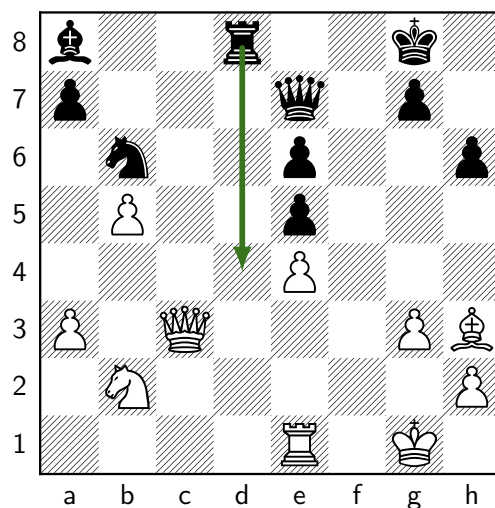
(a)



(b)

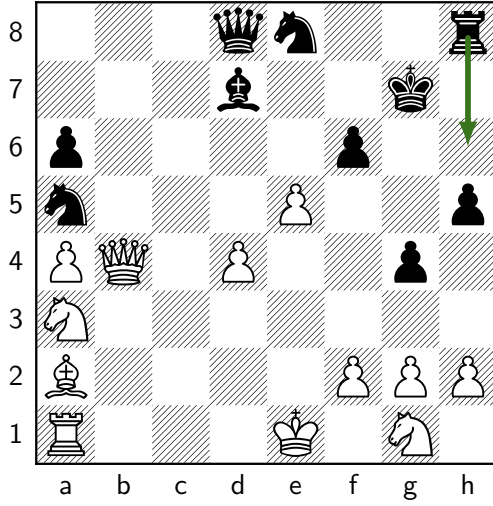


(c)

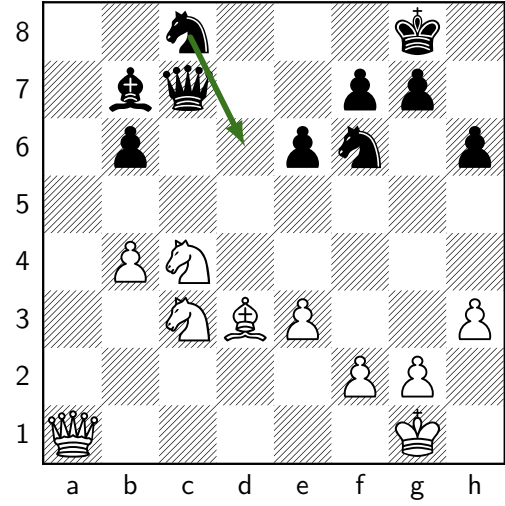


(d)

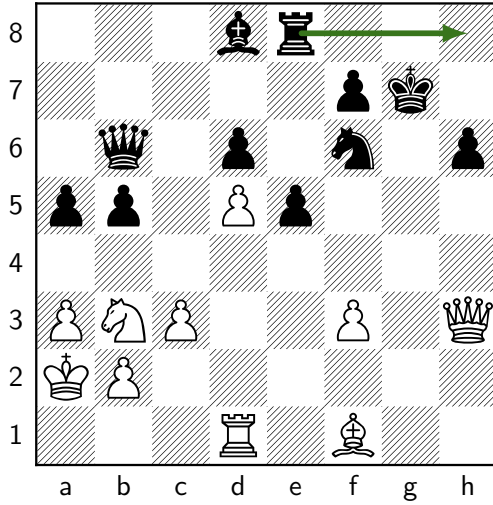
Figure 29: **Move 36, Black: g4**. (a) Our position; Melanie plays g4 (g5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1592` (ply 65) [structural distance $d=86$, QAP obj=11]: the master played Nc3+; under μ our Pg4 plays the role of that game’s Nb4 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Rd8, our Pa6 as their Ba7; transferred move g5g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1603` (ply 69) [structural distance $d=94$, QAP obj=11]: the master played Bxf3; under μ our Na4 plays the role of that game’s Ba1 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Rd8, our Bd7 as their Qc7; transferred move a5c6. (d) Retrieved from `ccr1_top_p0002#1602` (ply 73) [structural distance $d=91$, QAP obj=6]: the master played Rd4; under μ our Ne1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our pd4 as their Ba8, our Rh8 as their Pg7; transferred move e8d6. Synthesis: 1 retrieved analog transferred to g4 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



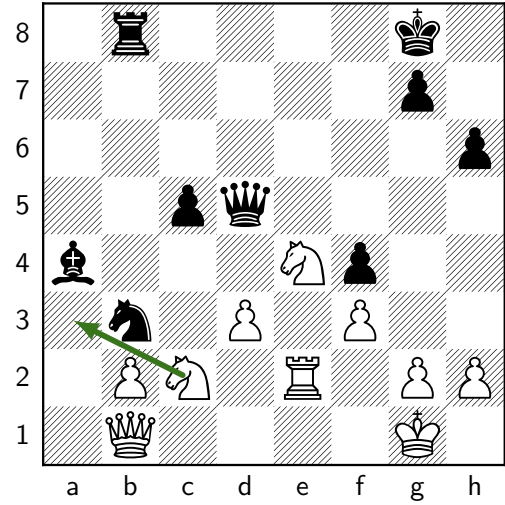
(a)



(b)

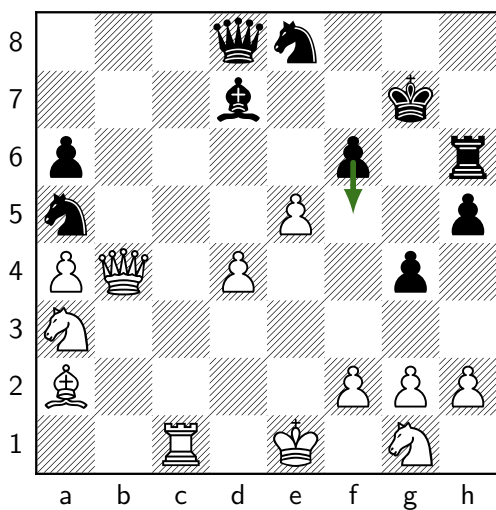


(c)

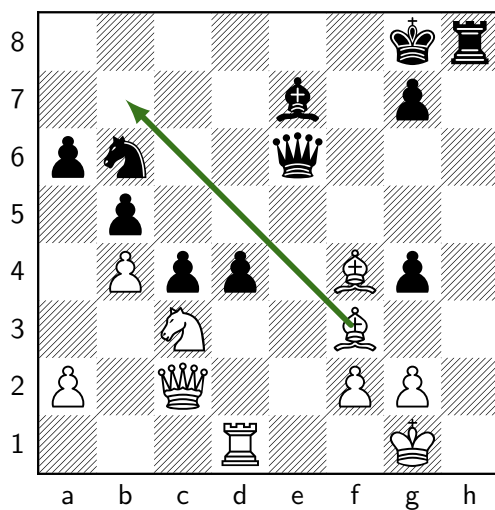


(d)

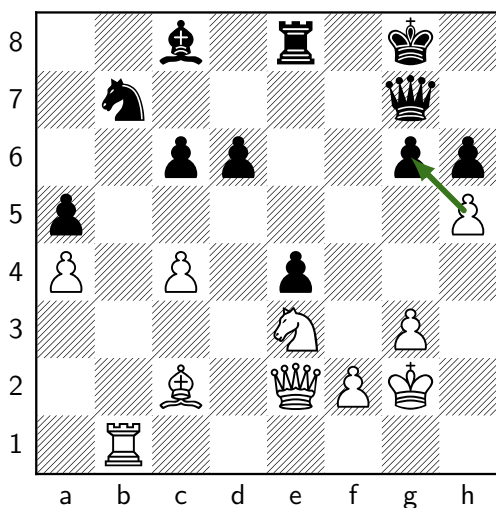
Figure 30: **Move 37, Black: Rh6.** (a) Our position; Melanie plays Rh6 (**h8h6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3636` (ply 71) [structural distance $d=92$, QAP obj=13]: the master played Nd6; under μ our Rh1 plays the role of that game’s Nc1 (a **cross-type** correspondence); cross-type roles: our na3 as their Bb7, our ba2 as their Pe6; transferred move **h8h6 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#2938` (ply 87) [structural distance $d=89$, QAP obj=4]: the master played Rh8; under μ our Rh1 plays the role of that game’s Re1; cross-type roles: our Qd8 as their Bd8, our Na5 as their Pf7; transferred move **h8h6 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#286` (ply 64) [structural distance $d=93$, QAP obj=27]: the master played Na3; under μ our Bd2 plays the role of that game’s Nc2 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Qb1, our pf2 as their Re2; transferred move **d7a4**. Synthesis: 2 retrieved analogs transferred to Rh6 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



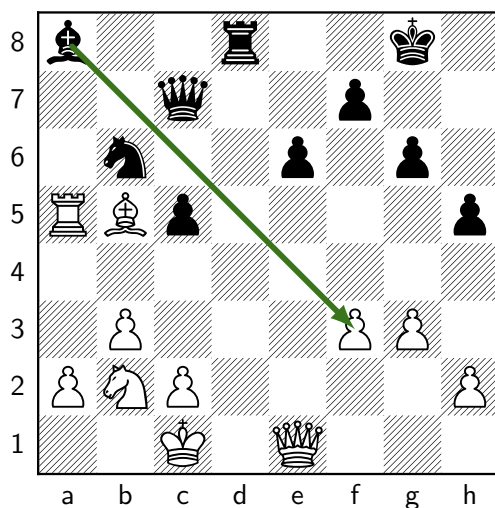
(a)



(b)

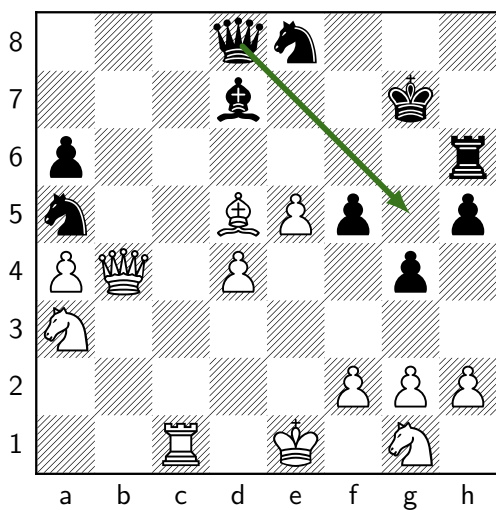


(c)

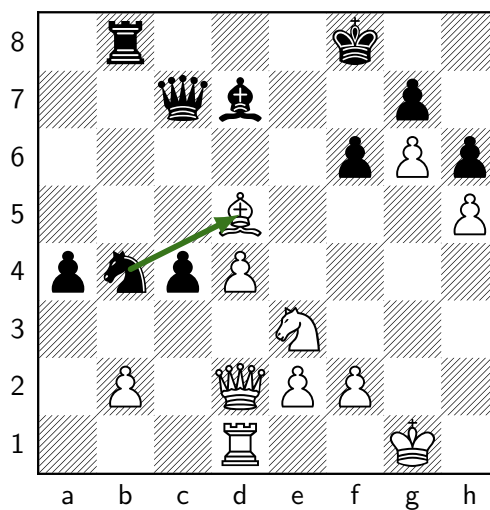


(d)

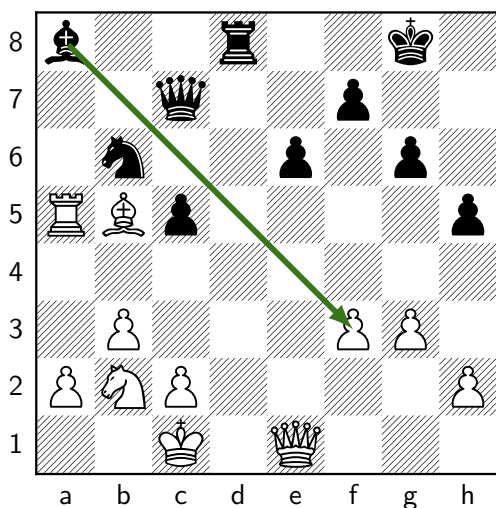
Figure 31: **Move 38, Black: f5**. (a) Our position; Melanie plays f5 (f6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#522` (ply 58) [structural distance $d=85$, QAP obj=35]: the master played Bb7; under μ our Pf3 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our ba2 as their Rd1, our ph2 as their Qc2; transferred move f6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1709` (ply 90) [structural distance $d=87$, QAP obj=18]: the master played hxg6; under μ our Pf3 plays the role of that game’s Ph5; cross-type roles: our Bd7 as their Rb1, our Rh6 as their Qe2; transferred move f6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1603` (ply 69) [structural distance $d=89$, QAP obj=13]: the master played Bxf3; under μ our Pf3 plays the role of that game’s Ba1 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Rd8, our Bd7 as their Qc7; transferred move f6f5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to f5 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



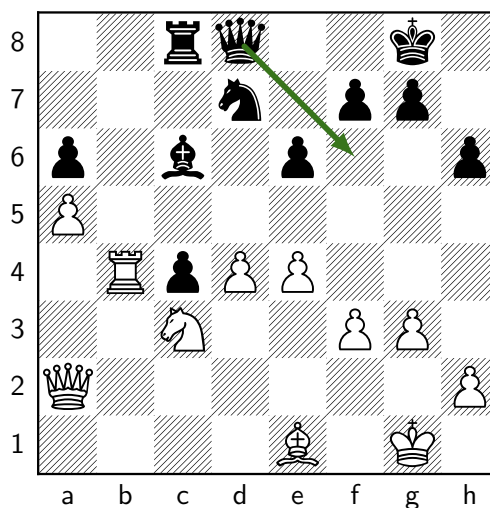
(a)



(b)

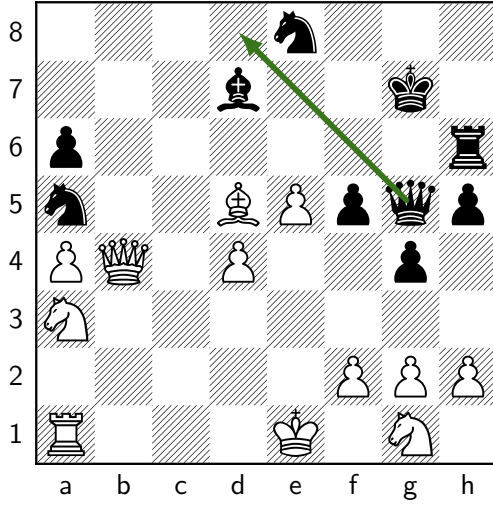


(c)

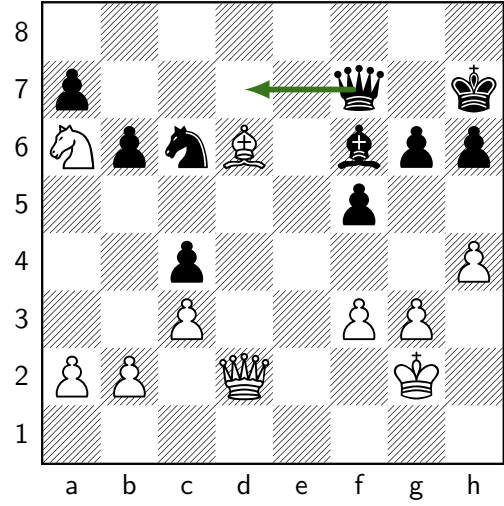


(d)

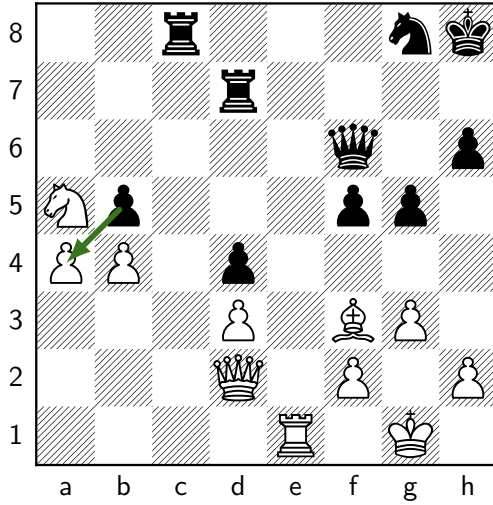
Figure 32: **Move 39, Black: Qg5.** (a) Our position; Melanie plays Qg5 (d8g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#392` (ply 77) [structural distance $d=87$, QAP obj=15]: the master played Nxd5; under μ our Na4 plays the role of that game’s Nb5; cross-type roles: our Ne8 as their Rb8, our Pg4 as their Qc7; transferred move a5b7. (c) Retrieved from `ccr1_top_p0003#1603` (ply 69) [structural distance $d=86$, QAP obj=12]: the master played Bxf3; under μ our Na4 plays the role of that game’s Ba1 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Rd8, our Pg4 as their Qc7; transferred move a5b3. (d) Retrieved from `ccr1_top_p0000#1025` (ply 69) [structural distance $d=95$, QAP obj=9]: the master played Qf6; under μ our Ne1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Bd7 as their Rc8, our pe5 as their Nd7; transferred move e8c7. Synthesis: no retrieved analog transferred to Qg5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



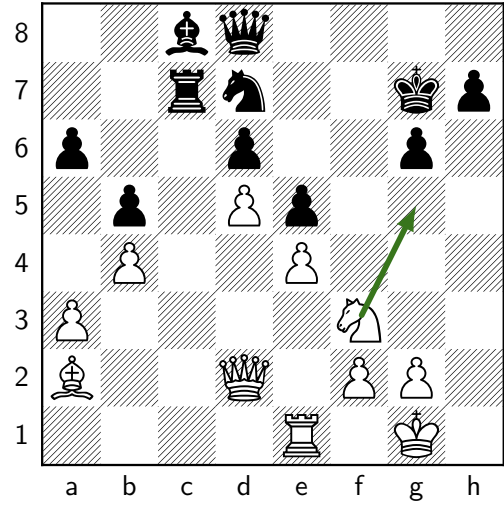
(a)



(b)



(c)



(d)

Figure 33: **Move 40, Black: Qd8.** (a) Our position; Melanie plays Qd8 (g5d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0001#1896` (ply 65) [structural distance $d=94$, QAP obj=7]: the master played Qd7; under μ our $Qg4$ plays the role of that game’s $Qf2$; cross-type roles: our $pg2$ as their $na6$, our $Bd7$ as their $Pb6$; transferred move g5d8 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#990` (ply 73) [structural distance $d=88$, QAP obj=12]: the master played bxa4; under μ our $Na4$ plays the role of that game’s $Pb4$ (a **cross-type** correspondence); cross-type roles: our $pd4$ as their $Rc8$, our $pe5$ as their $Ng8$; transferred move a5b3. (d) Retrieved from `cctl_top_p0002#1511` (ply 68) [structural distance $d=93$, QAP obj=6]: the master played Ng5; under μ our $Qg4$ plays the role of that game’s $Nf3$ (a **cross-type** correspondence); cross-type roles: our $Ne8$ as their $Re1$, our $pf2$ as their $Ba2$; transferred move g5h4. Synthesis: 1 retrieved analog transferred to Qd8 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

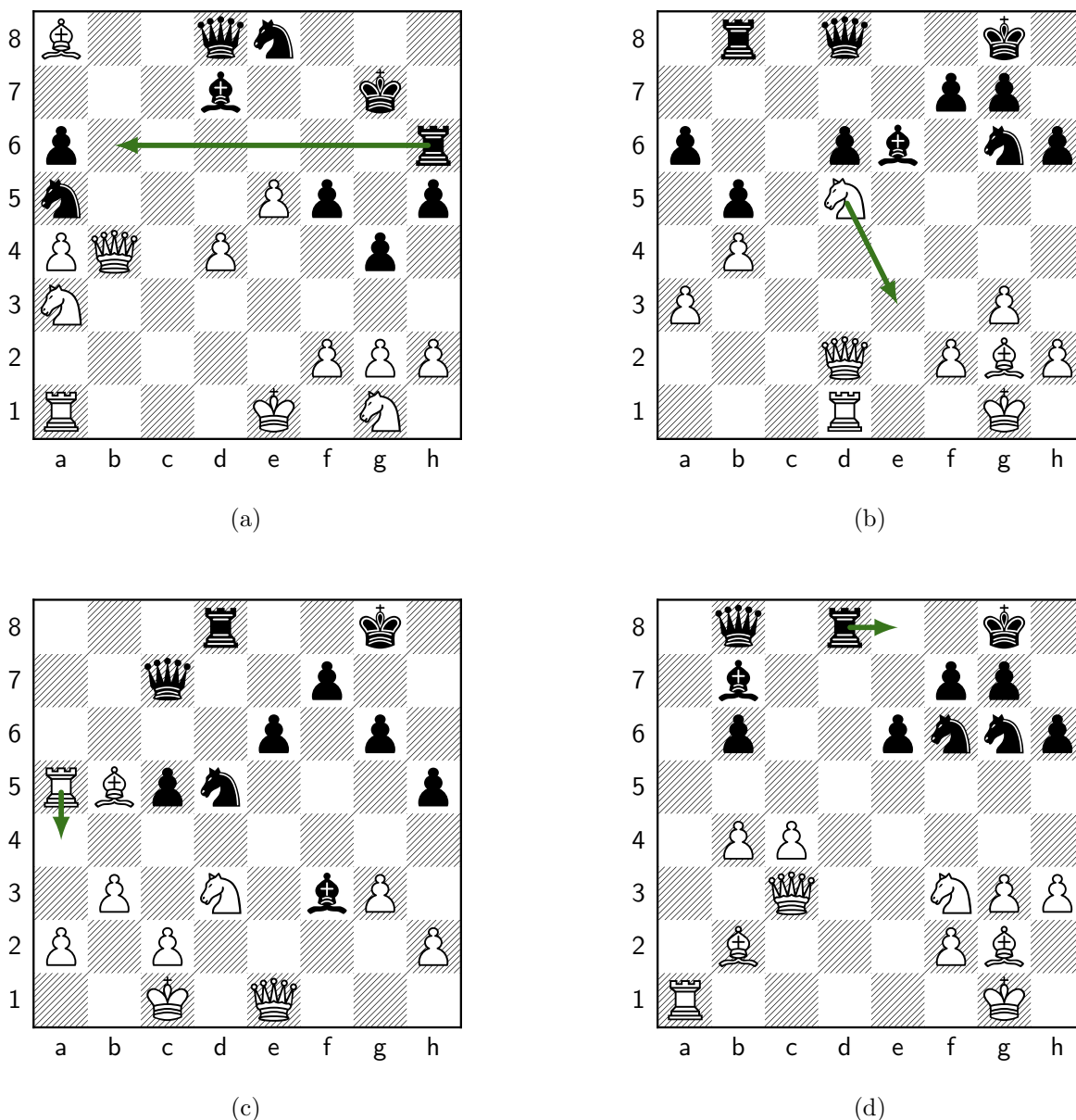
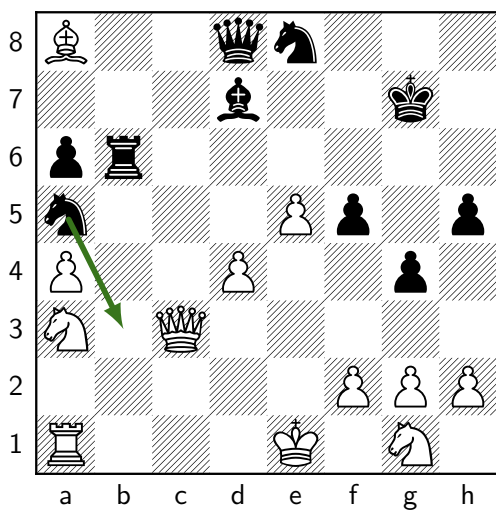
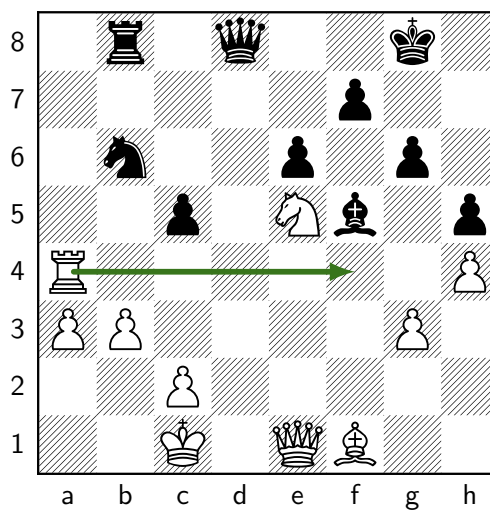


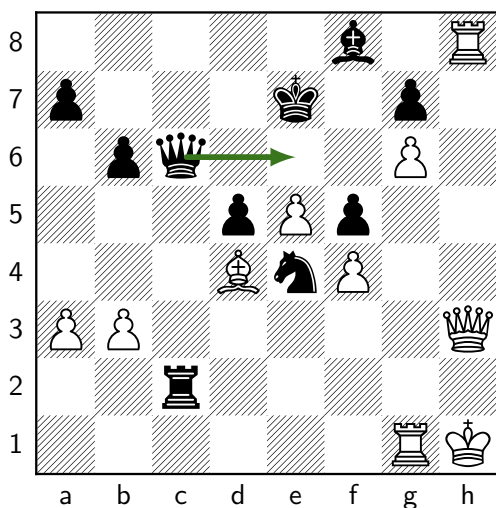
Figure 34: **Move 41, Black: Rb6.** (a) Our position; Melanie plays Rb6 (h6b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2406 (ply 70) [structural distance $d=81$, QAP obj=24]: the master played Ne3; under μ our Na4 plays the role of that game's Nd5; cross-type roles: our Ne8 as their Rd1, our ba8 as their Ph2; transferred move a5c6. (c) Retrieved from ccr1_top_p0003#1603 (ply 72) [structural distance $d=87$, QAP obj=22]: the master played Ra4; under μ our Na4 plays the role of that game's Ra5 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Qe1, our na3 as their Pa2; transferred move a5b3. (d) Retrieved from ccr1_top_p0001#3938 (ply 69) [structural distance $d=88$, QAP obj=15]: the master played Re8; under μ our Ne1 plays the role of that game's Rd1 (a **cross-type** correspondence); cross-type roles: our pf2 as their Qb8, our Qd8 as their Bb7; transferred move e8c7. Synthesis: no retrieved analog transferred to Rb6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



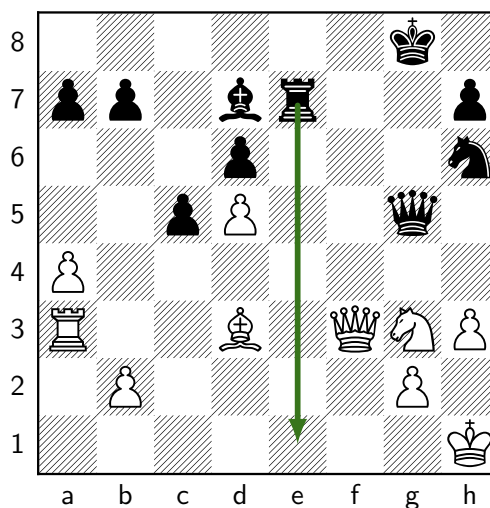
(a)



(b)

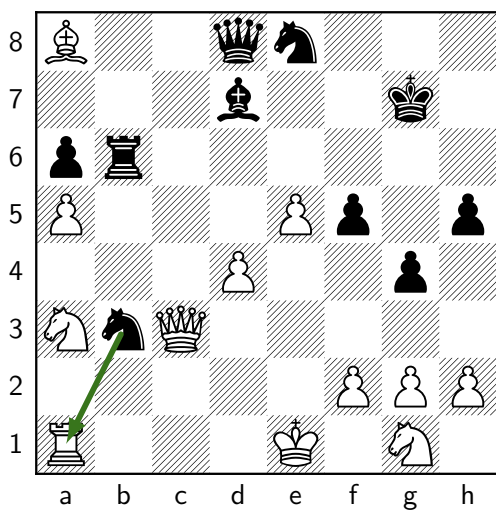


(c)

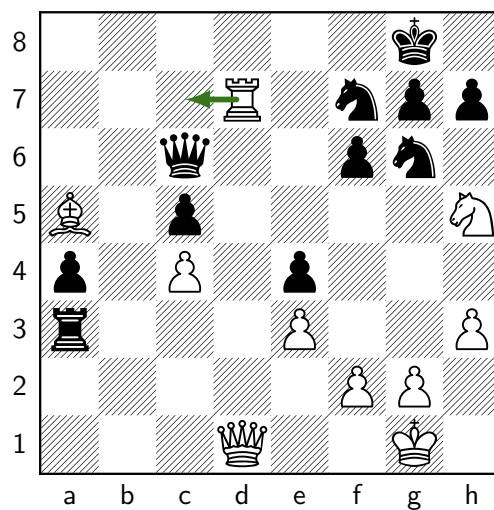


(d)

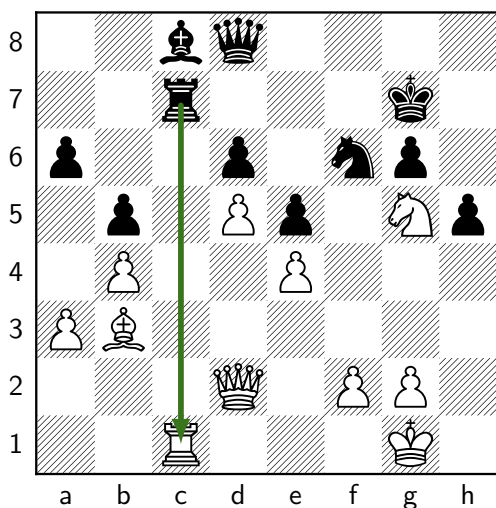
Figure 35: **Move 42, Black: Nb3.** (a) Our position; Melanie plays Nb3 (a5b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1603` (ply 82) [structural distance $d=99$, QAP obj=17]: the master played Rf4; under μ our Na4 plays the role of that game’s Ra4 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Qe1, our pe5 as their Bf1; transferred move a5b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3808` (ply 91) [structural distance $d=99$, QAP obj=31]: the master played Qe6; under μ our Pg5 plays the role of that game’s Qc3 (a **cross-type** correspondence); cross-type roles: our Na5 as their Bf8, our qc3 as their rh8; transferred move g4g3. (d) Retrieved from `ccr1_top_p0003#396` (ply 77) [structural distance $d=96$, QAP obj=26]: the master played Re1+; under μ our Qd1 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Pa6 as their Nh6, our Rb6 as their Pc5; transferred move d8e7. Synthesis: 1 retrieved analog transferred to Nb3 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



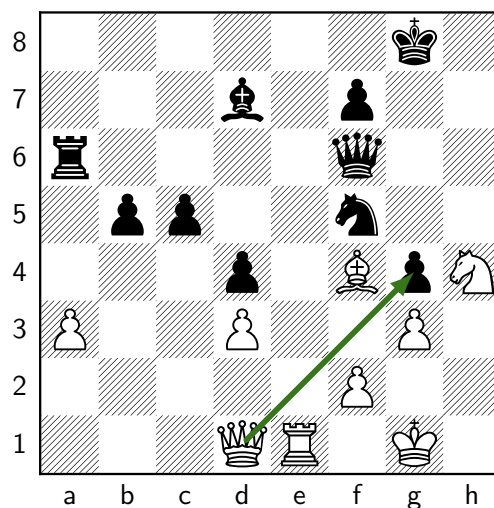
(a)



(b)

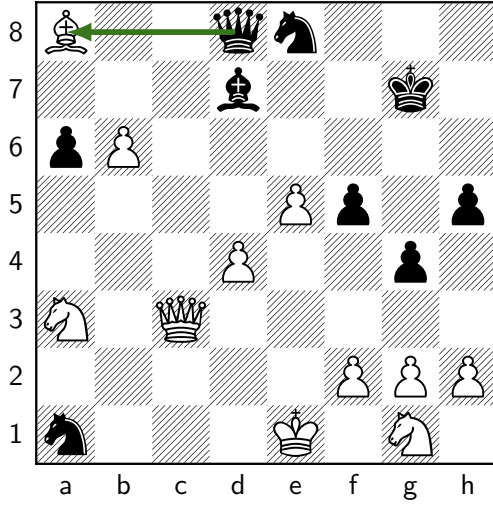


(c)

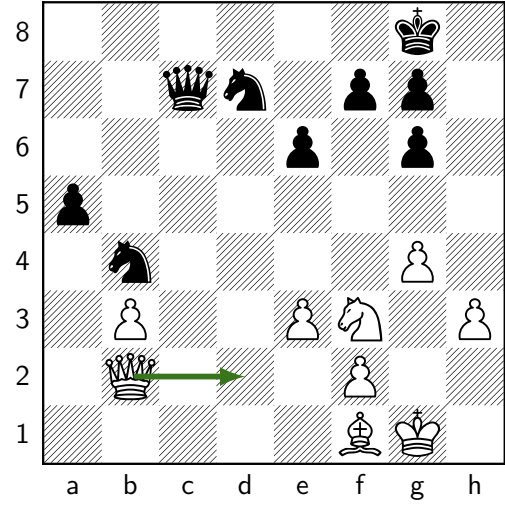


(d)

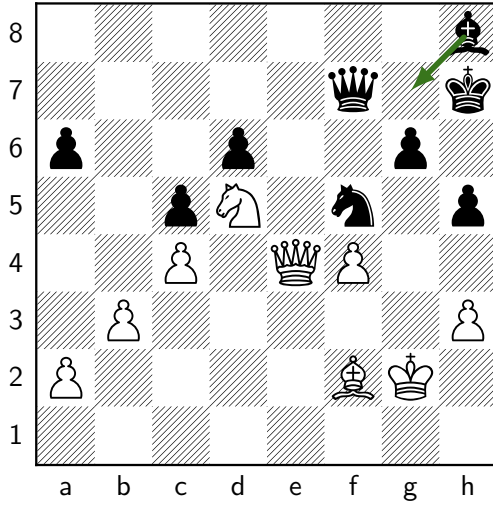
Figure 36: **Move 43, Black: Nxa1.** (a) Our position; Melanie plays Nxa1 (b3a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2627` (ply 76) [structural distance $d=98$, QAP obj=28]: the master played Rc7; under μ our Nb6 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Qd1, our Qd8 as their Pg2; transferred move b3a1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1511` (ply 73) [structural distance $d=100$, QAP obj=6]: the master played Rxc1+; under μ our Nb6 plays the role of that game’s Rc2 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Bc8, our Rb6 as their Qd8; transferred move b3a1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#846` (ply 68) [structural distance $d=92$, QAP obj=39]: the master played Qxg4+; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Ne8 as their Re1, our Bd7 as their Pd3; transferred move d8g5. Synthesis: 2 retrieved analogs transferred to Nxa1 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



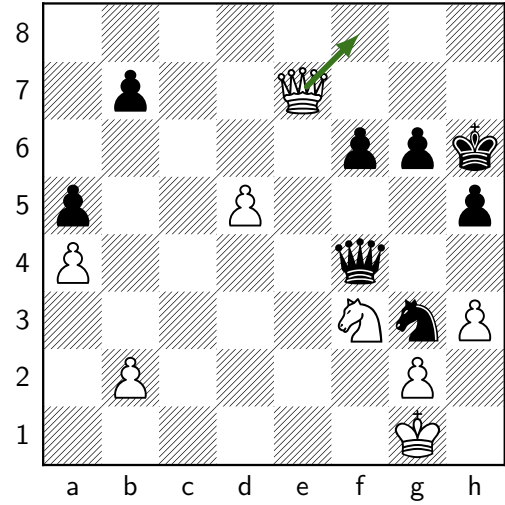
(a)



(b)

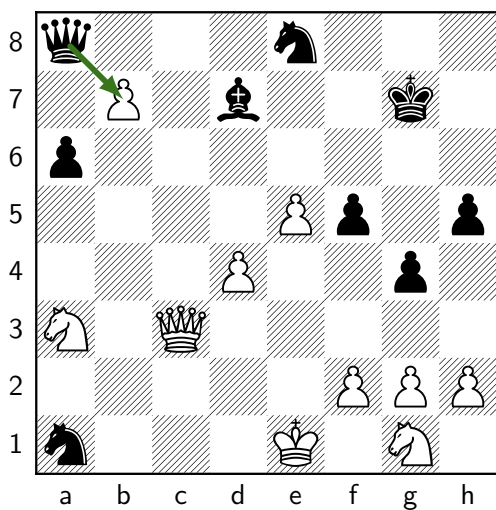


(c)

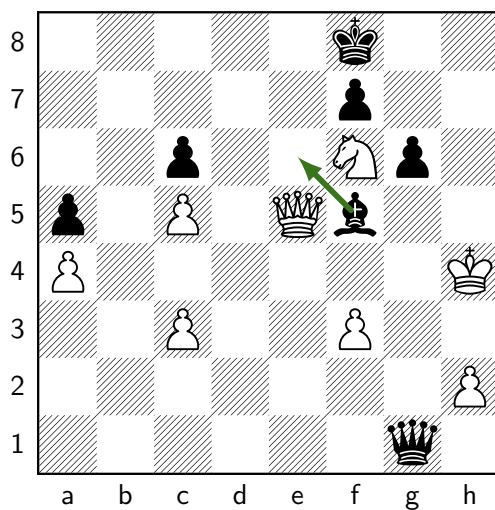


(d)

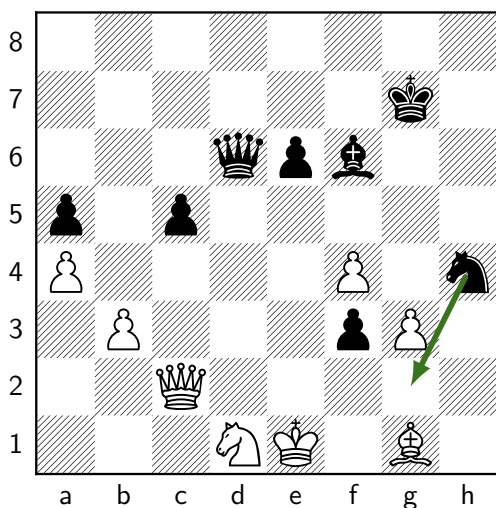
Figure 37: **Move 44, Black: Qxa8.** (a) Our position; Melanie plays Qxa8 (d8a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3545` (ply 74) [structural distance $d=79$, QAP obj=5]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qb2; cross-type roles: our pd4 as their Bf1, our Na1 as their Pf2; transferred move d8a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2175` (ply 101) [structural distance $d=73$, QAP obj=21]: the master played Bg7; under μ our Ne1 plays the role of that game’s Bh1 (a **cross-type** correspondence); cross-type roles: our Bd7 as their Pc5, our ba8 as their nd5; transferred move e8f6. (d) Retrieved from `ccr1_top_p0003#1738` (ply 84) [structural distance $d=76$, QAP obj=20]: the master played Qf8+; under μ our Qd1 plays the role of that game’s Qe7; cross-type roles: our pe5 as their ng3, our Bd7 as their Pa4; transferred move d8b6. Synthesis: 1 retrieved analog transferred to Qxa8 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



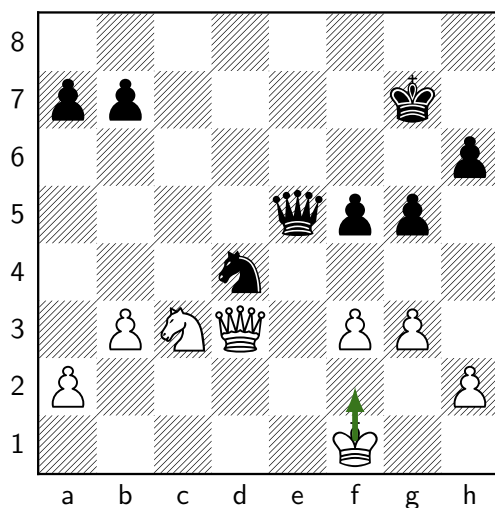
(a)



(b)

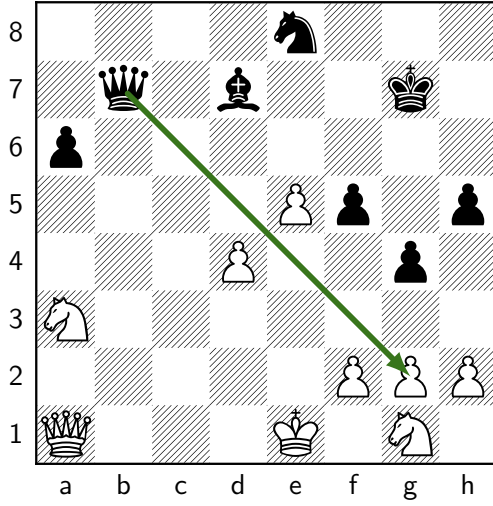


(c)

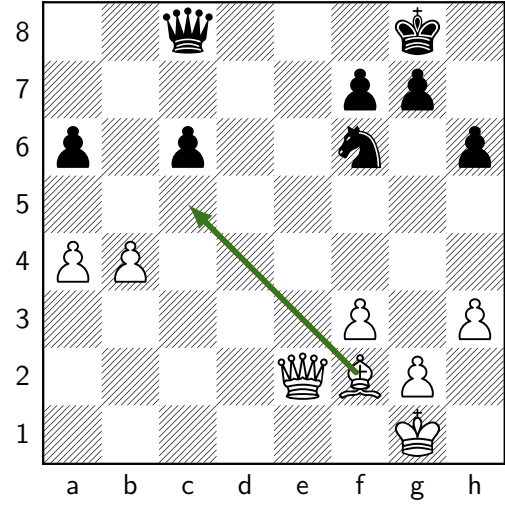


(d)

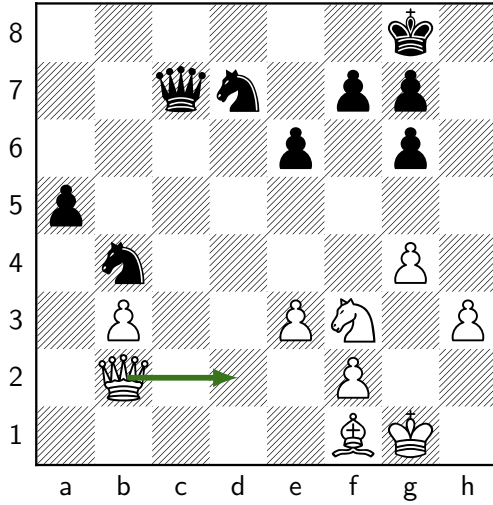
Figure 38: **Move 45, Black: Qxb7.** (a) Our position; Melanie plays Qxb7 (a8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2856` (ply 83) [structural distance $d=73$, QAP obj=9]: the master played Be6; under μ our Qa1 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our pe5 as their nf6, our Ne8 as their Pg6; transferred move a8b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1895` (ply 101) [structural distance $d=58$, QAP obj=11]: the master played Ng2+; under μ our Qa1 plays the role of that game’s Nh5 (a **cross-type** correspondence); cross-type roles: our Pf5 as their Qd6, our Ne8 as their Bf6; transferred move a8a7. (d) Retrieved from `ccr1_top_p0000#193` (ply 72) [structural distance $d=60$, QAP obj=11]: the master played Kf2; under μ our Kg2 plays the role of that game’s Kf1; cross-type roles: our Na1 as their Ph2, our pb7 as their Qd3; transferred move g7h7. Synthesis: 1 retrieved analog transferred to Qxb7 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



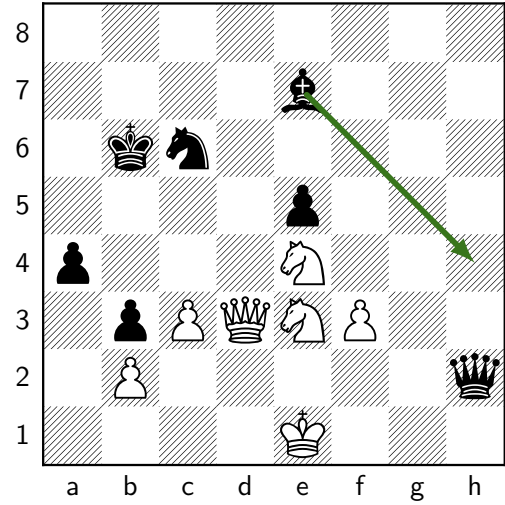
(a)



(b)

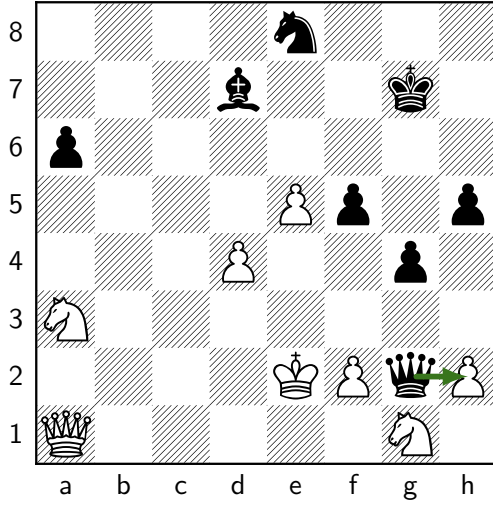


(c)

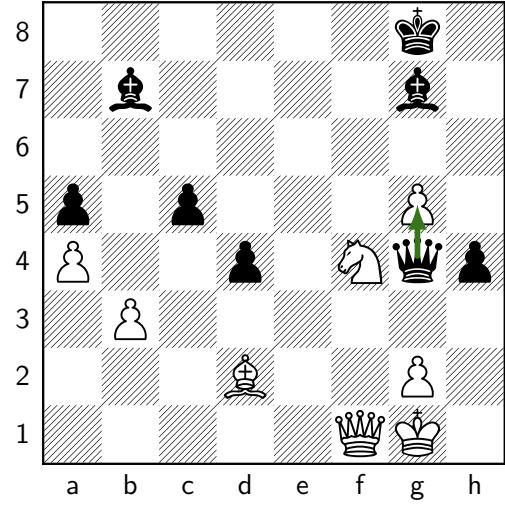


(d)

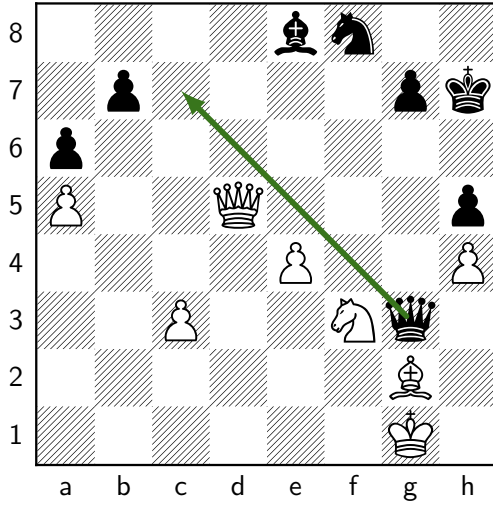
Figure 39: **Move 46, Black: Qxg2.** (a) Our position; Melanie plays Qxg2 (b7g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3357` (ply 102) [structural distance $d=47$, QAP obj=25]: the master played Bc5; under μ our *Ne1* plays the role of that game’s *Bf2* (a **cross-type** correspondence); cross-type roles: our *ng1* as their *Pg2*, our *Bd7* as their *pc6*; transferred move e8c7. (c) Retrieved from `ccr1_top_p0001#3545` (ply 74) [structural distance $d=44$, QAP obj=23]: the master played Qd2; under μ our *Qb2* plays the role of that game’s *Qb2*; cross-type roles: our *Ph5* as their *Bf1*, our *Bd7* as their *Pf2*; transferred move b7c7. (d) Retrieved from `ccr1_top_p0001#3867` (ply 107) [structural distance $d=51$, QAP obj=20]: the master played Bh4+; under μ our *Pa3* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Nc6*, our *Bd7* as their *Pe5*; transferred move a6a5. Synthesis: no retrieved analog transferred to Qxg2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



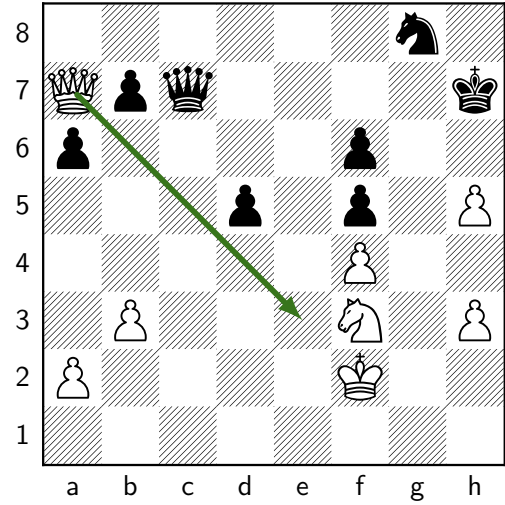
(a)



(b)

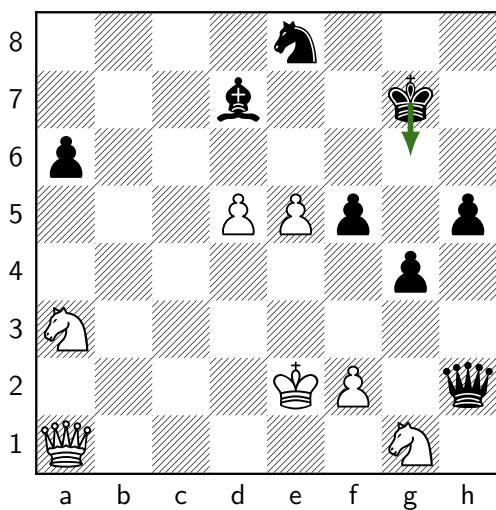


(c)

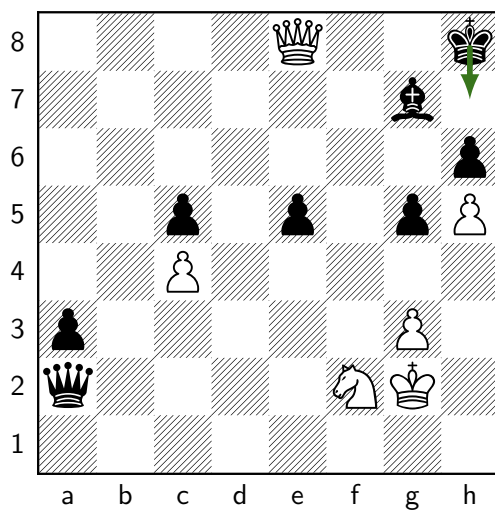


(d)

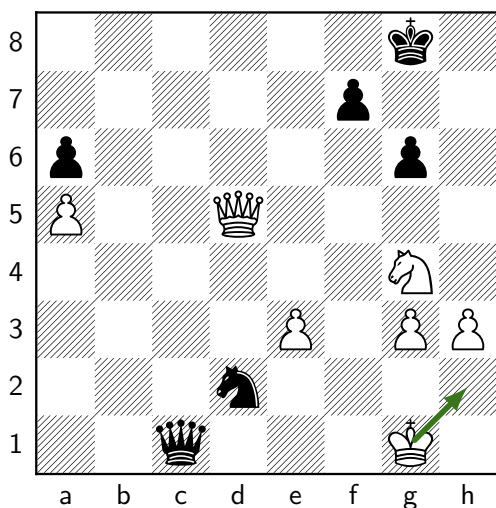
Figure 40: **Move 47, Black: Qxh2.** (a) Our position; Melanie plays Qxh2 (g2h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3420` (ply 95) [structural distance $d=55$, QAP obj=69]: the master played Qxg5; under μ our Qg7 plays the role of that game’s Qg5; cross-type roles: our Pf5 as their Bb7, our Ph5 as their Bg7; transferred move g2h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2391` (ply 83) [structural distance $d=58$, QAP obj=76]: the master played Qc7; under μ our Qg7 plays the role of that game’s Qg6; cross-type roles: our Pf5 as their Be8, our na3 as their pa5; transferred move g2b7. (d) Retrieved from `ccr1_top_p0002#231` (ply 80) [structural distance $d=59$, QAP obj=51]: the master played Qe3; under μ our Qg7 plays the role of that game’s Qa7; cross-type roles: our na3 as their pf5, our ng1 as their pa6; transferred move g2c6. Synthesis: 1 retrieved analog transferred to Qxh2 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



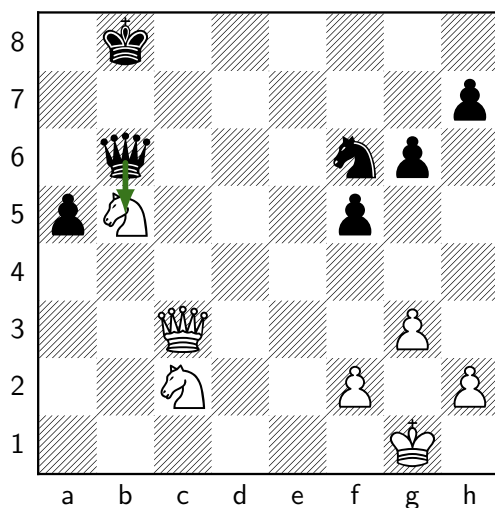
(a)



(b)

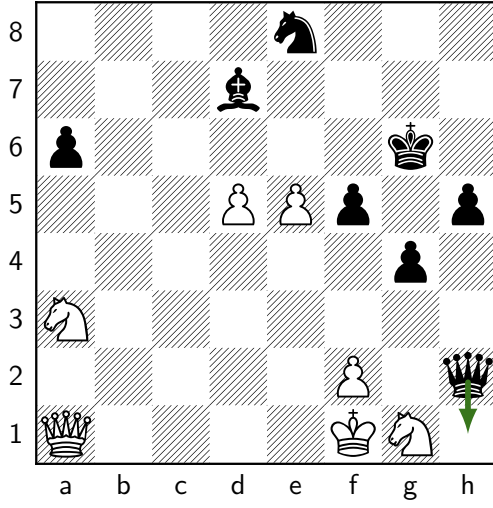


(c)

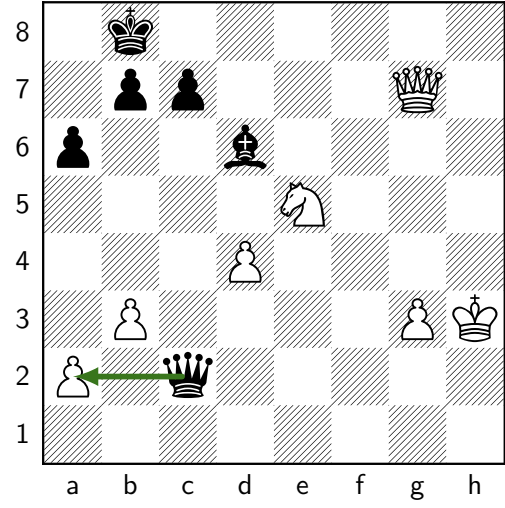


(d)

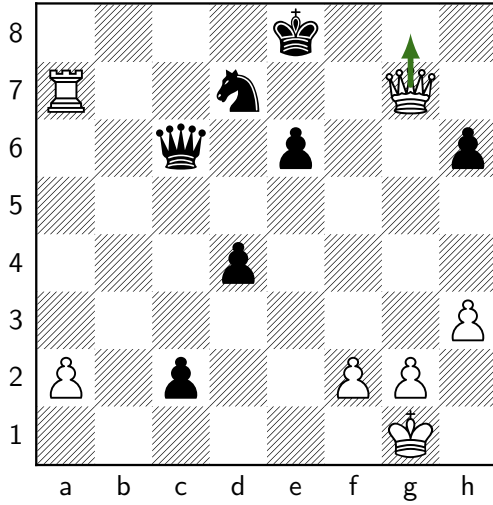
Figure 41: **Move 48, Black: Kg6**. (a) Our position; Melanie plays Kg6 (**g7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2237` (ply 83) [structural distance $d=60$, QAP obj=26]: the master played Kh7; under μ our Kg2 plays the role of that game’s Kh1; cross-type roles: our pe5 as their qe8, our Ne8 as their Bg7; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3175` (ply 80) [structural distance $d=58$, QAP obj=25]: the master played Kh2; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our pe5 as their nd2, our Ne8 as their Pe3; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2934` (ply 87) [structural distance $d=61$, QAP obj=40]: the master played Qxb5; under μ our Qh7 plays the role of that game’s Qb3; cross-type roles: our Ne8 as their Ph7, our Ph5 as their Nf6; transferred move h2h4. Synthesis: 3 retrieved analogs transferred to Kg6 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



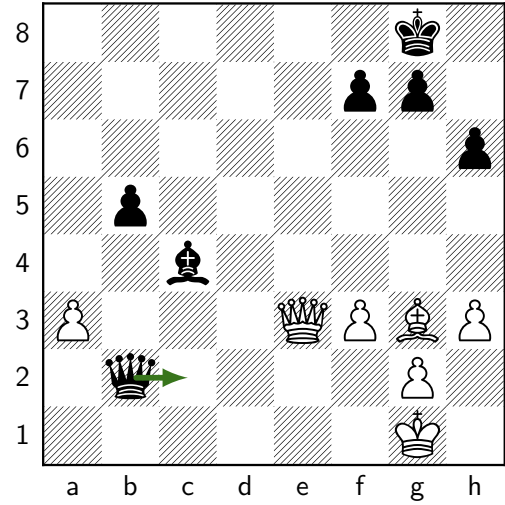
(a)



(b)

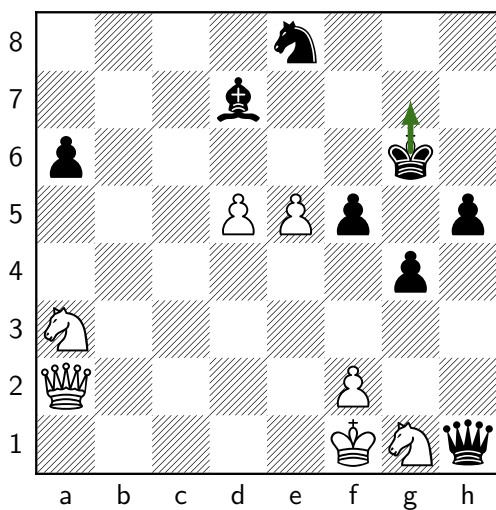


(c)

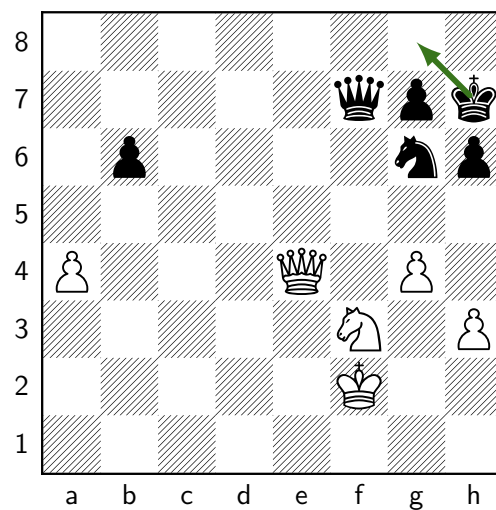


(d)

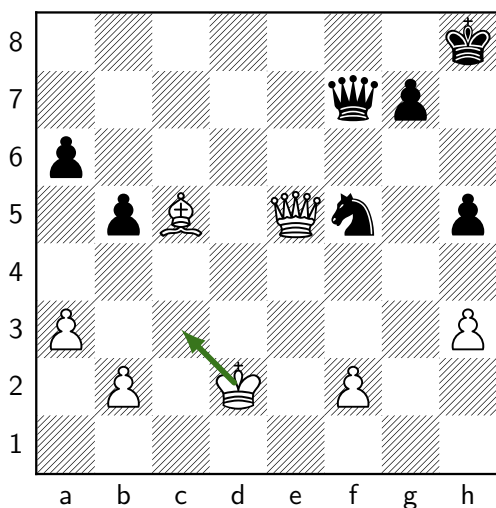
Figure 42: **Move 49, Black: Qh1.** (a) Our position; Melanie plays Qh1 (h2h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#111` (ply 97) [structural distance $d=42$, QAP obj=24]: the master played Qxa2; under μ our Qh7 plays the role of that game’s Qc7; cross-type roles: our Ne8 as their Pa6, our Qh2 as their Bd6; transferred move h2h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2590` (ply 98) [structural distance $d=52$, QAP obj=18]: the master played Qg8+; under μ our Qh7 plays the role of that game’s Qg7; cross-type roles: our na3 as their pc2, our Bd7 as their Ph3; transferred move h2h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3025` (ply 83) [structural distance $d=58$, QAP obj=10]: the master played Qc2; under μ our Qh7 plays the role of that game’s Qb7; cross-type roles: our Bd7 as their Pf7, our Ne8 as their Pg7; transferred move h2h1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qh1 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



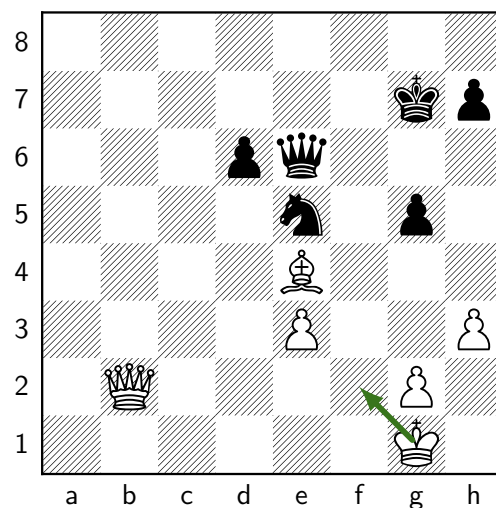
(a)



(b)

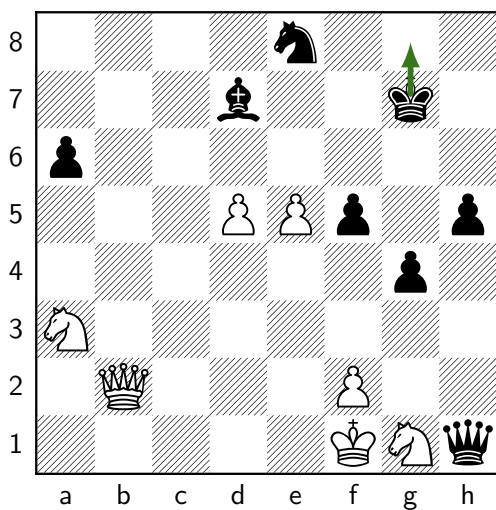


(c)

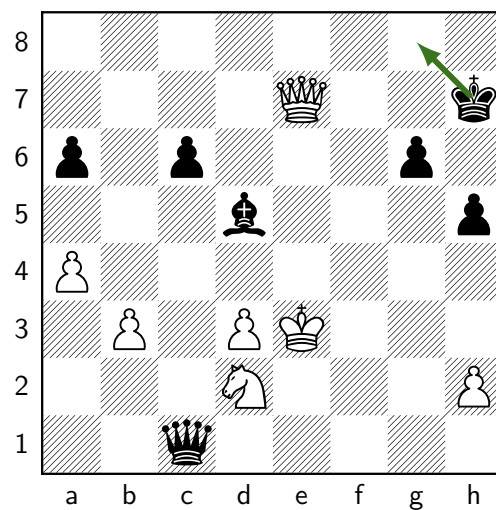


(d)

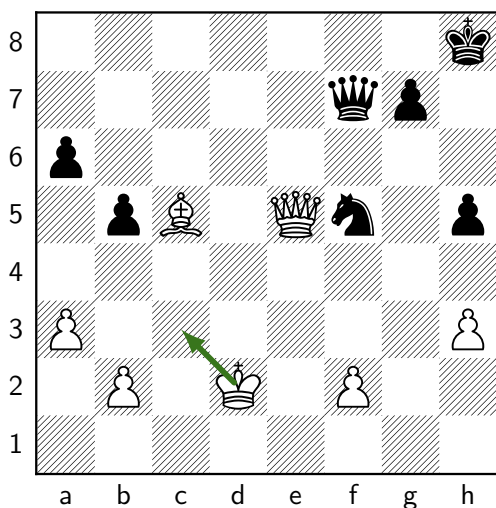
Figure 43: **Move 50, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#568` (ply 95) [structural distance $d=52$, QAP obj=26]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kh2; cross-type roles: our Ne8 as their Pb6, our Bd7 as their ph3; transferred move g6g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3823` (ply 114) [structural distance $d=41$, QAP obj=26]: the master played Kc3; under μ our Kg3 plays the role of that game’s Kd2; cross-type roles: our qa2 as their Pb2, our na3 as their Pa3; transferred move g6g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3651` (ply 88) [structural distance $d=49$, QAP obj=20]: the master played Kf2; under μ our Kg3 plays the role of that game’s Kg1; cross-type roles: our Bd7 as their Pe3, our Ne8 as their Be4; transferred move g6g7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg7 (summed QAP objective 72, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



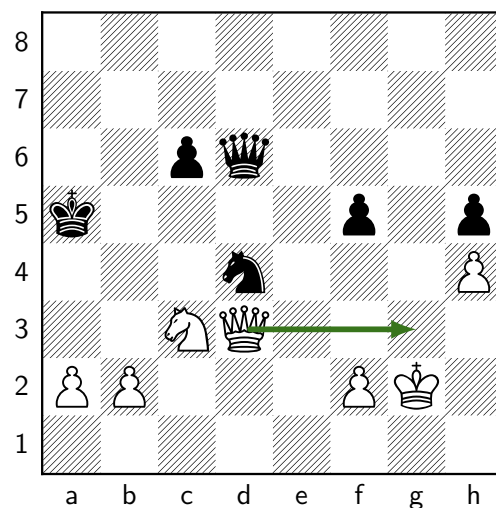
(a)



(b)

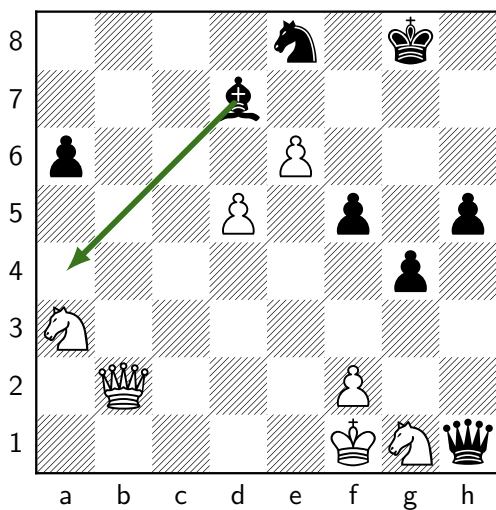


(c)

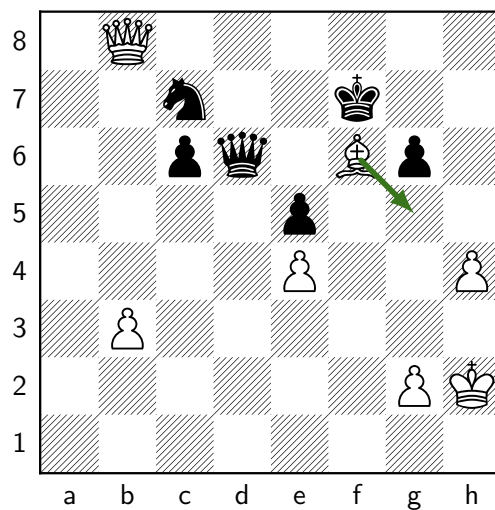


(d)

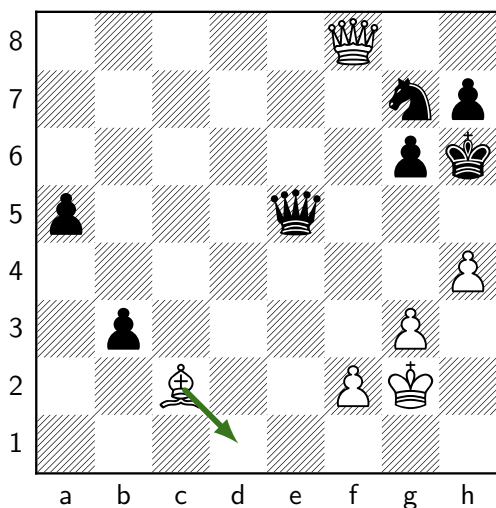
Figure 44: **Move 51, Black: Kg8**. (a) Our position; Melanie plays Kg8 (g7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#243` (ply 111) [structural distance $d=44$, QAP obj=28]: the master played Kg8; under μ our Kg2 plays the role of that game’s Kh2; cross-type roles: our na3 as their Pg6, our Pg4 as their Bd5; transferred move g7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3823` (ply 114) [structural distance $d=33$, QAP obj=25]: the master played Kc3; under μ our Kg2 plays the role of that game’s Kd2; cross-type roles: our na3 as their Pa3, our Ph5 as their Bc5; transferred move g7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3328` (ply 112) [structural distance $d=48$, QAP obj=45]: the master played Qg3; under μ our Qh8 plays the role of that game’s Qd3; cross-type roles: our Ne8 as their Pf2, our Pg4 as their Nc3; transferred move h1h3. Synthesis: 2 retrieved analogs transferred to Kg8 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



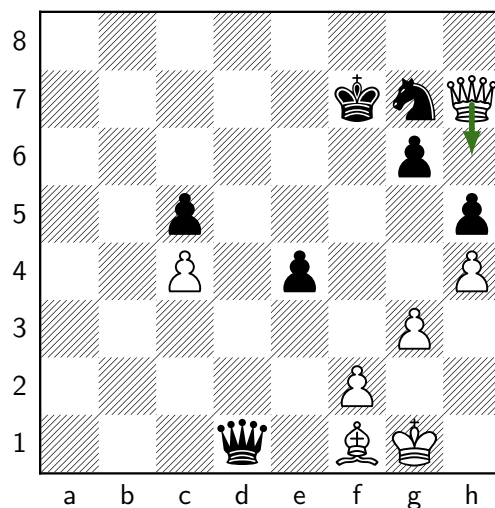
(a)



(b)

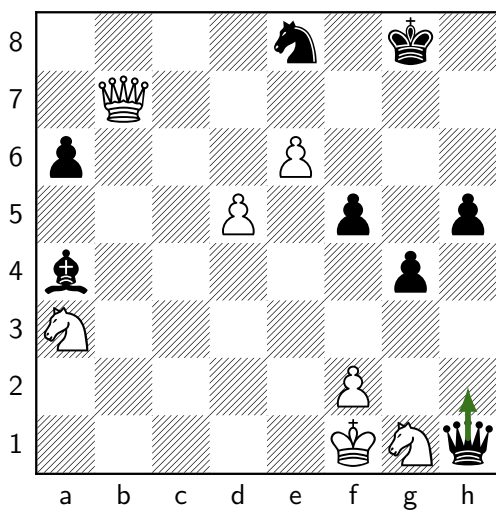


(c)

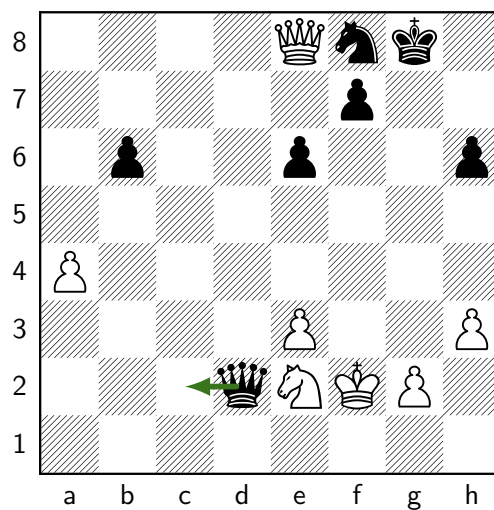


(d)

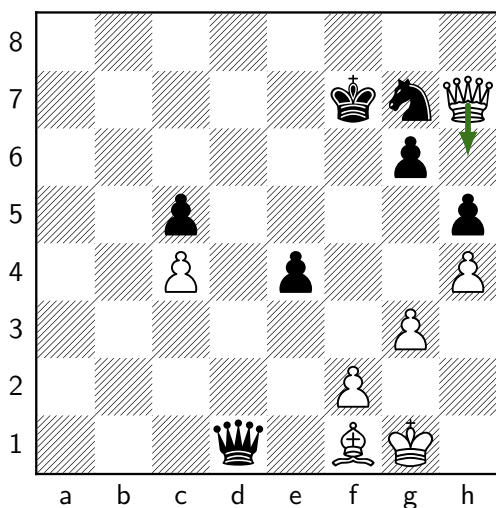
Figure 45: **Move 52, Black: Ba4.** (a) Our position; Melanie plays Ba4 (d7a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3754` (ply 108) [structural distance $d=35$, QAP obj=53]: the master played Bg5; under μ our Bd2 plays the role of that game’s Bf6; cross-type roles: our qb2 as their Pg2, our pe6 as their qd6; transferred move d7a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1350` (ply 94) [structural distance $d=43$, QAP obj=70]: the master played Bd1; under μ our Bd2 plays the role of that game’s Bc2; cross-type roles: our Ne8 as their pa5; transferred move d7c8. (d) Retrieved from `ccr1_top_p0002#1102` (ply 118) [structural distance $d=41$, QAP obj=63]: the master played Qh6; under μ our Qh8 plays the role of that game’s Qh7; cross-type roles: our ng1 as their Bf1, our Bd7 as their Pg3; transferred move h1h2. Synthesis: 1 retrieved analog transferred to Ba4 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



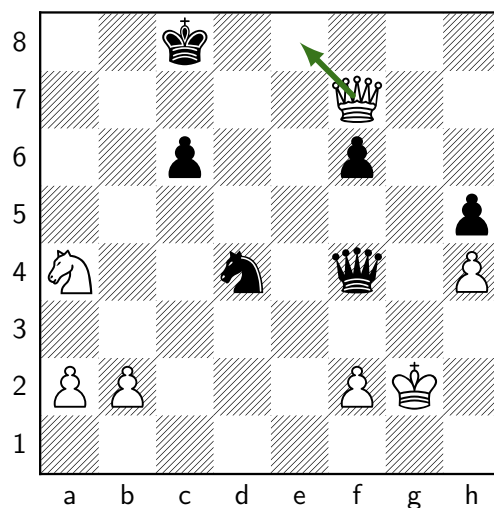
(a)



(b)

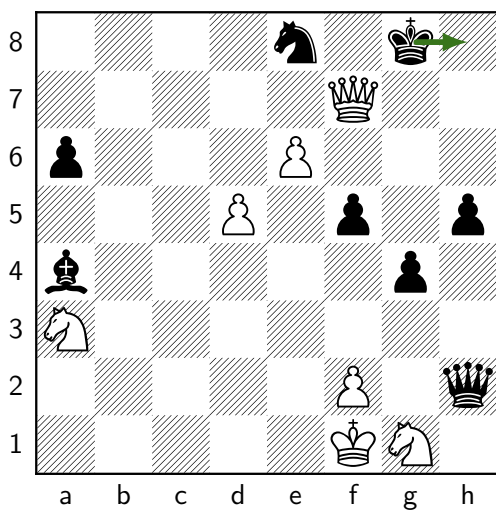


(c)

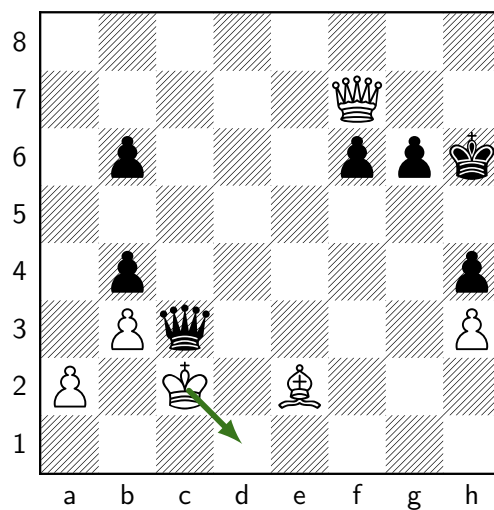


(d)

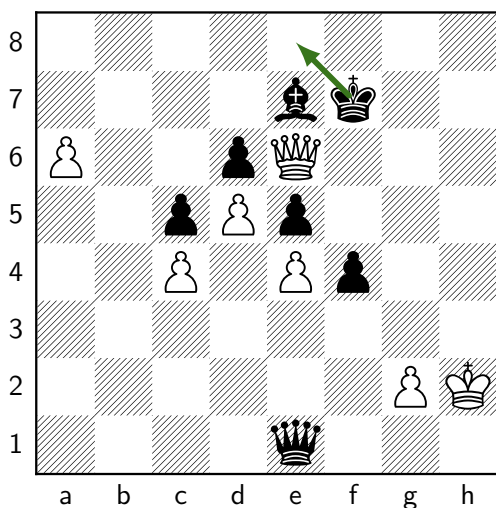
Figure 46: **Move 53, Black: Qh2**. (a) Our position; Melanie plays Qh2 (h1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2037` (ply 95) [structural distance $d=44$, QAP obj=87]: the master played Qc2; under μ our Qh8 plays the role of that game’s Qd7; cross-type roles: our ng1 as their Pf7, our qb7 as their Pb6; transferred move h1h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1102` (ply 118) [structural distance $d=43$, QAP obj=64]: the master played Qh6; under μ our Qh8 plays the role of that game’s Qh7; cross-type roles: our ng1 as their Bf1, our Ne8 as their Pg3; transferred move h1h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3328` (ply 100) [structural distance $d=45$, QAP obj=18]: the master played Qe8+; under μ our Qh8 plays the role of that game’s Qf7; cross-type roles: our Ne8 as their Pb2, our Ba4 as their Na4; transferred move h1h2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qh2 (summed QAP objective 183, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



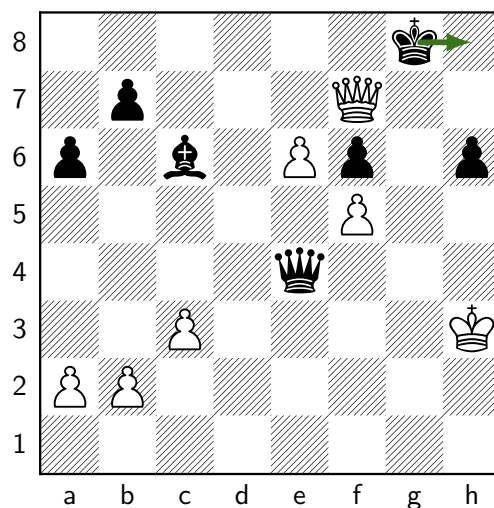
(a)



(b)

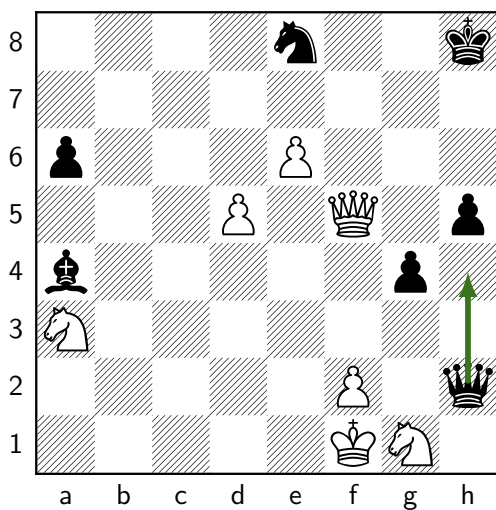


(c)

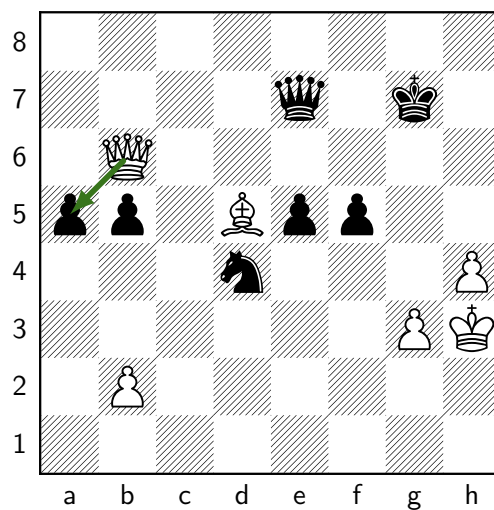


(d)

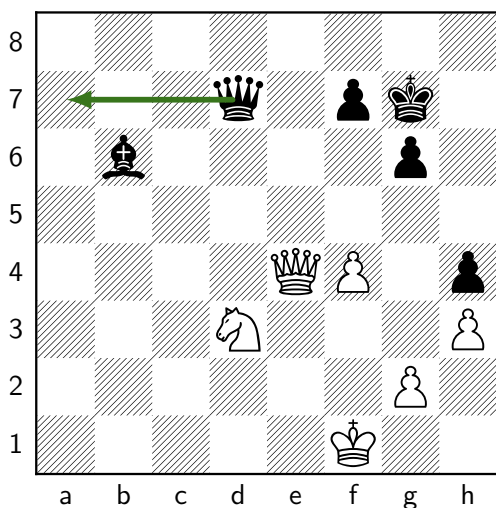
Figure 47: **Move 54, Black: Kh8 (g8h8)**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#312` (ply 120) [structural distance $d=40$, QAP obj=80]: the master played Kd1; under μ our Kg1 plays the role of that game’s Kc2; cross-type roles: our Ne8 as their Pa2, our ng1 as their Ph3; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1005` (ply 107) [structural distance $d=46$, QAP obj=78]: the master played Ke8; under μ our Kg1 plays the role of that game’s Kf2; cross-type roles: our Qh2 as their Be7, our Ne8 as their pa6; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#688` (ply 107) [structural distance $d=44$, QAP obj=77]: the master played Kh8; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our na3 as their Pb7, our Ne8 as their Ph6; transferred move g8h8 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kh8 (summed QAP objective 651, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



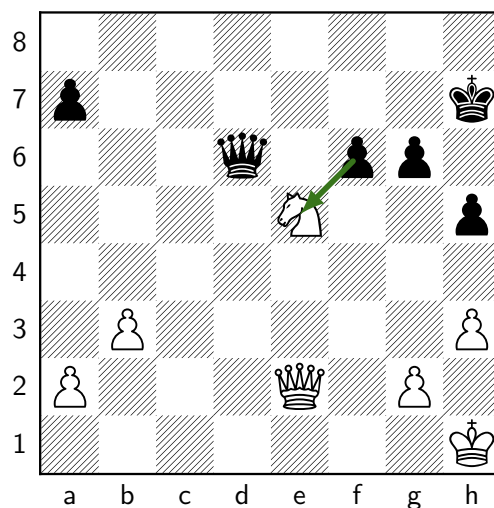
(a)



(b)

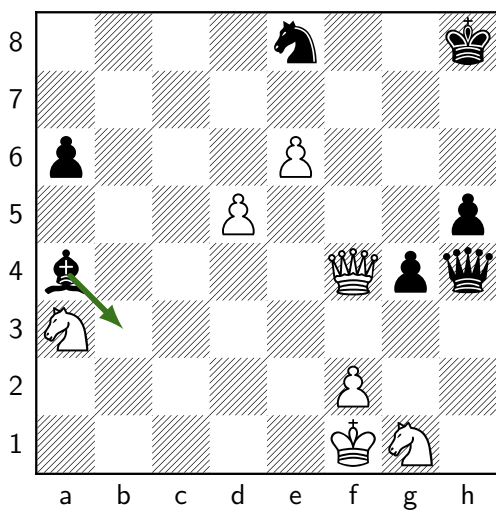


(c)

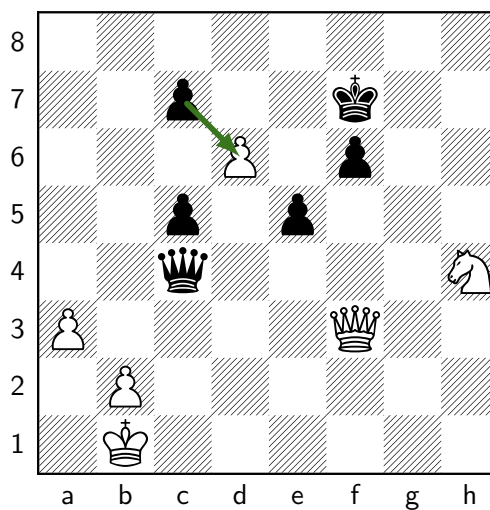


(d)

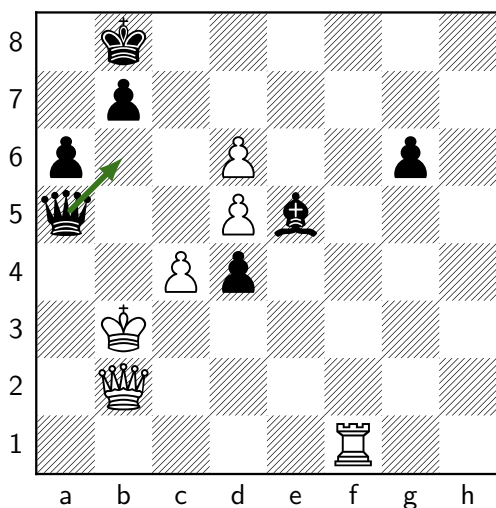
Figure 48: **Move 55, Black: Qh4 (h2h4)**. (a) Our position; Melanie plays Qh4 (h2h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2253` (ply 100) [structural distance $d=43$, QAP obj=24]: the master played Qxa5; under μ our Qh7 plays the role of that game’s Qb6; cross-type roles: our Ne8 as their Pb2, our Ba4 as their Pg3; transferred move h2h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2951` (ply 121) [structural distance $d=35$, QAP obj=14]: the master played Qa7; under μ our Qh7 plays the role of that game’s Qd2; cross-type roles: our Ne8 as their Bb6, our Ba4 as their Ph4; transferred move h2h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3423` (ply 111) [structural distance $d=27$, QAP obj=37]: the master played fxe5; under μ our Qh7 plays the role of that game’s Pf3 (a **cross-type** correspondence); cross-type roles: our Ne8 as their Pa7; transferred move h2f4. Synthesis: 2 retrieved analogs transferred to Qh4 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



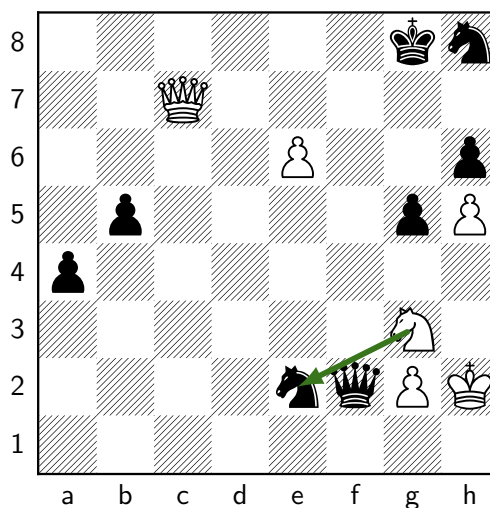
(a)



(b)

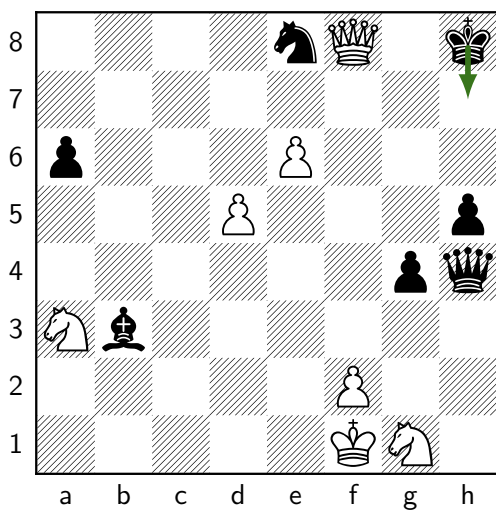


(c)

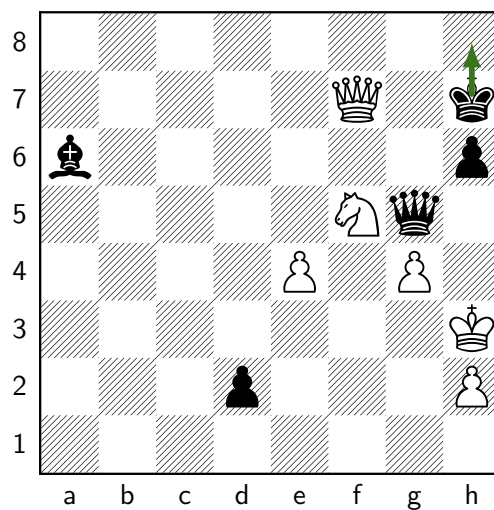


(d)

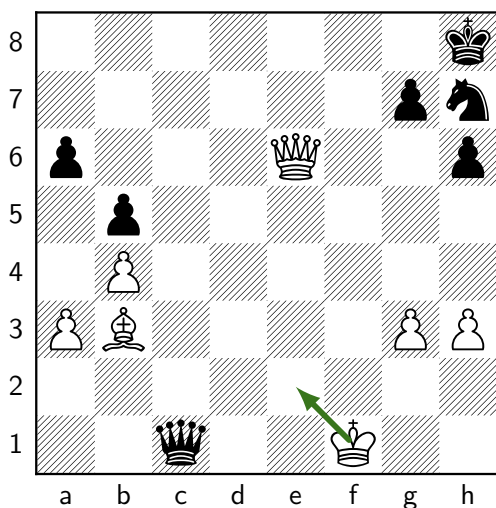
Figure 49: **Move 56, Black: Bb3.** (a) Our position; Melanie plays Bb3 (a4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2349` (ply 127) [structural distance $d=41$, QAP obj=37]: the master played `cx d6`; under μ our `Pg5` plays the role of that game’s `Pc2`; cross-type roles: our `qf4` as their `pd6`, our `Ne8` as their `Pe5`; transferred move `g4g3`. (c) Retrieved from `ccr1_top_p0001#1668` (ply 125) [structural distance $d=43$, QAP obj=36]: the master played `Qb6+`; under μ our `Qh5` plays the role of that game’s `Qa4`; cross-type roles: our `qf4` as their `Be5`, our `ng1` as their `pc4`; transferred move `h4g5`. (d) Retrieved from `ccr1_top_p0002#205` (ply 118) [structural distance $d=44$, QAP obj=35]: the master played `Nxe2`; under μ our `Pg5` plays the role of that game’s `Ng3` (a **cross-type** correspondence); cross-type roles: our `pf2` as their `ne2`, our `na3` as their `pa4`; transferred move `g4g3`. Synthesis: no retrieved analog transferred to Bb3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



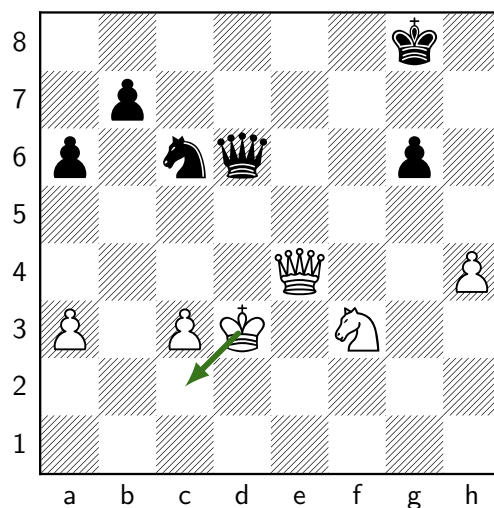
(a)



(b)



(c)



(d)

Figure 50: **Move 57, Black: Kh7 (h8h7)**. (a) Our position; Melanie plays Kh7 (h8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#117` (ply 97) [structural distance $d=41$, QAP obj=69]: the master played Kh8; under μ our Kh1 plays the role of that game’s Kh2; cross-type roles: our Ne8 as their Ba6, our pf2 as their nf5; transferred move h8h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2` (ply 106) [structural distance $d=38$, QAP obj=49]: the master played Ke2; under μ our Kh1 plays the role of that game’s Kf1; cross-type roles: our Ph5 as their Bb3, our Ne8 as their Pg3; transferred move h8h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3111` (ply 128) [structural distance $d=43$, QAP obj=47]: the master played Kc2; under μ our Kh1 plays the role of that game’s Kd3; cross-type roles: our Ne8 as their Pa3, our Bb3 as their Nf3; transferred move h8h7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh7 (summed QAP objective 323, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

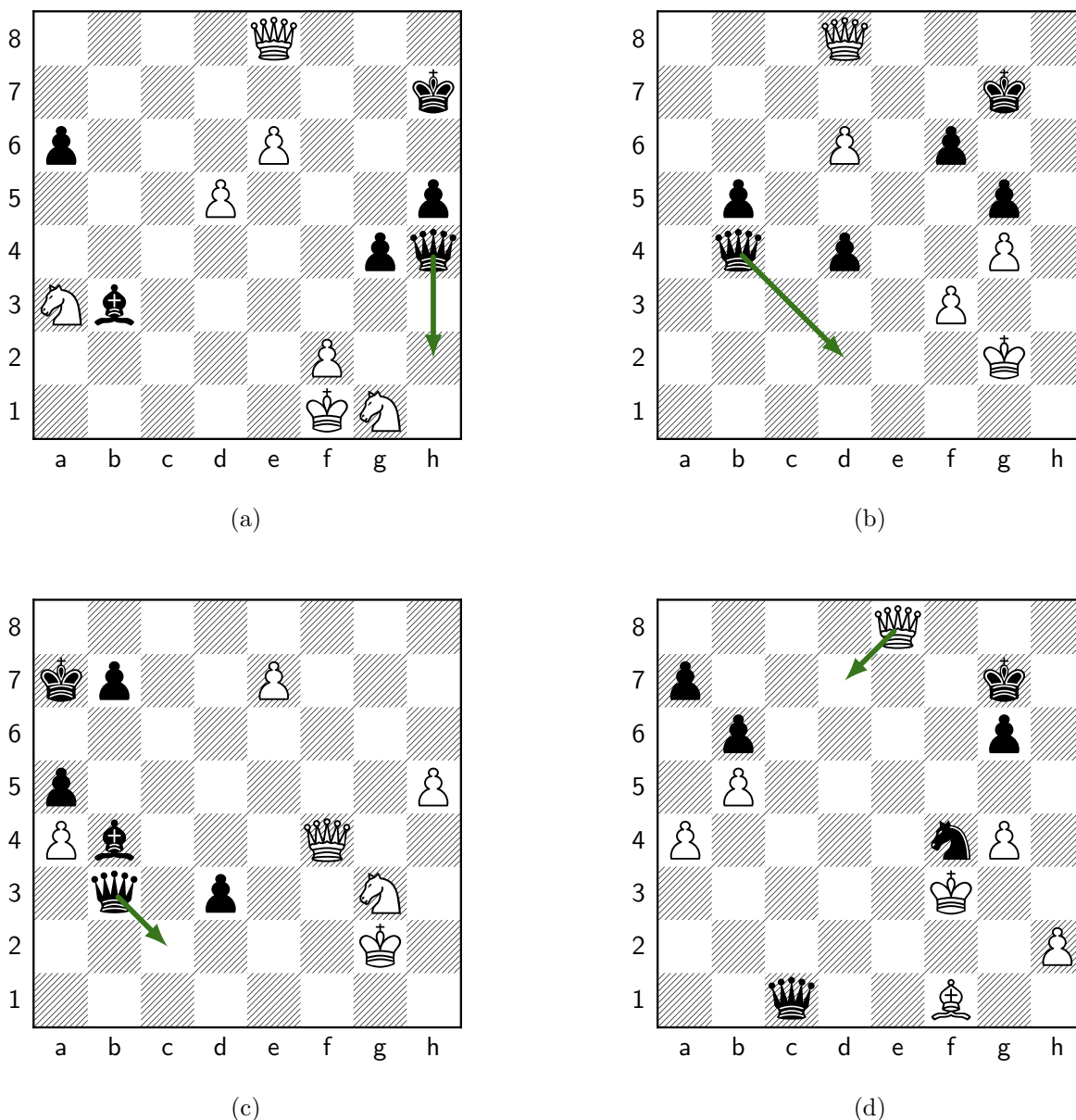
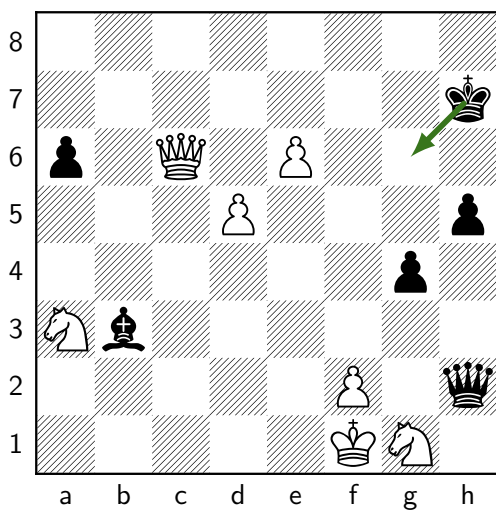
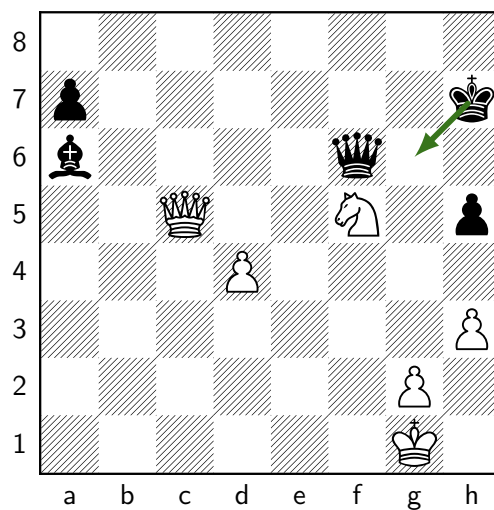


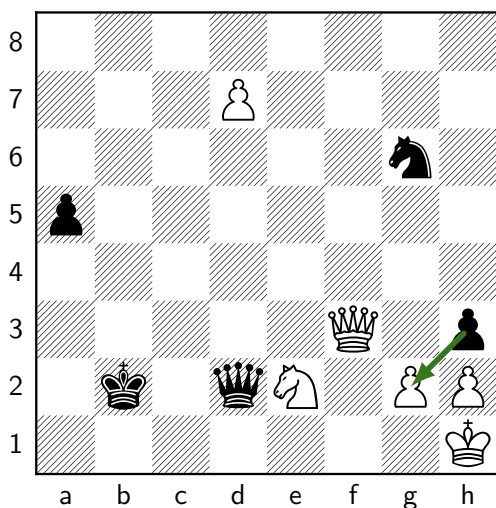
Figure 51: **Move 58, Black: Qh2.** (a) Our position; Melanie plays Qh2 (h4h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#122` (ply 113) [structural distance $d=31$, QAP obj=42]: the master played Qd2+; under μ our Qh5 plays the role of that game’s Qb5; cross-type roles: our Bb3 as their qd8, our ng1 as their pf3; transferred move h4h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2112` (ply 121) [structural distance $d=35$, QAP obj=21]: the master played Qc2+; under μ our Qh5 plays the role of that game’s Qb6; cross-type roles: our qe8 as their Pb7, our pd5 as their Bb4; transferred move h4h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3138` (ply 114) [structural distance $d=29$, QAP obj=4]: the master played Qd7+; under μ our Qh5 plays the role of that game’s Qe8; cross-type roles: our Pg4 as their Bf1, our Bb3 as their Pb5; transferred move h4h2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qh2 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



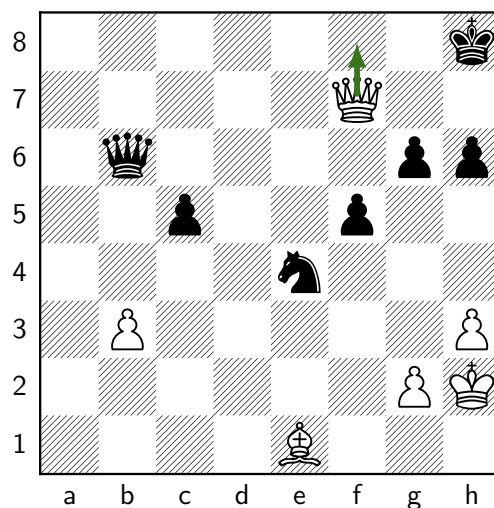
(a)



(b)

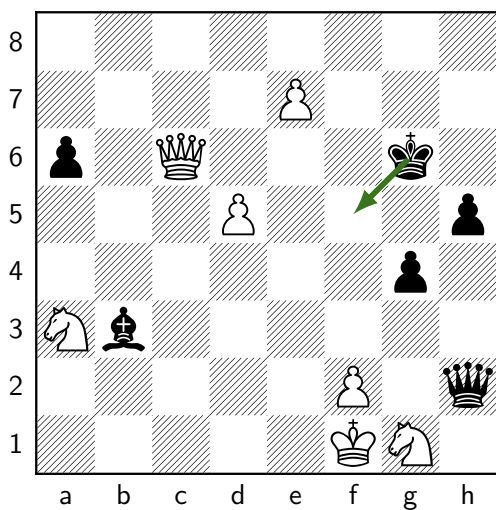


(c)

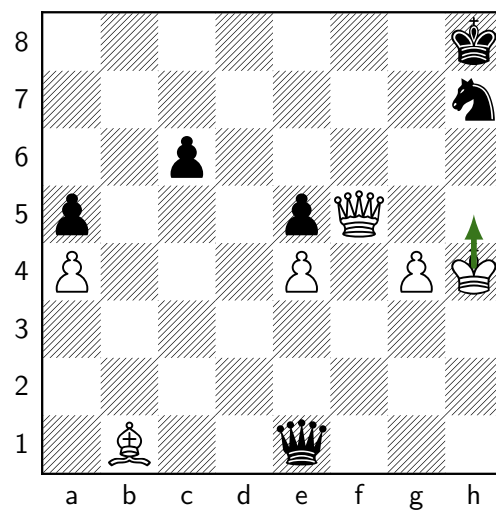


(d)

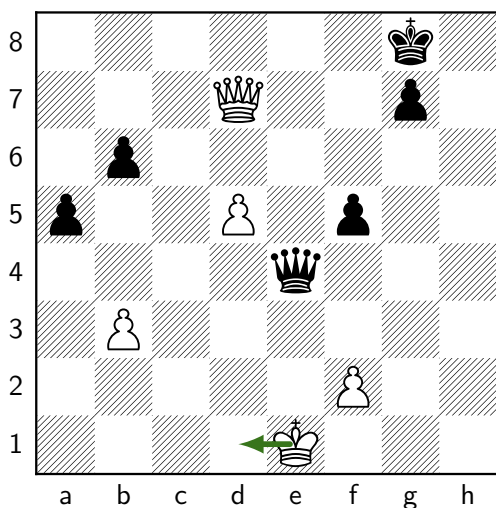
Figure 52: **Move 59, Black: Kg6.** (a) Our position; Melanie plays Kg6 (**h7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2871` (ply 117) [structural distance $d=31$, QAP obj=25]: the master played Kg6; under μ our *Kh2* plays the role of that game’s *Kh2*; cross-type roles: our *Ph5* as their *Ba6*, our *Bb3* as their *pg2*; transferred move **h7g6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3342` (ply 115) [structural distance $d=30$, QAP obj=21]: the master played **hxf2+**; under μ our *Bb6* plays the role of that game’s *Ph6* (a **cross-type** correspondence); cross-type roles: our *Pa6* as their *Ng6*, our *na3* as their *ph2*; transferred move **b3d5**. (d) Retrieved from `ccr1_top_p0000#2942` (ply 130) [structural distance $d=32$, QAP obj=16]: the master played **Qf8+**; under μ our *Qh7* plays the role of that game’s *Qf7*; cross-type roles: our *ng1* as their *pg6*; transferred move **h2h1**. Synthesis: 1 retrieved analog transferred to Kg6 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



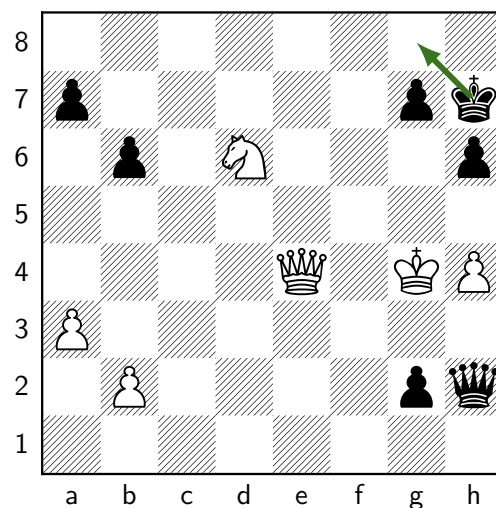
(a)



(b)

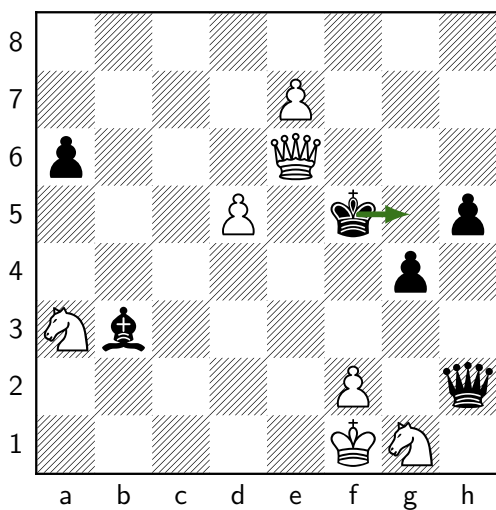


(c)

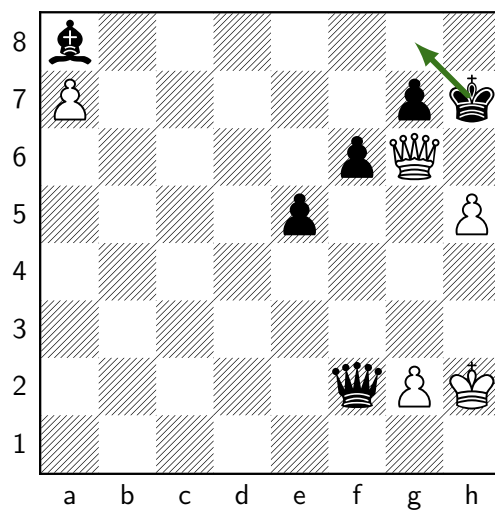


(d)

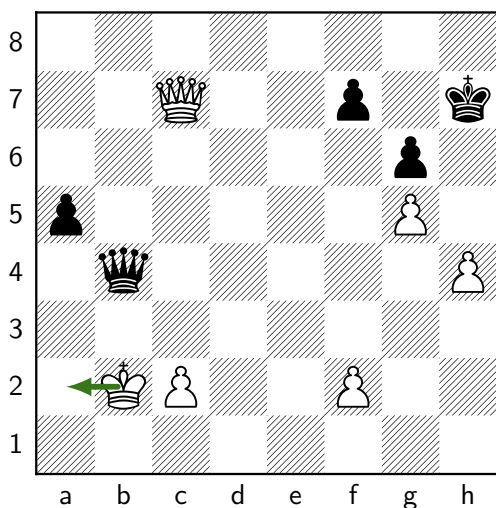
Figure 53: **Move 60, Black: Kf5.** (a) Our position; Melanie plays Kf5 (g6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1062` (ply 114) [structural distance $d=30$, QAP obj=57]: the master played Kh5; under μ our Kg3 plays the role of that game’s Kh4; cross-type roles: our Pa6 as their Bb1, our Bb3 as their Pa4; transferred move g6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#434` (ply 124) [structural distance $d=30$, QAP obj=58]: the master played Kd1; under μ our Kg3 plays the role of that game’s Ke1; cross-type roles: our ng1 as their pa5, our na3 as their pb6; transferred move g6h7. (d) Retrieved from `ccr1_top_p0002#158` (ply 115) [structural distance $d=35$, QAP obj=51]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kh2; cross-type roles: our na3 as their Pb6, our pf2 as their nd6; transferred move g6h7. Synthesis: 1 retrieved analog transferred to Kf5 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



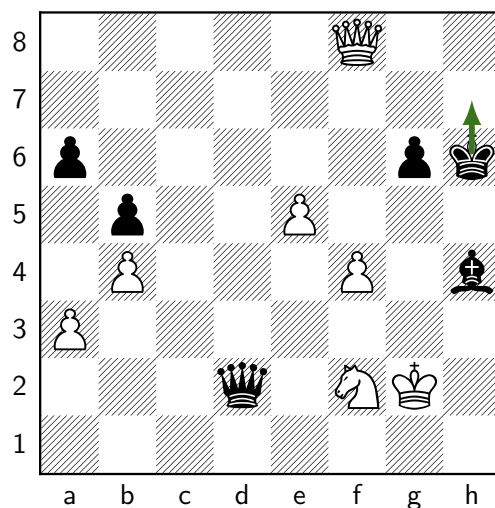
(a)



(b)



(c)



(d)

Figure 54: **Move 61, Black: Kg5.** (a) Our position; Melanie plays Kg5 (f5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1610` (ply 113) [structural distance $d=37$, QAP obj=86]: the master played Kg8; under μ our Kf4 plays the role of that game’s Kh2; cross-type roles: our Qh2 as their Ba8, our ng1 as their pa7; transferred move f5g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1289` (ply 104) [structural distance $d=38$, QAP obj=56]: the master played Ka2; under μ our Kf4 plays the role of that game’s Kb2; cross-type roles: our Qh2 as their Pg5, our ng1 as their pg6; transferred move f5g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#313` (ply 125) [structural distance $d=37$, QAP obj=48]: the master played Kh7; under μ our Kf4 plays the role of that game’s Kh3; cross-type roles: our na3 as their Pa6, our ng1 as their pf4; transferred move f5g5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg5 (summed QAP objective 393, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

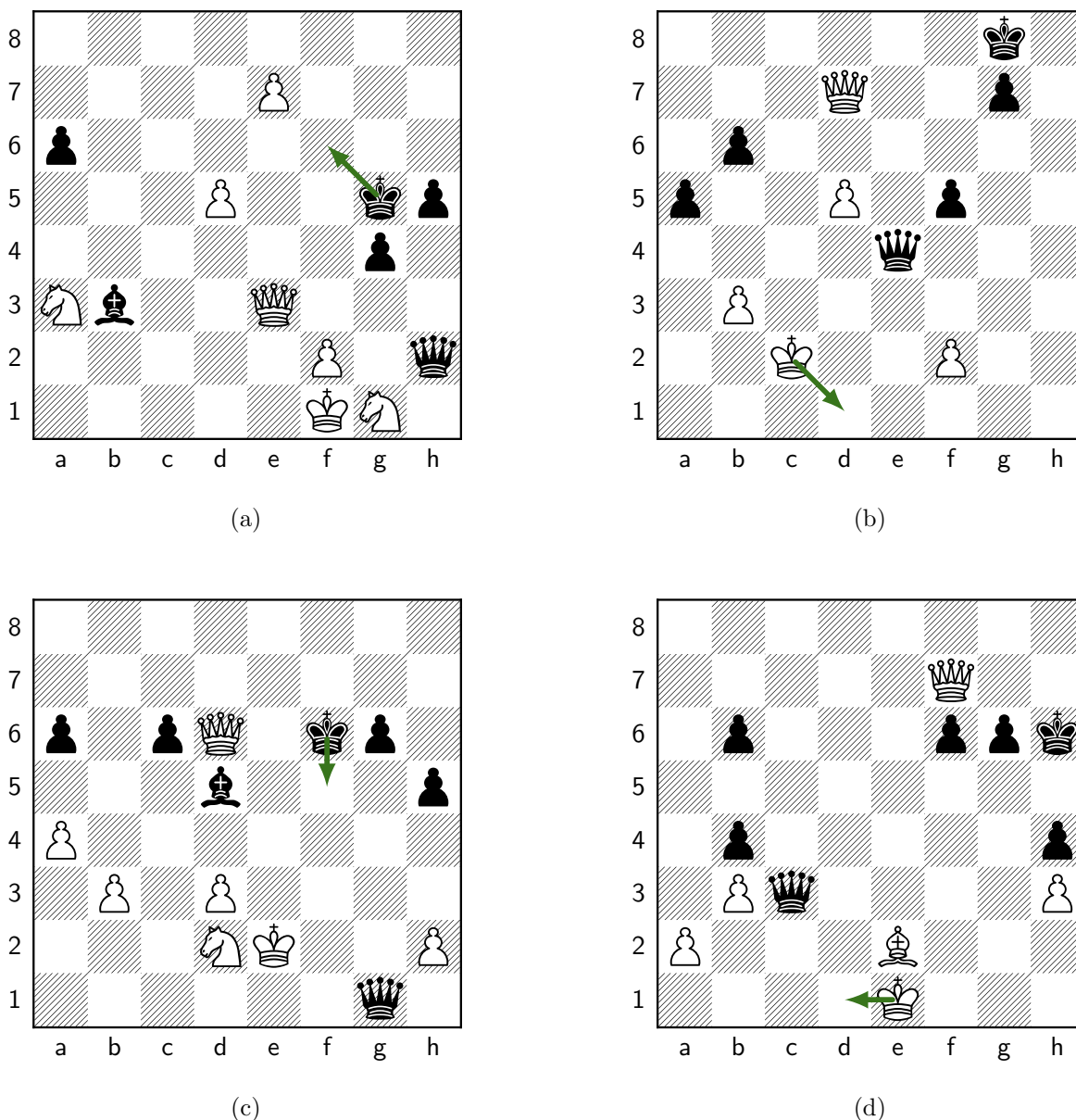
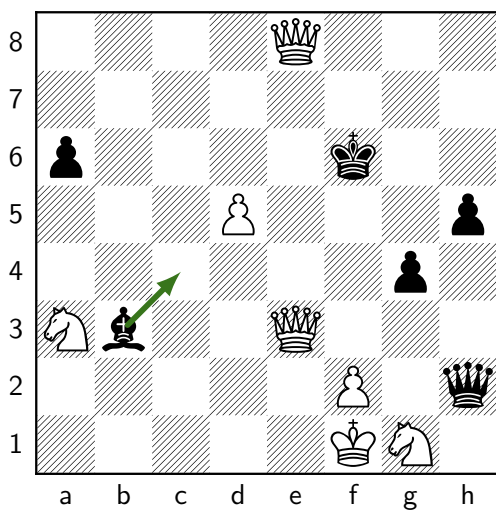
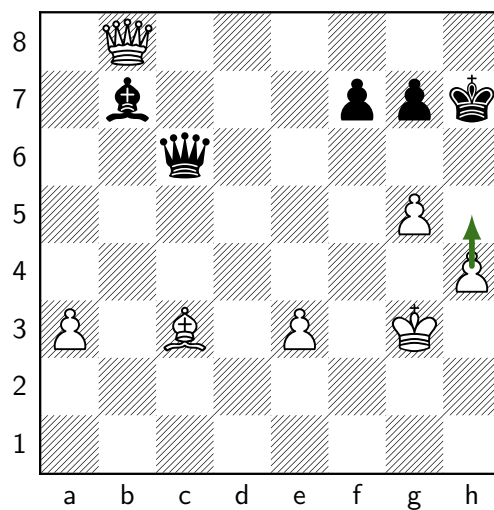


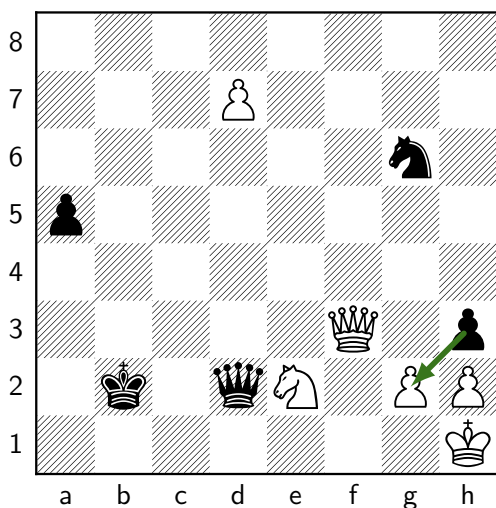
Figure 55: **Move 62, Black: Kf6**. (a) Our position; Melanie plays Kf6 (g5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#434` (ply 120) [structural distance $d=40$, QAP obj=77]: the master played Kd1; under μ our Kg4 plays the role of that game’s Kc2; cross-type roles: our Bb3 as their pb6, our ng1 as their pg7; transferred move g5f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#243` (ply 119) [structural distance $d=37$, QAP obj=80]: the master played Kf5; under μ our Kg4 plays the role of that game’s Kf3; cross-type roles: our na3 as their Bd5, our Bb3 as their ph2; transferred move g5f5. (d) Retrieved from `ccrl_top_p0002#312` (ply 112) [structural distance $d=40$, QAP obj=72]: the master played Kd1; under μ our Kg4 plays the role of that game’s Ke1; cross-type roles: our Pg4 as their Be2, our Bb3 as their Pb3; transferred move g5h4. Synthesis: 3 retrieved analogs transferred to Kf6 (summed QAP objective 210, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



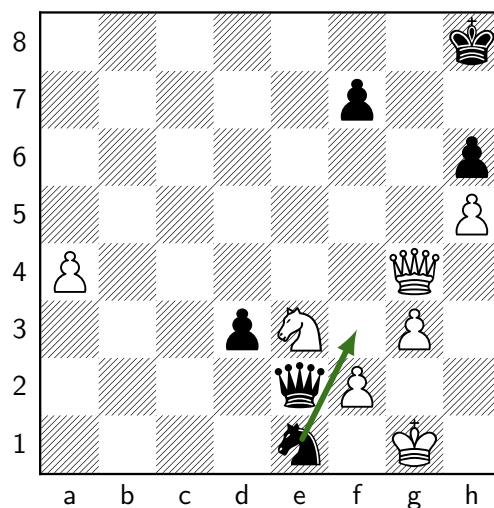
(a)



(b)

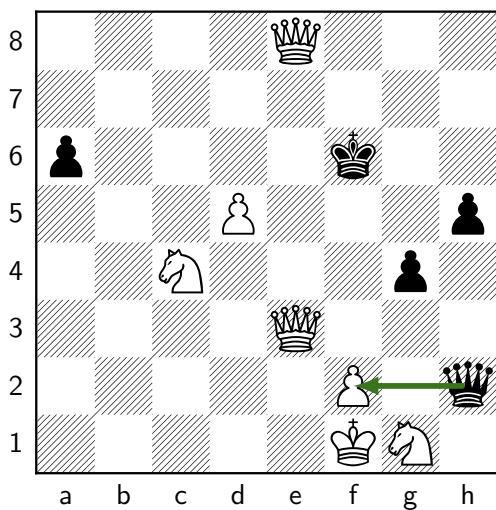


(c)

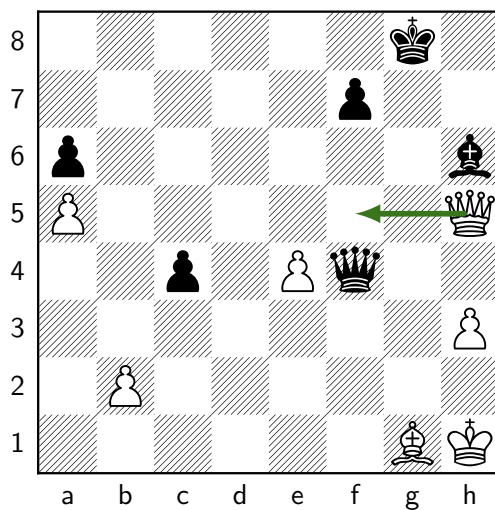


(d)

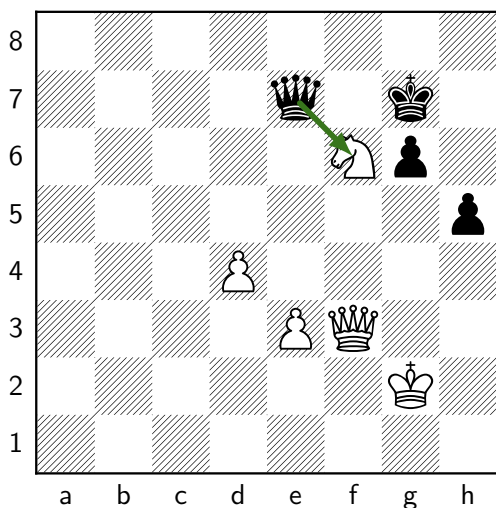
Figure 56: **Move 63, Black: Bc4+.** (a) Our position; Melanie plays Bc4+ (b3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1840` (ply 126) [structural distance $d=58$, QAP obj=35]: the master played h5; under μ our Ph_4 plays the role of that game’s Ph_4 ; cross-type roles: our $qe8$ as their $Pe3$, our $ng1$ as their $bb7$; transferred move h5h4. (c) Retrieved from `ccr1_top_p0000#3342` (ply 115) [structural distance $d=50$, QAP obj=29]: the master played hxg2+; under μ our $Bb6$ plays the role of that game’s $Ph6$ (a **cross-type** correspondence); cross-type roles: our $Pg4$ as their $Ng6$, our $na3$ as their $ph2$; transferred move b3c2. (d) Retrieved from `ccr1_top_p0000#3097` (ply 127) [structural distance $d=56$, QAP obj=29]: the master played Nf3+; under μ our $Pg5$ plays the role of that game’s $Ne8$ (a **cross-type** correspondence); cross-type roles: our $Bb3$ as their $Pf7$, our $na3$ as their $ph5$; transferred move g4g3. Synthesis: no retrieved analog transferred to Bc4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



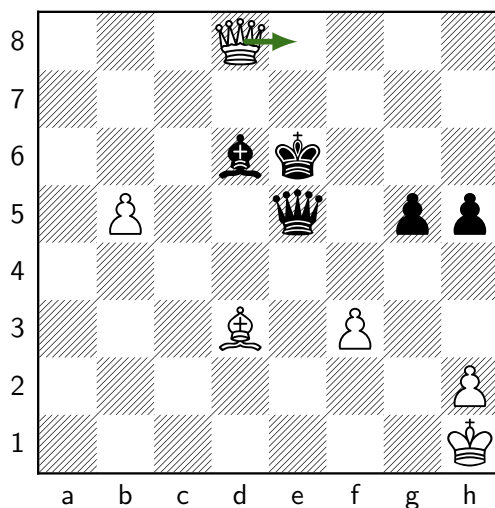
(a)



(b)

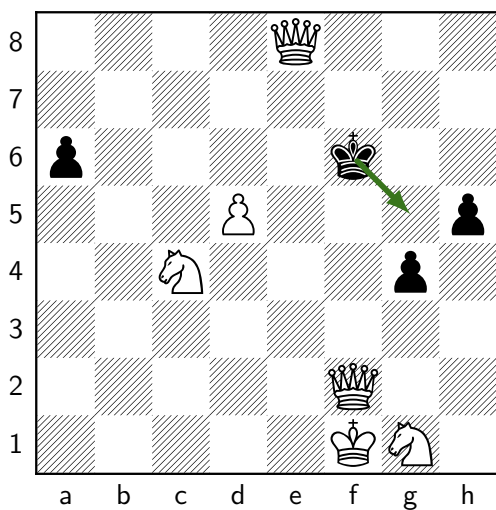


(c)

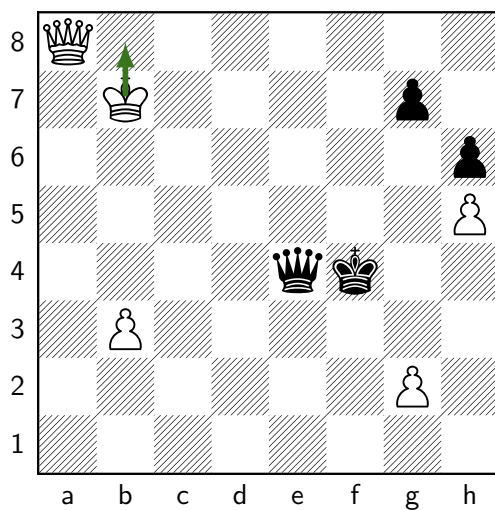


(d)

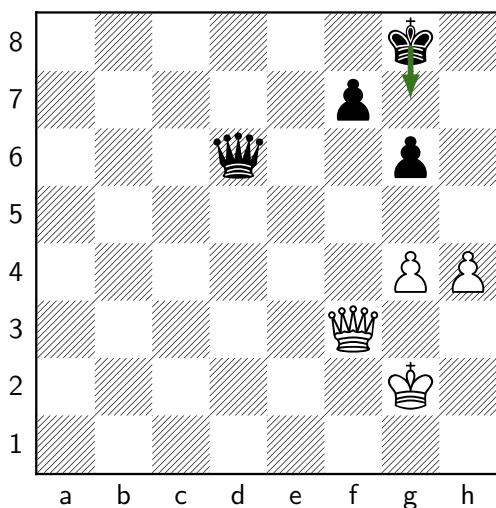
Figure 57: **Move 64, Black: Qxf2+.** (a) Our position; Melanie plays Qxf2+ (h2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#706` (ply 110) [structural distance $d=55$, QAP obj=27]: the master played Qf5; under μ our Qh7 plays the role of that game's Qh5; cross-type roles: our qe8 as their Pb2, our nc4 as their Pa5; transferred move h2f4. (c) Retrieved from `ccr1_top_p0000#1637` (ply 113) [structural distance $d=53$, QAP obj=23]: the master played Qxf6; under μ our Qh7 plays the role of that game's Qe2; transferred move h2e5. (d) Retrieved from `ccr1_top_p0001#3988` (ply 116) [structural distance $d=54$, QAP obj=17]: the master played Qe8+; under μ our Qh7 plays the role of that game's Qd8; cross-type roles: our qe8 as their Ph2, our Pa6 as their Bd3; transferred move h2h1. Synthesis: no retrieved analog transferred to Qxf2+ – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



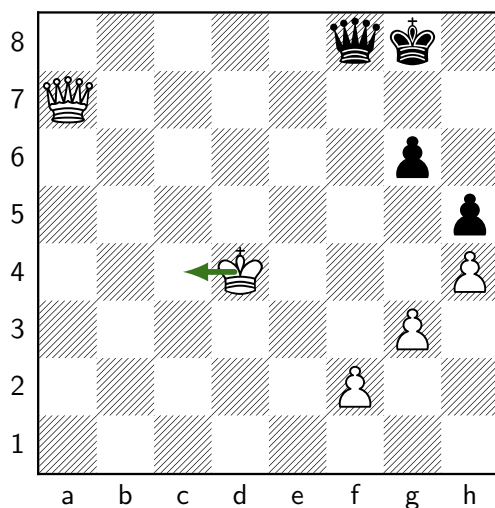
(a)



(b)



(c)



(d)

Figure 58: **Move 65, Black: Kg5.** (a) Our position; Melanie plays Kg5 (f6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1098` (ply 130) [structural distance $d=39$, QAP obj=52]: the master played Kb8; under μ our Kf3 plays the role of that game’s Kb7; cross-type roles: our qe8 as their ph6, our nc4 as their pg7; transferred move f6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2939` (ply 121) [structural distance $d=45$, QAP obj=4]: the master played Kg7; under μ our Kf3 plays the role of that game’s Kg1; cross-type roles: our qe8 as their Pf7, our Pa6 as their Qd6; transferred move f6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2856` (ply 116) [structural distance $d=44$, QAP obj=4]: the master played Kc4; under μ our Kf3 plays the role of that game’s Kd4; cross-type roles: our ng1 as their ph5, our qe8 as their pg6; transferred move f6g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg5 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

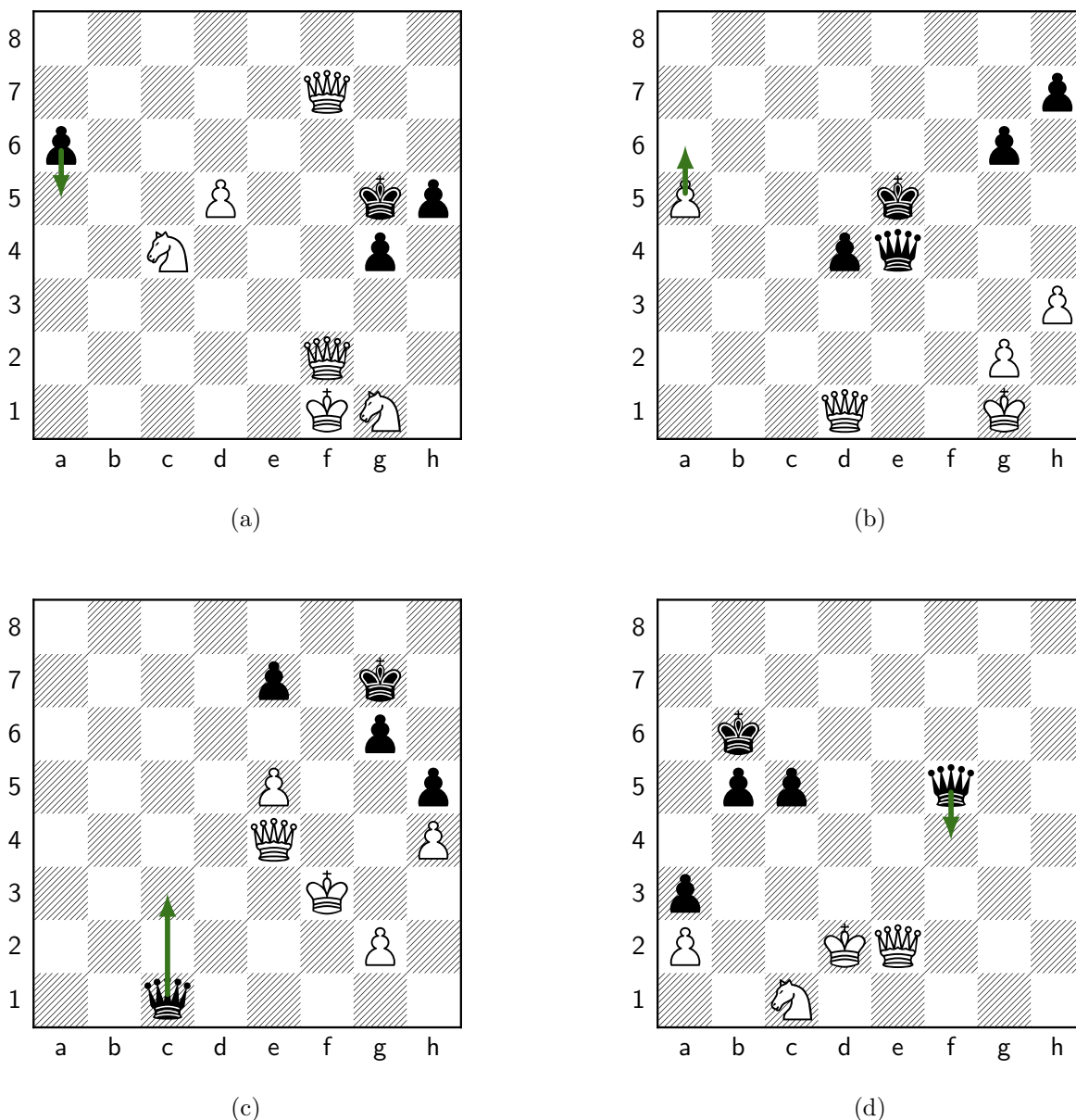


Figure 59: **Move 66, Black: a5.** (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#737` (ply 120) [structural distance $d=68$, QAP obj=23]: the master played a6; under μ our *Pa3* plays the role of that game’s *Pa5*; cross-type roles: our *qf2* as their *pg6*; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#156` (ply 123) [structural distance $d=67$, QAP obj=22]: the master played Qc3+; under μ our *Pa3* plays the role of that game’s *Qc8* (a **cross-type** correspondence); cross-type roles: our *nc4* as their *Pe7*, our *ng1* as their *ph4*; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1379` (ply 133) [structural distance $d=60$, QAP obj=15]: the master played Qf4+; under μ our *Pa3* plays the role of that game’s *Qf4* (a **cross-type** correspondence); cross-type roles: our *ng1* as their *pa2*, our *qf2* as their *nc1*; transferred move a6a5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to a5 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.